

ROUTERGFM: INDUCTIVE ROUTING AND LOCAL EXPERTISE TRANSFER FOR GRAPH LEARNING

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ABSTRACT

Reusing pretrained *graph neural networks* (GNNs) requires deciding which experts suit a new application and how to combine their predictions across graph contexts. Application-level selection does not resolve instance-specific contributions, while learning a new local router can require more target supervision than is available. We propose ROUTERGFM, a graph mixture-of-experts framework that combines inductive expert selection with context-guided local integration. A shared application-expert scorer over a heterogeneous context graph selects experts from metadata and historical evaluations before executing candidate GNNs on target data. After fitting the selected task heads, expert-conditioned retrieval transfers historical evidence about local expert performance into instance-dependent prediction weights, while the pretrained GNNs and router remain fixed. Our analysis separates expert coverage, local estimation error, and mixture weighting, and characterizes the limits of metadata-only expert insertion. Experiments on nine datasets cover node classification, link prediction, graph classification, and graph regression. In the fixed-team diagnosis, local integration reduces average and worst-context Brier loss by 11.7% and 17.3% relative to application-level weighting. Further analyses distinguish useful expert coverage from accurate integration and identify when unreliable historical evidence limits transfer.

1 INTRODUCTION

Graph foundation models aim to make pretrained knowledge reusable across graph domains and prediction tasks (Liu et al., 2024; Wang et al., 2024; 2025b). However, differences in node attributes, neighborhood structure, and supervision make the usefulness of a pretrained model application-dependent. A diverse pretrained expert pool preserves different model choices, but its use requires deciding both which experts to select and how to combine them.

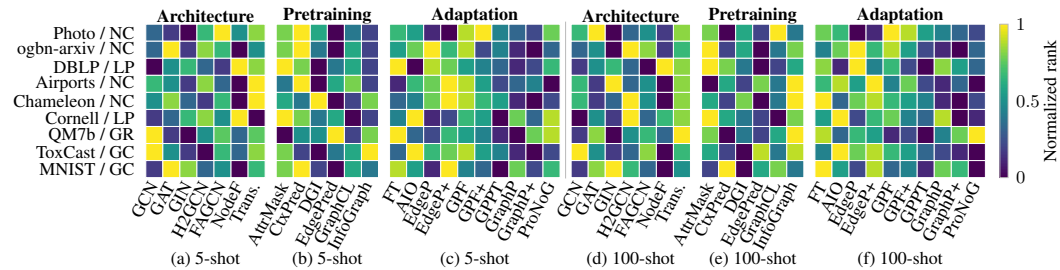


Figure 1: Configuration rankings across nine graph datasets. Each cell ranks the best reported configuration containing the indicated factor, with other factors varied and omitted results excluded. Ranks are normalized within each dataset and panel (higher: better; ties use average ranks), minimizing QM7b MAE and maximizing accuracy or ROC-AUC elsewhere. FT denotes full fine-tuning; LP panels follow the method-specific conventions in Appendix B.3.

Our study of seven architectures, six pretraining objectives, and ten adaptation strategies across nine datasets illustrates the difficulty of a fixed model choice (Figure 1). On Photo (Shchur et al., 2018), the architecture of the best observed configuration changes from FAGCN (Bo et al., 2021) at five shots (81.27%) to GAT (Veličković et al., 2018) at 100 shots (89.35%). On Chameleon (Rozemberczki et al., 2021), it changes from Transformer (Vaswani et al., 2017) (35.39%) to H2GCN (Zhu

et al., 2020) (42.30%). These configuration-level differences motivate application- and budget-dependent expert selection. A second decision remains after selection: an expert that performs well on average need not be the most useful expert in every local context. We therefore ask: *which pretrained experts should serve a new application, and how should their contributions depend on local graph context?*

The first question concerns *inductive expert selection*. Graph mixtures exploit different processing rules and source models (Wang et al., 2023; Xia & Huang, 2026; Liu et al., 2025a), while model-selection methods learn application-model relationships from historical evaluations (Park et al., 2023a; Cai et al., 2026). Both sides can change: new applications introduce prediction problems, while new checkpoints expand the candidate pool. A shared scorer can represent these additions through metadata and historical relations, before executing candidates on the target application.

The second question concerns *context-guided local integration*. Existing graph mixtures assign instance-dependent weights (Zeng et al., 2024; Ma et al., 2024; Meena et al., 2026), but a small target support set may not cover the contexts needed to learn a reliable gate. Historical evaluations record the graph contexts in which each expert performs well. Transferring this information requires comparing context-expert pairs across applications, without relearning a router for each target.

We propose ROUTERGFM, a graph mixture-of-experts framework that separates application-level expert selection from local prediction integration. A heterogeneous context graph links applications, experts, and their construction factors, supporting a shared scorer that selects a small expert team before target-side candidate execution. Expert-conditioned retrieval then transfers centered historical performance differences to determine each selected expert’s contribution to a query. Only the selected task heads are fitted on target support examples; pretrained GNNs and router parameters remain fixed. Application-masked training learns both decisions without exposing the historical evaluations being predicted.

Our contribution is the transfer of historical context-expert information across both selection and integration. The theoretical analysis relates prediction risk to expert coverage and estimation reliability, and distinguishes inserting an unseen expert from accurately predicting its behavior. Experiments on nine datasets span four task types, including multi-label molecular classification on ToxCast. The fixed-team diagnosis shows 11.7% lower average Brier loss and 17.3% lower worst-cell loss than application-level weighting, without changing the expert predictions. Analyses of team size, source calibration, and incomplete historical evidence explain when this transfer is useful and where it remains limited.

2 RELATED WORK

Graph foundation models. OFA, GFT, and GIT share knowledge across graph tasks through unified representations or transferable tree structures (Liu et al., 2024; Wang et al., 2024; 2025b). GraphAlign aligns features across graphs, MDGFM aligns topology, and BRIDGE combines feature alignment with mixture-based prompting (Hou et al., 2024; Wang et al., 2025a; Yuan et al., 2025). AutoGFM customizes architectures within a shared network (Chen et al., 2025). Graph prompting generally adapts a given pretrained GNN (Sun et al., 2022; Liu et al., 2023; Sun et al., 2023; Fang et al., 2023; Fu et al., 2025); ProNoG conditions prompts on non-homophilic patterns (Yu et al., 2025), while CTP supports cross-graph prompting without target-side parameter updates (Chen et al., 2027). ROUTERGFM instead reuses independently pretrained GNNs through expert selection and prediction integration after task-head fitting.

Graph mixtures and model reuse. GMoE and Node-MoE select aggregation ranges or graph filters, while Mowst combines feature- and structure-based experts (Wang et al., 2023; Han et al., 2024; Zeng et al., 2024). GraphMoRE and GeoMoE use geometric experts; TopExpert specializes by molecular topology, and GraphMETRO aligns experts for distribution shifts (Guo et al., 2025; Cao et al., 2026; Kim et al., 2023; Wu et al., 2024). AnyGraph routes across graphs, OMOG integrates source-specific models, and GMoPE learns prompt-conditioned experts (Xia & Huang, 2026; Liu et al., 2025a; Wang et al., 2025c). KDEM/PPEM improve expert merging, whereas OGMM uses generated graphs to merge and adapt heterogeneous models (Liu et al., 2025b; Wang et al., 2026). Our framework retains the pretrained parameters and transfers historical evidence into prediction weights rather than merging model parameters.

Inductive model selection. MetaGL learns evaluation-free graph-model selection, and GLEMOS benchmarks this problem (Park et al., 2023a;b). ModelLens recommends unseen models for unseen datasets using metadata and historical evaluations (Cai et al., 2026). GraphRouter uses a heterogeneous task-query-LLM graph, while RouteLLM learns from preference supervision (Feng et al., 2025; Ong et al., 2025). LogME and Model Spider assess pretrained-model transferability, and Quick-Tune also selects fine-tuning configurations (You et al., 2021; Zhang et al., 2023; Pineda Arango et al., 2024). These methods address which models to use; ROUTERGFM additionally transfers historical local evaluations to integrate the selected graph experts.

Local expert integration. Link-MoE learns node-pair-specific weights, and SAGMM uses topology-aware gating over diverse architectures, including a frozen-expert variant (Ma et al., 2024; Meena et al., 2026). META-DES learns local classifier competence for dynamic ensemble selection (Cruz et al., 2015). Our focus is cross-application transfer of local competence: an expert-conditioned retrieval map uses historical context-expert evaluations to weight an application-specific team without fitting a target-specific router.

3 METHODOLOGY

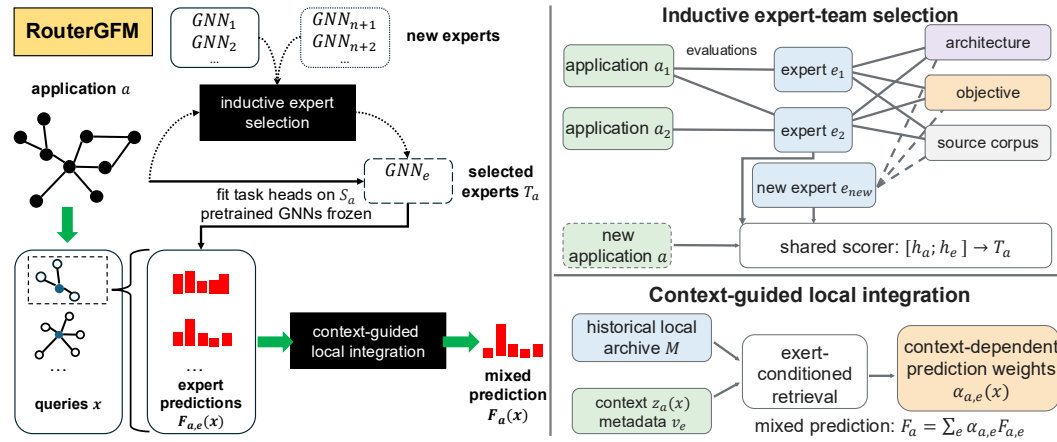


Figure 2: Overview of ROUTERGFM.

ROUTERGFM makes two decisions at different scales: which pretrained GNNs to use for an application, and how to combine their predictions for each node, node pair, or graph (Figure 2). Historical evaluations provide two complementary signals: application-average performance for selecting a team, and context-dependent performance differences for integrating its predictions. Only the selected task heads are fitted at deployment; pretrained GNNs and router parameters remain fixed.

3.1 PROBLEM FORMULATION

An application a specifies a graph dataset, a prediction task, and labeled training examples $S_a = \{(x_i, y_i)\}$. A graph $G = (V, E, \mathbf{X})$ has nodes V , edges E , and node features $\mathbf{X}$. For node or link prediction, an instance x contains a graph and a specified node or unordered node pair; for graph classification or regression, x is an entire graph. The output y is a node label, link-existence label, graph-level class label or multi-label vector, or numerical property, respectively. We call an instance with an unknown output a *query* and denote the query set by Q_a .

A node’s context is its surrounding subgraph with that node marked; a node pair’s context is an enclosing subgraph with both endpoints marked (Zhang & Chen, 2018). For graph-level prediction, the context is the input graph itself. A label-free descriptor $z_a(x)$ summarizes its structure, node features, and node or endpoint roles. The expert processes x , whereas the router uses $z_a(x)$ to relate its context to historical evaluations.

Given a pool of pretrained experts $\mathcal{E}$, our goal is to select a nonempty team T_a of at most K experts and combine their predictions for Q_a , without executing candidates for selection or training a target-

specific router. Each selected expert uses a frozen encoder f_{Θ_e} and a task head $g_{\omega_{a,e}}$ fitted on S_a . Writing $F_{a,e}(x) = g_{\omega_{a,e}}(f_{\Theta_e}(x))$, with the task readout included in f_{Θ_e} , the mixture is

$$F_a(x) = \sum_{e \in T_a} \alpha_{a,e}(x) F_{a,e}(x), \quad \alpha_{a,e}(x) \geq 0, \quad \sum_{e \in T_a} \alpha_{a,e}(x) = 1. \quad (1)$$

3.2 HISTORICAL EVALUATIONS AND CONTEXT GRAPH

For each historical application a , we fit task heads on S_a and evaluate them on a separate labeled diagnostic set D_a that is not used to fit the heads. For the nonempty subset $D_{a,e} \subseteq D_a$ evaluated by expert e , its average loss is

$$\bar{\mu}_{a,e} = \frac{1}{|D_{a,e}|} \sum_{(x,y) \in D_{a,e}} \ell_a(y, F_{a,e}(x)). \quad (2)$$

The task loss ℓ_a measures prediction error within a compatible task family (Appendix B). These averages supervise application-level selection, while the instance-level evaluations inform local integration.

We organize the averages in a heterogeneous context graph $\mathcal{H}$ with four node types: application/dataset, expert, architecture, and pretraining objective. This graph connects learning problems and models, rather than nodes of an input graph. Evaluation edges link applications to evaluated experts and carry $\bar{\mu}_{a,e}$ and $|D_{a,e}|$. Construction edges link experts to their architecture, objective, and source corpus, with source corpora using the application/dataset node type. These relations connect experts with shared construction factors while retaining their distinct empirical behavior.

3.3 INDUCTIVE EXPERT SELECTION

Each context-graph node v has features x_v combining a frozen text embedding of its description with numerical metadata. Application features describe the graph data, task, and supervision budget; expert features describe the architecture, objective, pretraining data, and adaptation rule. The same description and projection functions represent both historical and newly inserted applications or experts. A type-specific projection initializes $h_v^{(0)} = P_{\kappa(v)}(x_v)$, where $\kappa(v)$ is the node type.

A heterogeneous GNN passes messages in both directions along construction and evaluation relations, combining descriptions with historical performance:

$$h_v^{(l+1)} = h_v^{(l)} + \sigma \left(\sum_{r \in \mathcal{R}} \sum_{u \in N_r(v)} \psi_r^{(l)}(h_u^{(l)}, h_v^{(l)}, b_{uv}) \right). \quad (3)$$

Here $N_r(v)$ contains incoming neighbors under relation r , b_{uv} holds edge attributes, and $\psi_r^{(l)}$ is a relation-specific message function (Schlichtkrull et al., 2018; Hu et al., 2020c). The residual connection preserves the node’s own representation, σ is a nonlinear activation, and h_v denotes the final representation.

Selection is an application–expert edge-attribute prediction problem: estimate each candidate’s average loss after task-head fitting. For the nonempty compatible pool $\mathcal{E}_a \subseteq \mathcal{E}$ (Appendix B), a shared scorer selects

$$\hat{\mu}_{a,e} = \zeta(q_\theta([h_a; h_e])), \quad T_a = \text{TopK}_{e \in \mathcal{E}_a}(-\hat{\mu}_{a,e}). \quad (4)$$

Here brackets denote concatenation, $\zeta(t) = \log(1 + \exp t)$ ensures nonnegative estimates, and TopK retains up to K experts with the lowest estimates. The parameters θ comprise the projections, graph encoder, and scorer. A new application enters through metadata without evaluation edges; a new expert enters through metadata and construction links, adding architecture nodes when needed. Shared functions process these additions without router retraining or candidate GNN execution on the target. Selection returns the team T_a and its application-level loss estimates $\hat{\mu}_{a,e}$, which anchor the subsequent local correction.

3.4 CONTEXT-GUIDED LOCAL INTEGRATION

The selected team and fitted heads are shared across queries, but an expert’s usefulness can vary with graph context. For an instance and output (X, Y) from application a , define the conditional risk

$$r_{a,e}(z) = \mathbb{E}[\ell_a(Y, F_{a,e}(X)) \mid z_a(X) = z], \quad \mu_{a,e} = \mathbb{E}[r_{a,e}(z_a(X))].$$

These expectations condition on the support set, fitted experts, context graph, archive, and team. We estimate this context-conditional loss by transferring how historical contexts change an expert’s performance relative to its application average.

Let b distinguish a historical application from the target a . Clustering its diagnostic descriptors produces B_b cells $C_{b,j}$ with representatives $c_{b,j}$, shared across experts and grouping similar contexts rather than necessarily adjacent nodes. For $D_{b,j,e} = \{(x, y) \in D_{b,e} : z_b(x) \in C_{b,j}\}$ and $n_{b,j,e} = |D_{b,j,e}| > 0$, we store

$$\bar{r}_{b,j,e} = \frac{1}{n_{b,j,e}} \sum_{(x,y) \in D_{b,j,e}} \ell_b(y, F_{b,e}(x)). \quad (5)$$

The local archive $\mathcal{M}$ associates each representative with its expert, local and application-average losses, evaluation count, and source application. Descriptors use structural statistics, feature norms and spectra, node or endpoint roles, and task indicators; standardization uses historical training applications only. Link contexts exclude the candidate link and held-out positive links (Appendix B).

A learned key $k_\phi(z, v_e)$ represents context $z = z_a(x)$ together with expert metadata $v_e = h_e^{(0)}$, before evaluation-edge messages. For archive record i with context c_i and expert e_i , define $d_i(z, e) = \|k_\phi(z, v_e) - k_\phi(c_i, v_{e_i})\|_2$. We retrieve up to J nearest records from other applications with matching task/loss families, support budgets, and normalization conventions, with a per-application cap. For their indices $\mathcal{N}_J(z, e)$, the kernel weights are

$$w_i(z, e) = \frac{\exp(-d_i(z, e)^2/h^2)}{\sum_{i' \in \mathcal{N}_J(z, e)} \exp(-d_{i'}(z, e)^2/h^2)}, \quad h > 0. \quad (6)$$

Expert conditioning allows the same query context to retrieve different evidence for different team members, including evaluations of related checkpoints when an expert’s own records are unavailable.

Write $\bar{r}_i$ for a record’s local loss and $\bar{\mu}_i$ for the corresponding expert’s average on its source application. We transfer their difference rather than the raw loss, correcting the target application’s estimate as

$$\hat{r}_{a,e}(z) = \hat{\mu}_{a,e} + \rho \sum_{i \in \mathcal{N}_J(z, e)} w_i(z, e)(\bar{r}_i - \bar{\mu}_i), \quad \rho \in [0, 1]. \quad (7)$$

A negative difference indicates better-than-average local performance and lowers the estimated loss. Centering removes additive offsets, but not loss-scale differences or changes in expert behavior across applications. The coefficient ρ controls the correction; $\rho = 0$ or an empty retrieval set gives zero correction and recovers application-level weighting.

The resulting prediction weights use temperature $\tau > 0$:

$$\alpha_{a,e}(x) = \frac{\exp(-\hat{r}_{a,e}(z_a(x))/\tau)}{\sum_{e' \in T_a} \exp(-\hat{r}_{a,e'}(z_a(x))/\tau)}. \quad (8)$$

Over the probability simplex $\Delta(T_a)$, these weights minimize

$$\min_{\alpha \in \Delta(T_a)} \sum_{e \in T_a} \alpha_e \hat{r}_{a,e}(z) + \tau \sum_{e \in T_a} \alpha_e \log \alpha_e.$$

This entropy-regularized objective balances estimated individual-expert loss against concentrated weights; the final mixture loss is evaluated after combining predictions. Equation 1 combines aligned class probabilities, multi-label assay probabilities, or regression outputs in common units. Expert predictions are mixture inputs, not retrieval keys, and no parameters are updated for individual queries.

3.5 OPTIMIZATION AND INFERENCE

Training treats each historical application $b \in \mathcal{A}_{\text{tr}}$ as a new application by hiding its evaluation edges and reverses, local records, and label-derived aggregates before encoding and retrieval. The hidden losses serve only as supervision, requiring predictions from other applications' evidence. For evaluated experts $\Omega_b = \{e \in \mathcal{E}_b : |D_{b,e}| > 0\}$, we use

$$\mathcal{L}_{\text{glob}}(b) = \frac{1}{|\Omega_b|} \sum_{e \in \Omega_b} \text{Huber}(\hat{\mu}_{b,e}, \bar{\mu}_{b,e}) + \lambda_r \mathcal{L}_{\text{rank}}(b), \quad (9)$$

$$\mathcal{L}_{\text{loc}}(b) = \frac{1}{|\Omega_b|} \sum_{e \in \Omega_b} \frac{1}{|D_{b,e}|} \sum_{(x,y) \in D_{b,e}} (\hat{r}_{b,e}(z_b(x)) - \ell_b(y, F_{b,e}(x)))^2. \quad (10)$$

Huber regression fits average losses, ListMLE fits rankings (Xia et al., 2008), and squared local supervision fits context-dependent losses. With nonnegative loss weights, the selection and retrieval parameters minimize

$$\min_{\theta, \phi} \mathcal{L} = \frac{1}{|\mathcal{A}_{\text{tr}}|} \sum_{b \in \mathcal{A}_{\text{tr}}} [\mathcal{L}_{\text{glob}}(b) + \lambda_\ell \mathcal{L}_{\text{loc}}(b)]. \quad (11)$$

Historical heads, pretrained GNNs, and recorded evaluations remain fixed. Within an update, retrieved record identities are fixed but kernel weights are differentiable; search keys are refreshed between updates (Appendix B.3).

Router checkpoints and hyperparameters are selected on historical validation applications; tasks, budgets, and splits from one base dataset remain in the same group. Deployment selects T_a , fits its task heads on S_a , and predicts Q_a using context-dependent weights (Algorithm 1). Optional source calibration adds a new expert's evaluations on historical applications without target-query labels or router updates.

3.6 THEORETICAL ANALYSIS

Fix the application, fitted experts, archive, and team; suppress a and write $\mathcal{E} = \mathcal{E}_a$ and $T = T_a$. Using the conditional risks in Section 3.4, let $R(F) = \mathbb{E}[\ell(Y, F(X))]$ and $R_{\text{loc}} = \mathbb{E}[\min_{e \in \mathcal{E}} r_e(Z)]$, where $Z = z_a(X)$.

Theorem 1 (Expert coverage and local integration). *Assume convex loss, integrable risks, and mixture weights depending only on the permitted context. For $K = |T|$, let $\Gamma_K = \mathbb{E}[\min_{e \in T} r_e(Z) - \min_{e \in \mathcal{E}} r_e(Z)]$ and $\epsilon(Z) = \max_{e \in T} |\hat{r}_e(Z) - r_e(Z)|$. The mixture in Equation 1, with weights from Equation 8, satisfies*

$$R(F) - R_{\text{loc}} \leq \Gamma_K + 2\mathbb{E}[\epsilon(Z)] + \tau \log K. \quad (12)$$

The bound separates omitted expertise from local estimation error and mixture temperature. The comparator selects individual experts by context rather than an optimal convex ensemble, and the guarantee concerns convex prediction loss rather than accuracy or ROC-AUC. Theorem 3 further relates estimation error to retrieval distance, within-cell variation, cross-application changes, and evaluation noise. These results explain why expert coverage alone is insufficient; Appendix A provides the assumptions and proofs.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Applications and expert pool. We evaluate nine datasets across four tasks: node classification (NC) on Photo (Shchur et al., 2018), ogbn-arxiv (Hu et al., 2020a), Airports (Ribeiro et al., 2017), and Chameleon (Rozemberczki et al., 2021); link prediction (LP) on DBLP (Bojchevski & Günnemann, 2018) and Cornell (Craven et al., 1998); graph classification (GC) on ToxCast (Wu et al., 2018) and MNIST (LeCun et al., 1998; Dwivedi et al., 2023); and graph regression (GR) on QM7b (Montavon et al., 2013). The expert pool contains 588 pretrained GNNs from seven architectures, seven objectives, and twelve source datasets before task-compatibility filtering. Single-label classification uses five or 100 support examples per class, regression uses five or 100 support graphs,

and ToxCast uses multitask support partitions. LP uses edge-based supervision rather than per-class shot counts; its two displayed blocks are not a link-budget scaling experiment. Appendix B summarizes adaptation and evaluation, with detailed implementation and configurations in the accompanying code.

Comparisons and metrics. Table 1 compares graph mixtures and matched-pool methods on all nine datasets. SAGMM-PE uses the matched frozen-expert inventory (Meena et al., 2026), and MetaGL-U uniformly averages experts selected by MetaGL (Park et al., 2023a). ROUTERGFM-G shares ROUTERGFM’s selected team and fitted heads but uses application-level prediction weights; both select $K = 5$ experts. We report five-seed means and standard deviations, using accuracy for single-label classification, ROC-AUC for LP and ToxCast, and raw-unit MAE for QM7b. Accuracy and ROC-AUC are percentages. Hit@ K measures inclusion of a best eligible expert, and regret@ K is the loss gap between the best shortlisted and best eligible individual experts. The configuration study and task-scoped comparisons appear in Appendices D and E.

Table 1: Graph mixtures and matched-pool methods on nine datasets, with means and standard deviations over five seeds. Bold and underline identify the best and second distinct means within each support block. LP follows the shared edge-split convention in Appendix B.3; the two blocks retain their respective reported scores, and identical baseline entries are not additional runs.

Methods	Homophilic graphs			Datasets			Graph-level tasks		
	E-commerce	Academic	Academic	Transport	Web	Web	Chemistry	Chemistry	Vision
	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	GR / MAE ↓	GC / AUC ↑	GC / Acc. ↑
<i>5-shot: graph mixture-of-experts models</i>									
AnyGraph	40.29 $\pm$ 6.39	24.91 $\pm$ 4.18	83.81 $\pm$ 1.78	37.08 $\pm$ 1.85	25.92 $\pm$ 1.52	57.20 $\pm$ 8.08	15.41 $\pm$ 1.90	50.32 $\pm$ 1.32	24.59 $\pm$ 4.02
GMoE	71.03 $\pm$ 5.25	5.86 $\pm$ 6.24	54.83 $\pm$ 0.77	26.29 $\pm$ 1.78	22.27 $\pm$ 2.35	58.28 $\pm$ 10.40	115.69 $\pm$ 1.04	49.02 $\pm$ 2.86	11.48 $\pm$ 4.00
Mowst	75.67 $\pm$ 4.87	23.61 $\pm$ 2.09	55.32 $\pm$ 0.46	28.44 $\pm$ 3.52	29.73 $\pm$ 4.72	64.21 $\pm$ 9.67	114.77 $\pm$ 1.20	49.87 $\pm$ 2.13	22.88 $\pm$ 3.38
GraphMoRE	77.74 $\pm$ 1.35	21.70 $\pm$ 2.36	54.10 $\pm$ 0.40	33.97 $\pm$ 5.06	28.22 $\pm$ 5.89	57.19 $\pm$ 9.03	26.07 $\pm$ 46.29	48.79 $\pm$ 1.35	23.40 $\pm$ 2.67
GMoPE	81.48 $\pm$ 1.26	32.91 $\pm$ 1.20	81.22 $\pm$ 1.62	40.42 $\pm$ 3.11	36.78 $\pm$ 3.01	75.91 $\pm$ 6.10	5.64 $\pm$ 0.28	57.01 $\pm$ 1.24	30.66 $\pm$ 2.31
<i>5-shot: matched-pool routing and frozen-expert mixtures</i>									
SAGMM-PE	82.10 $\pm$ 0.92	33.15 $\pm$ 0.88	82.80 $\pm$ 1.23	41.35 $\pm$ 2.51	37.05 $\pm$ 2.74	76.80 $\pm$ 5.20	5.47 $\pm$ 0.21	57.00 $\pm$ 1.90	30.11 $\pm$ 1.91
MetaGL-U	80.48 $\pm$ 0.92	31.20 $\pm$ 0.88	80.43 $\pm$ 1.40	39.95 $\pm$ 2.52	34.82 $\pm$ 2.42	73.36 $\pm$ 4.79	5.73 $\pm$ 0.22	56.35 $\pm$ 1.19	28.90 $\pm$ 1.73
ROUTERGFM-G	82.48 $\pm$ 1.06	33.47 $\pm$ 0.88	83.31 $\pm$ 1.26	41.55 $\pm$ 2.46	36.43 $\pm$ 2.74	76.33 $\pm$ 4.65	5.40 $\pm$ 0.26	58.29 $\pm$ 1.69	30.37 $\pm$ 1.96
ROUTERGFM	83.74 $\pm$ 0.89	42.92 $\pm$ 1.55	91.08 $\pm$ 0.45	47.09 $\pm$ 1.87	39.53 $\pm$ 2.68	79.50 $\pm$ 4.48	5.19 $\pm$ 0.26	60.98 $\pm$ 0.96	32.28 $\pm$ 1.84
<i>100-shot: graph mixture-of-experts models</i>									
AnyGraph	84.00 $\pm$ 1.90	50.24 $\pm$ 1.76	83.81 $\pm$ 1.78	40.40 $\pm$ 2.89	36.63 $\pm$ 2.21	57.20 $\pm$ 8.08	11.92 $\pm$ 0.73	53.15 $\pm$ 0.65	33.23 $\pm$ 1.60
GMoE	85.76 $\pm$ 2.14	25.68 $\pm$ 0.57	54.83 $\pm$ 0.77	38.18 $\pm$ 3.77	33.09 $\pm$ 1.69	58.28 $\pm$ 10.40	113.74 $\pm$ 2.41	55.51 $\pm$ 2.71	34.65 $\pm$ 3.08
Mowst	87.19 $\pm$ 0.77	38.97 $\pm$ 0.64	55.32 $\pm$ 0.46	42.68 $\pm$ 1.89	35.24 $\pm$ 0.93	64.21 $\pm$ 9.67	103.16 $\pm$ 1.53	52.23 $\pm$ 2.52	28.32 $\pm$ 6.74
GraphMoRE	85.56 $\pm$ 0.72	33.85 $\pm$ 0.92	54.10 $\pm$ 0.40	38.94 $\pm$ 1.90	35.06 $\pm$ 0.70	57.19 $\pm$ 9.03	22.93 $\pm$ 39.40	52.78 $\pm$ 0.47	29.49 $\pm$ 0.44
GMoPE	89.66 $\pm$ 0.57	50.26 $\pm$ 1.03	81.22 $\pm$ 1.62	48.11 $\pm$ 1.91	44.52 $\pm$ 1.73	75.91 $\pm$ 6.10	5.04 $\pm$ 0.16	59.85 $\pm$ 0.56	44.63 $\pm$ 1.52
<i>100-shot: matched-pool routing and frozen-expert mixtures</i>									
SAGMM-PE	89.43 $\pm$ 0.47	49.72 $\pm$ 0.80	82.80 $\pm$ 1.23	48.54 $\pm$ 1.44	44.35 $\pm$ 1.17	76.80 $\pm$ 5.20	4.99 $\pm$ 0.11	60.10 $\pm$ 1.46	43.62 $\pm$ 1.13
MetaGL-U	89.03 $\pm$ 0.45	48.06 $\pm$ 0.84	80.43 $\pm$ 1.40	47.48 $\pm$ 1.76	42.33 $\pm$ 1.16	73.36 $\pm$ 4.79	5.14 $\pm$ 0.12	59.84 $\pm$ 1.50	42.51 $\pm$ 1.05
ROUTERGFM-G	89.71 $\pm$ 0.40	50.79 $\pm$ 0.81	83.31 $\pm$ 1.26	49.23 $\pm$ 1.53	44.05 $\pm$ 1.45	76.33 $\pm$ 4.65	4.91 $\pm$ 0.12	60.99 $\pm$ 1.22	44.16 $\pm$ 1.26
ROUTERGFM	94.89 $\pm$ 0.34	57.60 $\pm$ 0.60	92.04 $\pm$ 0.75	62.18 $\pm$ 1.20	59.14 $\pm$ 1.90	79.81 $\pm$ 4.55	4.42 $\pm$ 0.43	62.64 $\pm$ 1.13	46.18 $\pm$ 1.12

4.2 PREDICTIVE PERFORMANCE ACROSS APPLICATIONS

Local integration complements application-level selection. ROUTERGFM has the best mean among the methods in Table 1 in all eighteen displayed dataset–setting entries. Relative to ROUTERGFM-G, five-shot local integration adds 9.45 accuracy points on ogbn-arxiv, 5.54 on Airports, 3.10 on Chameleon, and 1.91 on MNIST, while reducing QM7b MAE from 5.40 to 5.19. ToxCast ROC-AUC increases from 58.29% to 60.98% at five shots and from 60.99% to 62.64% at 100 shots, giving gains of 2.69 and 1.65 points. The benefit therefore extends to multi-label molecular classification with the same selected predictors. The broader adaptation catalog remains task-dependent: at 100 shots, ProNoG obtains lower QM7b MAE than ROUTERGFM (4.06 versus 4.42; Table 3).

Stronger task heads do not remove the need for local weighting. The gain over ROUTERGFM-G increases from 1.26 to 5.18 accuracy points on Photo, from 5.54 to 12.95 on Airports, and from 3.10 to 15.09 on Chameleon. Across the five single-label classification datasets, the mean accuracy gain increases from 4.25 to 8.41 points; LP, ToxCast ROC-AUC, and regression are excluded from this accuracy-only average. However, the gains narrow on ogbn-arxiv (9.45 to 6.81 points) and

ToxCast (2.69 to 1.65 points). The integration benefit persists with more target supervision, but its change is application-dependent.

4.3 INDUCTIVE SELECTION AND EXPERT INSERTION

Historical relations improve expert recommendation. Figure 3(a) evaluates selection separately from prediction integration. At five shots, ROUTERGFM increases hit@5 from MetaGL’s 72% to 84% and reduces regret@5 from 0.032 to 0.019 (40.6%). Model Spider matches MetaGL’s hit@5 but has lower regret (0.029), distinguishing inclusion from the cost of missed selections. At 100 shots, ROUTERGFM reaches 88% hit@5 and 0.013 regret, compared with the best competing values of 80% and 0.022.

Expert insertion and competence transfer are distinct. With router parameters fixed, ROUTERGFM improves on MetaGL+metadata in all four insertion conditions, but loss rises from 0.225 for new configurations to 0.259 for unseen architectures and 0.263 under joint novelty (Figure 3(b)). This separates the ability to represent a new expert from the availability of historical evidence about its behavior. Adding evaluations on four source applications reduces joint-novelty loss to 0.227 and residual-estimation error from 0.086 to 0.046; eight applications give loss 0.224 (Table 11). The first four account for 92.3% of the observed loss reduction. Calibration supplies additional source evidence without target-query labels or router updates.

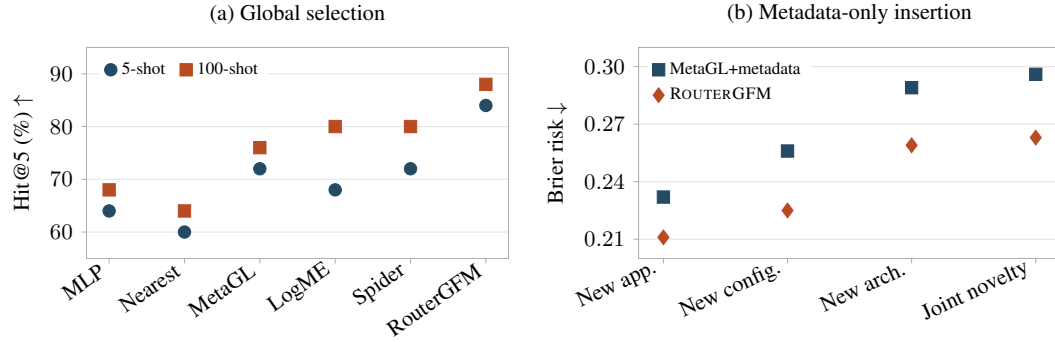


Figure 3: Application-level selection and metadata-only expert insertion. Hit@5 measures best-expert inclusion; insertion uses fixed router parameters without source calibration (Tables 9 and 10).

4.4 LOCAL INTEGRATION WITH EXPERT PREDICTIONS FIXED

Transferred local information improves a fixed set of predictions. Figure 4 fixes five expert checkpoints, their fitted heads, and their query predictions, and varies only the integration rule. ROUTERGFM reduces average Brier loss from 0.239 to 0.211 (11.7%) and worst-cell loss from 0.388 to 0.321 (17.3%) relative to application-level weighting. Simplex stacking reaches 0.235 average loss, whereas Local-MLP reaches 0.226; ROUTERGFM also lowers their worst-cell losses from 0.381 and 0.354 to 0.321. Because the gates differ in historical supervision, this comparison evaluates complete integration procedures on fixed predictions.

The correspondence between context and expert performance matters. The no-centering control gives Brier loss 0.228, while shuffling context–evaluation correspondence gives 0.244, exceeding the global-weighting loss of 0.239. Local-winner agreement increases from 43.6% under global weighting to 67.3% under local transfer (Table 4). The agreement and prediction results support context-sensitive contributions, while shuffling identifies a failure under unreliable correspondence.

4.5 EXPERT COVERAGE AND INTEGRATION LIMITS

Additional coverage has diminishing predictive value. Increasing K from one to five reduces Brier loss from 0.269 to 0.211 and raises local-winner coverage from 40.4% to 90.6% (Figure 5). Between $K = 8$ and $K = 16$, coverage increases from 96.1% to 98.8%, while loss changes from 0.209 to 0.212. The point estimates therefore show a prediction plateau despite increasing coverage,

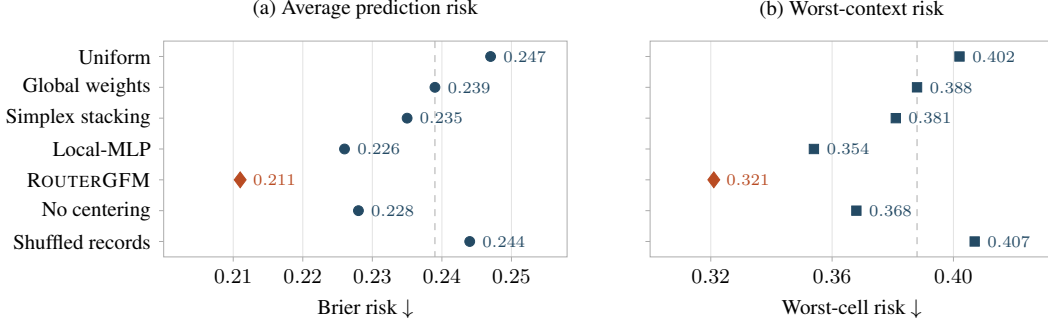


Figure 4: Context-guided integration with a fixed expert team. All methods share five fitted expert predictors; diamonds highlight ROUTERGFM, and dashed lines mark application-level weighting (Table 4). The two panels separate average prediction loss from the largest evaluated cell loss.

consistent with the distinction between omitted expertise and estimation error in Theorem 1. Five experts recover 96.7% of the observed loss reduction between one and eight experts with 37.5% fewer expert contributions, an execution-count comparison rather than a measured latency gain.

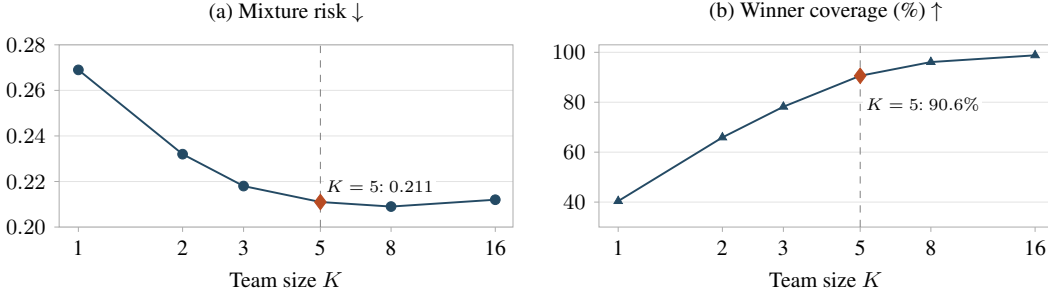


Figure 5: Expert coverage and prediction quality are distinct. Coverage increases with team size while prediction loss reaches a plateau; the marked $K = 5$ team captures most of the observed loss reduction (Table 12).

Shrinkage moderates unreliable evidence but does not remove it. Local integration raises loss from 0.216 to 0.220 under low specialization, but lowers it by 0.028 and 0.053 in the medium and high conditions (Figure 7(a)). With half the historical cells removed, source-validated shrinkage gives loss 0.227, compared with 0.235 for full correction ($\rho = 1$) and 0.239 for global weighting. Missing a context family or reversing residual relationships instead gives losses 0.242 and 0.251, both above global weighting, although below the unshrunk losses 0.281 and 0.334 (Figure 7(b)). Thus partial evidence can remain useful, whereas incompatible local behavior can outweigh the benefit of context matching. The shift-oriented comparisons in Appendix D.6 further show that no single method dominates every evaluated shift condition.

5 CONCLUDING REMARKS

ROUTERGFM reuses heterogeneous pretrained graph experts through inductive selection and context-guided local integration. Historical evaluations support both an application-level shortlist and query-specific prediction weights, while target adaptation is limited to the selected task heads. Experiments across nine datasets show that local integration improves the selected predictors across four task types, and the fixed-team diagnosis separates this benefit from changes in expert inventory. The theory and existing diagnostics identify two complementary requirements: a team must contain useful expertise, and its local contributions must be estimated reliably. Larger teams and metadata-based insertion alone do not meet both requirements; historical context coverage and transferable expert behavior remain central to effective model reuse.

AI USE STATEMENT

Generative AI tools assisted with literature checking, manuscript editing, methodology organization, figure editing, and preparation of numerical-audit code. The authors are responsible for verifying the final text, mathematical claims, experimental evidence, and bibliography.

ETHICS STATEMENT

Historical expert evaluations may underrepresent particular graph populations or structural contexts, and routing decisions can reproduce these coverage biases. We examine missing-context behavior and worst-context loss alongside average predictive performance. The results do not establish suitability for high-stakes deployment, and this work does not involve a new human-subject study.

REPRODUCIBILITY STATEMENT

Appendix A provides the theoretical assumptions and proofs, and Appendix B summarizes the algorithm, expert adaptation, and evaluation conventions. Appendices C–E describe the comparisons and report the detailed results. Implementation details and experiment configurations are provided with the accompanying code at <https://anonymous.4open.science/r/RouterGFM-B7B9>.

REFERENCES

- Deyu Bo, Xiao Wang, Chuan Shi, and Huawei Shen. Beyond low-frequency information in graph convolutional networks. In *Proceedings of the 2021 AAAI Conference on Artificial Intelligence (AAAI)*, pp. 3950–3957, 2021.
- Aleksandar Bojchevski and Stephan Günnemann. Deep gaussian embedding of graphs: Unsupervised inductive learning via ranking. In *Proceedings of the 2018 International Conference on Learning Representations (ICLR)*, 2018.
- Rui Cai, Weijie Jacky Mo, Xiaofei Wen, Qiyao Ma, Wenhui Zhu, Xiwen Chen, Muhao Chen, and Zhe Zhao. ModelLens: Finding the best for your task from myriads of models. *arXiv preprint arXiv:2605.07075*, 2026.
- Haifang Cao, Yu Wang, Timing Li, Xinjie Yao, and Pengfei Zhu. Geometric mixture-of-experts with curvature-guided adaptive routing for graph representation learning. *CoRR*, abs/2603.22317, 2026.
- Haibo Chen, Xin Wang, Zeyang Zhang, Haoyang Li, Ling Feng, and Wenwu Zhu. AutoGFM: Automated graph foundation model with adaptive architecture customization. In *Proceedings of the 2025 International Conference on Machine Learning (ICML)*, volume 267, pp. 9295–9315. PMLR, 2025.
- Yaqi Chen, Shixun Huang, Lei Wang, Ryan Twemlow, John Le, Sheng Wang, Willy Susilo, Jun Yan, and Jun Shen. A cross-graph tuning-free GNN prompting framework. *Neural Networks*, 205:109466, 2027.
- Mark Craven, Dan DiPasquo, Dayne Freitag, Andrew McCallum, Tom M. Mitchell, Kamal Nigam, and Seán Slattery. Learning to extract symbolic knowledge from the world wide web. In *Proceedings of the 1998 AAAI Conference on Artificial Intelligence (AAAI)*, pp. 509–516, 1998.
- Rafael M. O. Cruz, Robert Sabourin, George D. C. Cavalcanti, and Tsang Ing Ren. META-DES: A dynamic ensemble selection framework using meta-learning. *Pattern Recognition*, 48(5):1925–1935, 2015.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 4171–4186. ACL, 2019.

- Vijay Prakash Dwivedi, Chaitanya K Joshi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Benchmarking graph neural networks. *Journal of Machine Learning Research (JMLR)*, 24(43):1–48, 2023.
- Taoran Fang, Yunchao Zhang, Yang Yang, Chunping Wang, and Lei Chen. Universal prompt tuning for graph neural networks. In *Proceedings of the 2023 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 52464–52489, 2023.
- Tao Feng, Yanzhen Shen, and Jiaxuan You. GraphRouter: A graph-based router for LLM selections. In *Proceedings of the 2025 International Conference on Learning Representations (ICLR)*, 2025.
- Matthias Fey and Jan E. Lenssen. Fast graph representation learning with pytorch geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
- Xingbo Fu, Yinhan He, and Jundong Li. Edge prompt tuning for graph neural networks. In *Proceedings of the 2025 International Conference on Learning Representations (ICLR)*, 2025.
- Zihao Guo, Qingyun Sun, Haonan Yuan, Xingcheng Fu, Min Zhou, Yisen Gao, and Jianxin Li. GraphMoRE: Mitigating topological heterogeneity via mixture of riemannian experts. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 11754–11762, 2025.
- Haoyu Han, Juanhui Li, Wei Huang, Xianfeng Tang, Hanqing Lu, Chen Luo, Hui Liu, and Jiliang Tang. Node-wise filtering in graph neural networks: A mixture of experts approach. *CoRR*, abs/2406.03464, 2024.
- Zhenyu Hou, Haozhan Li, Yukuo Cen, Jie Tang, and Yuxiao Dong. GraphAlign: Pretraining one graph neural network on multiple graphs via feature alignment. *arXiv preprint arXiv:2406.02953*, 2024.
- Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. Open graph benchmark: Datasets for machine learning on graphs. In *Proceedings of the 2020 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 22118–22133, 2020a.
- Weihua Hu, Bowen Liu, Joseph Gomes, Marinka Zitnik, Percy Liang, Vijay S. Pande, and Jure Leskovec. Strategies for pre-training graph neural networks. In *Proceedings of the 2020 International Conference on Learning Representations (ICLR)*, 2020b.
- Ziniu Hu, Yuxiao Dong, Kuansan Wang, and Yizhou Sun. Heterogeneous graph transformer. In *Proceedings of the 2020 International Conference on World Wide Web (WWW)*, pp. 2704–2710. ACM, 2020c.
- Suyeon Kim, Dongha Lee, SeongKu Kang, Seonghyeon Lee, and Hwanjo Yu. Learning topology-specific experts for molecular property prediction. In *Proceedings of the 2023 AAAI Conference on Artificial Intelligence (AAAI)*, pp. 8291–8299, 2023.
- Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *Proceedings of the 2015 International Conference on Learning Representations (ICLR)*, 2015.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *Proceedings of the 2017 International Conference on Learning Representations (ICLR)*, 2017.
- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Hao Liu, Jiarui Feng, Lecheng Kong, Ningyue Liang, Dacheng Tao, Yixin Chen, and Muhan Zhang. One for all: Towards training one graph model for all classification tasks. In *Proceedings of the 2024 International Conference on Learning Representations (ICLR)*, 2024.
- Jingzhe Liu, Haitao Mao, Zhikai Chen, Bingheng Li, Wenqi Fan, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. One model for one graph: A new perspective for pretraining with cross-domain graphs. *arXiv preprint arXiv:2412.00315*, 2025a.

- Lei Liu, Xingyu Xia, Qianqian Xie, Ben Liu, Wenjie Xu, and Min Peng. Enhanced expert merging for mixture-of-experts in graph foundation models. In *Proceedings of the 2025 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 22173–22207, 2025b.
- Zemin Liu, Xingtong Yu, Yuan Fang, and Xinming Zhang. Graphprompt: Unifying pre-training and downstream tasks for graph neural networks. In *Proceedings of the 2023 International Conference on World Wide Web (WWW)*, pp. 417–428. ACM, 2023.
- Li Ma, Haoyu Han, Juanhui Li, Harry Shomer, Hui Liu, Xiaofeng Gao, and Jiliang Tang. Mixture of link predictors on graphs. In *Advances in Neural Information Processing Systems*, volume 37, pp. 16043–16070, 2024.
- Mohit Meena, Yash Punjabi, Abhishek A, Vishal Sharma, and Mahesh Chandran. Self-adaptive graph mixture of models. In *Proceedings of the 2026 AAAI Conference on Artificial Intelligence (AAAI)*, volume 40, pp. 24344–24352, 2026.
- Grégoire Montavon, Matthias Rupp, Vivekanand Gobre, Alvaro Vazquez-Mayagoitia, Katja Hansen, Alexandre Tkatchenko, Klaus-Robert Müller, and O. Anatole von Lilienfeld. Machine learning of molecular electronic properties in chemical compound space. *New Journal of Physics*, 15(9): 095003, 2013.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs from preference data. In *Proceedings of the 2025 International Conference on Learning Representations (ICLR)*, 2025.
- Namyong Park, Ryan A. Rossi, Nesreen Ahmed, and Christos Faloutsos. MetaGL: Evaluation-free selection of graph learning models via meta-learning. In *Proceedings of the 2023 International Conference on Learning Representations (ICLR)*, 2023a.
- Namyong Park, Ryan A. Rossi, Xing Wang, Antoine Simoulin, Nesreen K. Ahmed, and Christos Faloutsos. GLEMOS: Benchmark for instantaneous graph learning model selection. In *Advances in Neural Information Processing Systems*, volume 36, pp. 69887–69899, 2023b.
- Sebastian Pineda Arango, Fabio Ferreira, Arlind Kadra, Frank Hutter, and Josif Grabocka. Quick-tune: Quickly learning which pretrained model to finetune and how. In *Proceedings of the 2024 International Conference on Learning Representations (ICLR)*, 2024.
- Leonardo F. R. Ribeiro, Pedro H. P. Savarese, and Daniel R. Figueiredo. *struc2vec*: Learning node representations from structural identity. In *Proceedings of the 2017 ACM Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 385–394, 2017.
- Benedek Rozemberczki, Carl Allen, and Rik Sarkar. Multi-scale attributed node embedding. *Journal of Complex Networks*, 9(2), 2021.
- Michael Sejr Schlichtkrull, Thomas N. Kipf, Peter Bloem, Rianne van den Berg, Ivan Titov, and Max Welling. Modeling relational data with graph convolutional networks. In *European Semantic Web Conference (ESWC)*, pp. 593–607. Springer, 2018.
- Oleksandr Shchur, Maximilian Mumme, Aleksandar Bojchevski, and Stephan Günnemann. Pitfalls of graph neural network evaluation. *CoRR*, abs/1811.05868, 2018.
- Fan-Yun Sun, Jordan Hoffmann, Vikas Verma, and Jian Tang. Infograph: Unsupervised and semi-supervised graph-level representation learning via mutual information maximization. In *Proceedings of the 2020 International Conference on Learning Representations (ICLR)*, 2020.
- Mingchen Sun, Kaixiong Zhou, Xin He, Ying Wang, and Xin Wang. GPPT: Graph pre-training and prompt tuning to generalize graph neural networks. In *Proceedings of the 2022 ACM Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1717–1727. ACM, 2022.
- Xiangguo Sun, Hong Cheng, Jia Li, Bo Liu, and Jihong Guan. All in one: Multi-task prompting for graph neural networks. In *Proceedings of the 2023 ACM Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 2120–2131. ACM, 2023.

- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Proceedings of the 2017 Annual Conference on Neural Information Processing Systems (NIPS)*, pp. 5998–6008, 2017.
- Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *Proceedings of the 2018 International Conference on Learning Representations (ICLR)*, 2018.
- Petar Velickovic, William Fedus, William L. Hamilton, Pietro Liò, Yoshua Bengio, and R. Devon Hjelm. Deep graph infomax. In *Proceedings of the 2019 International Conference on Learning Representations (ICLR)*, 2019.
- Haotao Wang, Ziyu Jiang, Yuning You, Yan Han, Gaowen Liu, Jayanth Srinivasa, Ramana Rao Kompella, and Zhangyang Wang. Graph mixture of experts: Learning on large-scale graphs with explicit diversity modeling. In *Proceedings of the 2023 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 50825–50837, 2023.
- Shuo Wang, Bokui Wang, Zhixiang Shen, Boyan Deng, and Zhao Kang. Multi-domain graph foundation models: Robust knowledge transfer via topology alignment. In *Proceedings of the 2025 International Conference on Machine Learning (ICML)*, pp. 64806–64821. PMLR, 2025a.
- Yidi Wang, Ziyue Qiao, Jiawei Gu, Xubin Zheng, Pengyang Wang, Xiaobing Pei, and Xiao Luo. Out-of-distribution graph models merging. In *Proceedings of the 2026 International Conference on Learning Representations (ICLR)*, 2026.
- Zehong Wang, Zheyuan Zhang, Nitesh V Chawla, Chuxu Zhang, and Yanfang Ye. GFT: Graph foundation model with transferable tree vocabulary. In *Proceedings of the 2024 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 107403–107443, 2024.
- Zehong Wang, Zheyuan Zhang, Tianyi Ma, Nitesh V. Chawla, Chuxu Zhang, and Yanfang Ye. Towards graph foundation models: Learning generalities across graphs via task-trees. In *Proceedings of the 2025 International Conference on Machine Learning (ICML)*, volume 267, pp. 65518–65555. PMLR, 2025b.
- Zhibin Wang, Zhixing Zhang, Shuqi Wang, Xuanning Xie, and Zhao Kang. GMoPE: A prompt-expert mixture framework for graph foundation models. *CoRR*, abs/2511.03251, 2025c.
- Qitian Wu, Wentao Zhao, Zenan Li, David P. Wipf, and Junchi Yan. Nodeformer: A scalable graph structure learning transformer for node classification. In *Proceedings of the 2022 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 27387–27401, 2022.
- Shirley Wu, Kaidi Cao, Bruno Ribeiro, James Y. Zou, and Jure Leskovec. GraphMETRO: Mitigating complex graph distribution shifts via mixture of aligned experts. In *Proceedings of the 2024 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 9358–9387, 2024.
- Zhenqin Wu, Bharath Ramsundar, Evan N Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S Pappu, Karl Leswing, and Vijay Pande. Moleculenet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018.
- Fen Xia, Tie-Yan Liu, Jue Wang, Wensheng Zhang, and Hang Li. Listwise approach to learning to rank: theory and algorithm. In *Proceedings of the 2008 International Conference on Machine Learning (ICML)*, pp. 1192–1199. ACM, 2008.
- Lianghao Xia and Chao Huang. AnyGraph: Graph foundation model in the wild. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 882–896. Association for Computational Linguistics, 2026.
- Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *Proceedings of the 2019 International Conference on Learning Representations (ICLR)*, 2019.
- Kaichao You, Yong Liu, Jianmin Wang, and Mingsheng Long. LogME: Practical assessment of pre-trained models for transfer learning. In *Proceedings of the 2021 International Conference on Machine Learning (ICML)*, volume 139, pp. 12133–12143. PMLR, 2021.

- Yuning You, Tianlong Chen, Yongduo Sui, Ting Chen, Zhangyang Wang, and Yang Shen. Graph contrastive learning with augmentations. In *Proceedings of the 2020 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 5812–5823, 2020.
- Xingtong Yu, Zhenghao Liu, Yuan Fang, Zemin Liu, Sihong Chen, and Xinming Zhang. Generalized graph prompt: Toward a unification of pre-training and downstream tasks on graphs. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 36(11):6237–6250, 2024.
- Xingtong Yu, Jie Zhang, Yuan Fang, and Renhe Jiang. Non-homophilic graph pre-training and prompt learning. In *Proceedings of the 2025 ACM Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1844–1854. ACM, 2025.
- Haonan Yuan, Qingyun Sun, Junhua Shi, Xingcheng Fu, Bryan Hooi, Jianxin Li, and Philip S. Yu. How much can transfer? BRIDGE: bounded multi-domain graph foundation model with generalization guarantees. In *Proceedings of the 2025 International Conference on Machine Learning (ICML)*, pp. 73604–73644. PMLR, 2025.
- Hanqing Zeng, Hanjia Lyu, Diyi Hu, Yinglong Xia, and Jiebo Luo. Mixture of weak and strong experts on graphs. In *Proceedings of the 2024 International Conference on Learning Representations (ICLR)*, 2024.
- Muhan Zhang and Yixin Chen. Link prediction based on graph neural networks. In *Proceedings of the 2018 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 5171–5181, 2018.
- Yi-Kai Zhang, Ting-Ji Huang, Yao-Xiang Ding, De-Chuan Zhan, and Han-Jia Ye. Model spider: Learning to rank pre-trained models efficiently. In *Proceedings of the 2023 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 13692–13719, 2023.
- Jiong Zhu, Yujun Yan, Lingxiao Zhao, Mark Heimann, Leman Akoglu, and Danai Koutra. Beyond homophily in graph neural networks: Current limitations and effective designs. In *Proceedings of the 2020 Annual Conference on Neural Information Processing Systems (NeurIPS)*, pp. 7793–7804, 2020.

A COMPLETE THEORETICAL ANALYSIS

This appendix analyzes inductive expert selection and context-guided prediction integration, with the assumptions needed for each guarantee. The statements are conditional on the realized support set, the fitted experts, and the information permitted at deployment.

A.1 NOTATION AND MEASURABILITY

Let (X, Y) follow the target application’s query distribution, let $Z = z_a(X)$, and let $\mathcal{E}$ be a nonempty finite eligible pool. For each $e \in \mathcal{E}$, $F_e(X)$ is a prediction in a common convex set and $r_e(Z) = \mathbb{E}[\ell(Y, F_e(X)) \mid Z]$ is an integrable conditional risk. For any application a , define its mean risk and centered performance residual by

$$\mu_{a,e} = \mathbb{E}[r_{a,e}(Z)], \quad \delta_{a,e}(z) = \mu_{a,e} - r_{a,e}(z). \quad (13)$$

The sign convention makes $\delta_{a,e}(z)$ positive when expert e has lower loss in context z than its own application average. For the measured local and application-average losses of record i , define $\bar{\delta}_i = \bar{\mu}_i - \bar{r}_i$. With the retrieval weights of Equation 6, let

$$\hat{\delta}_{a,e}(z) = \sum_{i \in \mathcal{N}_J(z,e)} w_i(z,e) \bar{\delta}_i, \quad \hat{r}_{a,e}(z) = \hat{\mu}_{a,e} - \rho \hat{\delta}_{a,e}(z). \quad (14)$$

Equation 14 is the residual form of Equation 7; both produce the same local loss estimates and mixture weights. The support set, archive, application metadata, and shortlist are fixed when these conditional expectations are formed. Equivalently, they can be included in the conditioning sigma-field. All comparisons use the same support-adapted experts; changing adaptation algorithms changes the comparison problem. A local gate is measurable with respect to Z and the permitted fixed information, so it cannot inspect the realized query label. This requirement is essential when moving its weights outside a conditional expectation. For squared local-risk supervision, we additionally assume finite second moments of the diagnostic loss.

A.2 THE OPPORTUNITY FOR LOCAL EXPERT SELECTION

Proposition 2 (Value of local expertise). *For a finite eligible expert pool, define $R_{\text{fixed}} = \min_e \mathbb{E}[r_e(Z)]$ and $R_{\text{loc}} = \mathbb{E}[\min_e r_e(Z)]$. Then $\Delta_{\text{loc}} = R_{\text{fixed}} - R_{\text{loc}} \geq 0$, with equality if and only if a fixed expert minimizes $r_e(Z)$ almost surely.*

The difference measures the opportunity for local individual-expert selection, not improvement over the best constant convex mixture.

Proof. Choose $e_0 \in \arg \min_e \mathbb{E} r_e(Z)$, which exists because the pool is finite. For every z , $r_{e_0}(z) - \min_e r_e(z) \geq 0$. Taking expectations proves

$$\Delta_{\text{loc}} = \mathbb{E}[r_{e_0}(Z) - \min_e r_e(Z)] \geq 0. \quad (15)$$

If one expert e_* minimizes conditional risk almost surely, its mean risk equals $\mathbb{E} \min_e r_e(Z)$, so equality follows. Conversely, if $\Delta_{\text{loc}} = 0$, the nonnegative integrable random variable $r_{e_0}(Z) - \min_e r_e(Z)$ has zero expectation and therefore equals zero almost surely. Thus e_0 minimizes conditional risk almost surely, proving both directions. $\square$

Example with equal global scores. Let two context regions have equal probability and let two experts have risks (a, b) and (b, a) in those regions, where $0 \leq a < b$. Both global risks equal $(a + b)/2$, but $R_{\text{loc}} = a$ and $\Delta_{\text{loc}} = (b - a)/2$. A global ranking cannot reveal which expert to prefer in either region. The construction isolates local specialization while keeping the two global risks equal.

An exact probabilistic example. Figure 6 evaluates an exact two-region probability model. In two equally likely regions, let Y have Bernoulli means 0.8 and 0.2 and let the experts return opposite fixed predictions 0.8 and 0.2. Their local squared probability losses are $(0.16, 0.52)$ and $(0.52, 0.16)$, so both global risks equal 0.34 and the local oracle risk equals 0.16. In this example only, $\sigma(t) =$

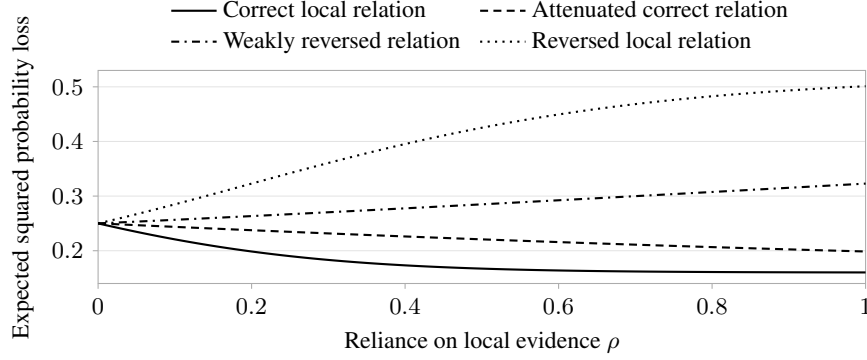


Figure 6: Exact analytical two-region example of residual orientation and shrinkage. With equally likely regions and label probabilities 0.8 and 0.2, two experts exchange predictions 0.8 and 0.2. Increasing ρ helps only when the transferred residual relation has the correct orientation. The global uniform mixture has loss 0.25 and the conditional individual-expert oracle has loss 0.16.

$1/(1 + e^{-t})$ denotes the logistic sigmoid. Scale the true local residual by $1 - 2q$, where q is a declared mismatch parameter, and use temperature 0.1. In the first region the first expert’s weight is $\sigma(3.6\rho(1 - 2q))$, so the exact mixture risk is

$$R(\rho, q) = 0.16 + [0.2 + 0.6\sigma(3.6\rho(1 - 2q)) - 0.8]^2. \quad (16)$$

Symmetry gives the same loss in the other region. Thus stronger local weighting reduces risk for $q < 1/2$, increases it for $q > 1/2$, and changes nothing for $q = 1/2$. The closed-form curves separate useful local expertise from the reliability of the transferred residuals.

Why global shortlisting can fail. Let a region have probability $p \in (0, 1)$ and consider a generalist with constant risk c and a specialist with risk zero inside that region and one outside it. When $0 < c < 1 - p$, the generalist has lower global risk and a size-one shortlist excludes the specialist. Its local omission cost is nevertheless $pc > 0$. Hence accurate global risk prediction alone does not make Γ_K vanish. Additional experts with the same generalist behavior extend the example to any fixed shortlist size. This example motivates measuring shortlist coverage in addition to dataset-level ranking accuracy.

A.3 PROOF OF THEOREM 1

Write $T = T_a$, $\mathcal{E} = \mathcal{E}_a$, and $K = |T|$. The local omission cost and maximum within-team risk-estimation error are

$$\Gamma_K = \mathbb{E}[\min_{e \in T} r_e(Z) - \min_{e \in \mathcal{E}} r_e(Z)], \quad \epsilon(Z) = \max_{e \in T} |\hat{r}_e(Z) - r_e(Z)|. \quad (17)$$

For fixed z , write $q_e = \hat{r}_e(z)$ and let $\alpha_e = \exp(-q_e/\tau) / \sum_{j \in T} \exp(-q_j/\tau)$. For any distribution β on T , nonnegativity of $\text{KL}(\beta||\alpha)$ gives

$$\langle \alpha, q \rangle - \tau H(\alpha) \leq \langle \beta, q \rangle - \tau H(\beta), \quad H(\beta) = - \sum_{e \in T} \beta_e \log \beta_e. \quad (18)$$

This also follows by expanding the KL divergence and canceling the common log-normalizer. Choose β concentrated on a minimizer of q and use $H(\alpha) \leq \log K$ to obtain

$$\sum_{e \in T} \alpha_e \hat{r}_e(z) \leq \min_{e \in T} \hat{r}_e(z) + \tau \log K. \quad (19)$$

By the definition of $\epsilon(z)$, both replacing a weighted estimated risk by its true risk and replacing its minimum by the true minimum cost at most $\epsilon(z)$. Consequently,

$$\sum_{e \in T} \alpha_e r_e(z) \leq \sum_{e \in T} \alpha_e \hat{r}_e(z) + \epsilon(z) \quad (20)$$

$$\leq \min_{e \in T} \hat{r}_e(z) + \tau \log K + \epsilon(z) \quad (21)$$

$$\leq \min_{e \in T} r_e(z) + \tau \log K + 2\epsilon(z). \quad (22)$$

Convexity of ℓ in the prediction gives the pointwise inequality

$$\ell\left(Y, \sum_{e \in T} \alpha_e(Z) F_e(X)\right) \leq \sum_{e \in T} \alpha_e(Z) \ell(Y, F_e(X)). \quad (23)$$

Conditional expectation given $Z = z$ is legitimate because the weights are measurable with respect to Z . Combining the resulting inequality with Equation 22, integrating over Z , and subtracting R_{loc} proves

$$R(F) - R_{\text{loc}} \leq \Gamma_K + 2\mathbb{E}\epsilon(Z) + \tau \log K. \quad (24)$$

If $\mathbb{E}\epsilon$ is infinite, the same statement is valid but uninformative. In particular, $\Gamma_K + 2\mathbb{E}[\epsilon(Z)] + \tau \log K < \Delta_{\text{loc}}$ suffices to outperform the best fixed individual expert. $\square$

Interpretation of the comparator. The conditional single-expert oracle is not an upper limit on the performance of a convex mixture. A mixture can have lower loss than every individual expert through error cancellation. The theorem provides an upper bound on excess risk relative to that oracle, without requiring this excess to be nonnegative. For hard selection $\hat{e}(z) \in \arg \min_{e \in T} \hat{r}_e(z)$, the same proof removes the $\tau \log K$ term and does not require convexity of the loss. However, a dataset-level best-expert score remains a valid ceiling only for selecting one fixed expert from exactly that pool on exactly that evaluation split. Neither an oracle from a different checkpoint pool nor a test-selected team is a deployable baseline.

A.4 RELIABILITY OF CONTEXT-GUIDED ESTIMATION

Theorem 3 (Context-conditioned estimation and its failure conditions). *Assume that compatible centered performance residuals satisfy*

$$|\delta_{a,e}(z) - \delta_{b,f}(z')| \leq L \|k_\phi(z, v_e) - k_\phi(z', v_f)\|_2 + \eta, \quad (25)$$

where η permits application-dependent residual shift. For record i , let s_i be its mean within-cell key radius and suppose its residual estimate differs from its population cell mean by at most ε_i . For normalized nonnegative weights, set $d_i = \|k_\phi(z, v_e) - k_\phi(c_i, v_{e_i})\|_2$. Then

$$|\hat{\delta}_{a,e}(z) - \delta_{a,e}(z)| \leq L \sum_i w_i (d_i + s_i) + \eta + \sum_i w_i \varepsilon_i. \quad (26)$$

If $|\hat{\mu}_{a,e} - \mu_{a,e}| \leq \varepsilon_g$ and $|\delta_{a,e}(z)| \leq D$, the local-risk error is at most

$$\varepsilon_g + (1 - \rho)D + \rho \left[L \sum_i w_i (d_i + s_i) + \eta + \sum_i w_i \varepsilon_i \right]. \quad (27)$$

Proof. Let record i originate from application b_i , expert f_i , and cell C_i . Define its population centered residual by $\delta_i^{\text{cell}} = \mathbb{E}[\delta_{b_i, f_i}(Z') \mid Z' \in C_i]$ and assume $|\bar{\delta}_i - \delta_i^{\text{cell}}| \leq \varepsilon_i$. Set

$$s_i = \mathbb{E}[\|k_\phi(Z', v_{f_i}) - k_\phi(c_i, v_{f_i})\|_2 \mid Z' \in C_i]. \quad (28)$$

The representative need not be a Euclidean centroid after the nonlinear key map; the radius above accounts for that mismatch. Apply Equation 25 and the triangle inequality inside this conditional expectation to get

$$|\delta_i^{\text{cell}} - \delta_{a,e}(z)| \leq \mathbb{E}[|\delta_{b_i, f_i}(Z') - \delta_{a,e}(z)| \mid Z' \in C_i] \quad (29)$$

$$\leq L(d_i + s_i) + \eta. \quad (30)$$

Since $w_i \geq 0$ and $\sum_i w_i = 1$, another triangle inequality yields

$$|\hat{\delta}_{a,e}(z) - \delta_{a,e}(z)| \leq \sum_i w_i (|\bar{\delta}_i - \delta_i^{\text{cell}}| + |\delta_i^{\text{cell}} - \delta_{a,e}(z)|) \quad (31)$$

$$\leq \sum_i w_i \varepsilon_i + L \sum_i w_i (d_i + s_i) + \eta. \quad (32)$$

Finally, expand the difference between estimated and true local risks:

$$\hat{r} - r = (\hat{\mu} - \mu) + (1 - \rho)\delta + \rho(\delta - \hat{\delta}). \quad (33)$$

Taking absolute values and applying the stated bounds proves Equation 27. No step treats neighboring graph queries as independent observations. $\square$

Centering and finite diagnostic evidence. Let $\bar{\mu} = \sum_j n_j \bar{r}_j / \sum_j n_j$, using exactly the same valid observations and weights as the cell risks. Then $\sum_j n_j (\bar{\mu} - \bar{r}_j) = 0$ by algebra. When labels are missing, the counts must match the loss’s valid-query weighting rather than the raw number of nodes or targets. If the global diagnostic mean and cell risk have errors at most u and v_j , the residual error is at most $u + v_j$. Small cells can therefore have large ε_i even when their centroids are close to a target query. Empirical counts alone do not imply a $1/\sqrt{n_i}$ rate on a dependent graph. Such a rate requires an additional sampling or dependence model and is not asserted here.

A.5 INDUCTIVE INSERTION: EQUIVARIANCE AND AMBIGUITY

Proposition 4 (Type-preserving equivariance). *Suppose node feature maps are shared within each type, neighborhood aggregation is permutation invariant, and scoring and retrieval use shared functions with permutation-equivariant tie handling. Relabeling nodes within each type, while relabeling their features, relations, and evidence consistently, relabels all scores and mixture weights in the same way.*

Proof. The initial node states commute with such a relabeling because $P_{\kappa(v)}$ depends on type and features, not identity. Assume equivariance at layer l . Relabeling induces a bijection between the typed neighborhoods of corresponding nodes, preserving each message’s inputs. Permutation-invariant aggregation and the shared residual update then give equivariance at layer $l + 1$. Induction proves the claim for the final embeddings. The shared pair head preserves corresponding scores, and metadata-based distance computations preserve corresponding retrieval weights. Top- K selection is equivariant away from boundary ties; ties can be handled by a consistently coupled random order or by equal-weight treatment rather than by arbitrary integer node IDs. The prediction mixture is a sum over corresponding experts, so it is unchanged as a numerical prediction. $\square$

The proposition supports processing additional nodes with existing type-specific parameters. It does not imply that inserting a new node leaves every old embedding or ranking unchanged, because its messages can alter the graph neighborhood. It also does not imply prediction consistency for a new architecture outside the observed metadata distribution.

Proposition 5 (Metadata-only ambiguity). *There exist two newly inserted experts and two target risk assignments that induce identical permitted router inputs but require opposite optimal choices. For any randomized size-one selection rule, the larger expected regret across these assignments is at least $\Delta/2$, where $\Delta > 0$ is their risk gap.*

Proof. Give the two experts identical observable metadata and no diagnostic observations that distinguish them. In the first assignment their risks are $(0, \Delta)$; in the second they are $(\Delta, 0)$. Both assignments produce the same router input. If the first expert is chosen with probability p , the two expected regrets are $(1 - p)\Delta$ and $p\Delta$. Their maximum is at least their average, namely $\Delta/2$. $\square$

This is an information limit on metadata-only selection, not a claim that all real experts have ambiguous descriptions. Source calibration may supply distinguishing evidence, but its value depends on its relevance to the target application.

B EXPERIMENTAL DETAILS

Algorithm 1 ROUTERGFM: historical router learning and target prediction

Require: Pretrained experts and metadata; historical sets (S_b, D_b) ; target application (S_a, Q_a) ; routing hyperparameters.

Ensure: Predictions $F_a(x)$ for $x \in Q_a$.

Historical evaluation

- 1: **for** each historical application b **do**
- 2: Fit evaluated expert heads on S_b with encoders frozen; record losses on disjoint D_b .
- 3: Compute application averages by Equation 2; group label-free contexts and compute local averages by Equation 5.
- 4: **end for**
- 5: Build the context graph $\mathcal{H}$ and local evaluation archive $\mathcal{M}$.

Router learning

- 6: **for** each training episode with application b **do**
 - 7: Mask b 's evaluation edges and their reverses, local records, and label-derived aggregates.
 - 8: Encode the masked graph and predict average performance for $e \in \Omega_b$.
 - 9: Retrieve evaluations from other applications and estimate local losses using Equation 7.
 - 10: Update (θ, ϕ) by Equation 11, keeping historical predictions and losses fixed.
 - 11: **end for**
 - 12: Select the router checkpoint and hyperparameters on separate historical validation applications.
 - Target prediction: router parameters fixed**
 - 13: Insert target and new-expert metadata with construction edges; filter $\mathcal{E}$ to $\mathcal{E}_a$.
 - 14: Select T_a by Equation 4, without executing candidate encoders on the target.
 - 15: Fit selected task heads once on S_a with pretrained GNNs frozen.
 - 16: **for** each query $x \in Q_a$ **do**
 - 17: Compute context $z_a(x)$ and expert predictions $F_{a,e}(x)$ for $e \in T_a$.
 - 18: Retrieve compatible evaluations using Equation 6.
 - 19: Compute local loss estimates and prediction weights using Equations 7 and 8.
 - 20: Return the weighted prediction $F_a(x)$ from Equation 1.
 - 21: **end for**
-

B.1 EXPERTS AND ADAPTATION

The pool contains $7 \times 7 \times 12 = 588$ pretrained checkpoints before task-compatibility filtering. The architectures are GCN, GAT, GIN, H2GCN, FAGCN, NodeFormer, and Transformer (Kipf & Welling, 2017; Veličković et al., 2018; Xu et al., 2019; Zhu et al., 2020; Bo et al., 2021; Wu et al., 2022; Vaswani et al., 2017). Pretraining uses attribute masking, context prediction, DGI, edge prediction, GraphCL, InfoGraph, and supervised objectives (Hu et al., 2020b; Veličković et al., 2019; You et al., 2020; Sun et al., 2020). The twelve source datasets are Cora, PubMed, Computers, Flickr, Actor, Squirrel, Email, BBBP, Tox21, QM9, PROTEINS, and CIFAR10. Each expert retains its checkpoint-specific preprocessing and task readout; matched-pool comparisons use the same eligible inventory and support observations. Node tasks read the marked node, link tasks use an enclosing-subgraph and symmetric endpoint readout, and graph tasks use pooled representations. Selected encoders remain frozen while task heads are fitted on target support examples, with output coordinates aligned before mixing. The implementation uses PyTorch Geometric (Fey & Lenssen, 2019), frozen BERT description embeddings (Devlin et al., 2019), and Adam optimization (Kingma & Ba, 2015). Detailed model, preprocessing, and optimizer settings are provided with the accompanying code.

B.2 TASK LOSSES AND LOCAL EVIDENCE

For single-label classification, the routing loss is $\ell(y, p) = \frac{1}{2} \|p - \text{onehot}(y)\|_2^2$; binary link prediction uses squared probability error. Multi-label predictions retain separate assay probabilities, and unobserved targets are masked in task losses. For regression, targets are normalized using support medians and median absolute deviations with a 10^{-6} scale floor; predictions are transformed back to raw units for MAE reporting. Archive records retain local and application-average losses,

valid evaluation counts, context representatives, and expert identities. Retrieval matches task/loss families, support budgets, and normalization conventions, and caps records from any one source application. Counts determine diagnostic averages but do not multiply the distance-based weights in Equation 6. Source-trained descriptor standardization is reused at deployment, and an empty compatible retrieval set gives zero local correction.

B.3 TRAINING AND EVALUATION

Historical support sets fit expert heads, while disjoint diagnostic observations supervise the router. Application-masked training removes the pseudo-target’s evaluation edges, reverse edges, local records, and label-derived aggregates before its losses are predicted. Router selection uses source validation, and tasks, budgets, and splits of one base dataset remain grouped. Target support labels fit the selected heads; query labels are used only for evaluation. Single-label classification uses examples per class, regression uses support graphs in total, and ToxCast uses the multitask partitions specified in the code. Prediction results use seeds 42, 0, 100, 123, and 2024; mechanism tables report point estimates rather than significance tests.

The routed LP comparisons use the 10%–5%–10% positive-edge convention, with remaining positives unused and held-out positives excluded from message passing. The two Table 1 blocks retain their reported LP results; repeated baseline entries are shared results, not additional runs or evidence of link-budget scaling. The configuration catalog uses 5%–10%–10% at five shots and 10%–5%–10% at 100 shots; ProNoG additionally uses remaining positives as message-only context. These LP groups are interpreted within their respective information conditions. Within each configuration, validation selects the checkpoint; the adaptation catalog then reports the best observed test mean across completed configurations with that configuration’s standard deviation. Dataset loaders, support partitions, and complete configurations are documented in the code rather than repeated here.

B.4 DIAGNOSTIC MEASURES AND SCOPE

Worst-cell risk is the largest cell-average mixture loss within an evaluation episode, using the same context partition across integration rules. Local-winner coverage measures the mass of cells whose best eligible expert is in the shortlist, while winner agreement compares the highest-weight expert with a cellwise minimizing expert. Mixture losses are computed from the mixed predictions, not from weighted individual-expert losses. These empirical diagnostics are distinct from the population conditional risks in Theorem 1.

Selection scores metadata pairs without executing every candidate GNN on target data; prediction still uses the selected experts. For N queries, K experts, R compatible archive records, and key dimension d_k , exhaustive retrieval requires $O(NKRd_k)$ distance work and $O(Rd_k)$ cached-key storage. These counts do not establish a measured latency or memory advantage. Local transfer depends on compatible historical evidence, and metadata alone need not distinguish experts with different behavior.

C BASELINE COMPARISONS

Selection and integration. The metadata MLP, nearest-application rule, MetaGL, LogME, and Model Spider compare application-level expert recommendation (Park et al., 2023a; You et al., 2021; Zhang et al., 2023). MetaGL+metadata denotes the metadata extension used for expert insertion. MetaGL-U averages its selected predictions, while ROUTERGFM-G uses ROUTERGFM’s team and fitted heads with application-level weights. The fixed-team study compares uniform weighting, global weighting, simplex stacking, Local-MLP, and historical local transfer on the same expert predictions. The comparison isolates the complete integration procedure from changes in predictors; it does not separately match the historical supervision of every learned integration rule. The no-centering and shuffled-record rows assess sensitivity to changes in the archived evidence.

Graph mixtures and adaptation. SAGMM-PE uses the matched frozen-expert pool, whereas AnyGraph, GMoE, Mowst, GraphMoRE, and GMoPE retain their method-specific expert organizations (Meena et al., 2026; Xia & Huang, 2026; Wang et al., 2023; Zeng et al., 2024; Guo et al., 2025; Wang et al., 2025c). Node-MoE is evaluated on node tasks, Link-MoE on link tasks, and

META-DES on classification; KDEM/PPEM use compatible architectures for expert merging (Han et al., 2024; Ma et al., 2024; Cruz et al., 2015; Liu et al., 2025b). Task-aware adaptations outside a method’s original setting follow the implementations in the code. GraphMETRO, OGMM, and GeoMoE supply shift-oriented comparisons with their own expert inventories (Wu et al., 2024; Wang et al., 2026; Cao et al., 2026). The configuration catalog includes full fine-tuning and nine prompting variants, with GraphPrompt+ denoting the generalized extension of GraphPrompt (Yu et al., 2024). These catalog and method-specific results complement the matched-pool comparison rather than define equal-compute or identical-adaptation comparisons.

D ADDITIONAL RESULTS AND ANALYSES

D.1 APPLICATION-LEVEL EFFECTS OF LOCAL INTEGRATION

Table 2 gives the differences between ROUTERGFM and ROUTERGFM-G in Table 1, retaining each task’s reporting metric. The mean accuracy improvement increases from 4.25 to 8.41 points across the five single-label classification datasets, while ToxCast’s ROC-AUC improvement decreases from 2.69 to 1.65 points. For QM7b, the MAE reductions are 0.21 and 0.49, corresponding to 3.9% and 10.0% relative improvements. The gains therefore vary across applications and metrics rather than defining one universal supervision-response curve. The LP differences summarize the displayed comparisons without attributing their change to different link-label budgets.

Table 2: Application-specific improvement from local integration, computed from the displayed ROUTERGFM and ROUTERGFM-G means in Table 1. Accuracy and ROC-AUC gains are percentage points; regression gains are reductions in raw-unit MAE. LP differences retain the two reported blocks rather than merging them into one result. The accuracy mean excludes ToxCast, LP, and regression.

Application	Improvement measure	5-shot	100-shot
Photo	Accuracy increase	1.26	5.18
ogbn-arxiv	Accuracy increase	9.45	6.81
Airports	Accuracy increase	5.54	12.95
Chameleon	Accuracy increase	3.10	15.09
MNIST	Accuracy increase	1.91	2.02
Accuracy mean	Five single-label datasets	4.25	8.41
ToxCast	ROC-AUC increase	2.69	1.65
QM7b	MAE reduction	0.21	0.49
DBLP	ROC-AUC increase by reported block	7.77	8.73
Cornell	ROC-AUC increase by reported block	3.17	3.48

Table 3: 100-shot performance summary on nine datasets. Each adaptation entry is the best observed test mean over the architecture-objective grid, with the standard deviation of the same configuration. Accuracy and ROC-AUC are percentages; QM7b uses MAE. Bold and underline identify the best and second distinct means; LP conventions are specified in Appendix B.3. LP rankings are computed separately for the configuration and routed comparison groups. Both ROUTERGFM and ROUTERGFM-G fine-tune only task heads.

Models	Homophilic graphs			Heterophilic graphs			Graph-level tasks		
	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. $\uparrow$	NC / Acc. $\uparrow$	LP / AUC $\uparrow$	NC / Acc. $\uparrow$	NC / Acc. $\uparrow$	LP / AUC $\uparrow$	GR / MAE $\downarrow$	GC / AUC $\uparrow$	GC / Acc. $\uparrow$
<i>Supervised graph models</i>									
GCN	87.37 $\pm$ 0.78	37.72 $\pm$ 0.68	55.60 $\pm$ 0.65	39.82 $\pm$ 3.13	34.27 $\pm$ 1.32	62.78 $\pm$ 5.43	5.08 $\pm$ 0.08	52.49 $\pm$ 1.21	24.37 $\pm$ 2.61
GAT	86.41 $\pm$ 0.57	37.84 $\pm$ 0.53	56.01 $\pm$ 0.59	35.57 $\pm$ 1.87	35.26 $\pm$ 1.35	69.72 $\pm$ 9.48	21.51 $\pm$ 0.59	53.17 $\pm$ 1.03	27.96 $\pm$ 0.88
GIN	87.05 $\pm$ 0.63	31.82 $\pm$ 0.91	55.24 $\pm$ 1.26	40.68 $\pm$ 1.53	37.59 $\pm$ 0.38	69.74 $\pm$ 8.13	36.71 $\pm$ 2.31	51.45 $\pm$ 1.86	38.59 $\pm$ 1.96
H2GCN	87.01 $\pm$ 0.79	38.89 $\pm$ 1.00	55.47 $\pm$ 0.73	27.14 $\pm$ 0.84	41.26 $\pm$ 1.68	73.42 $\pm$ 6.76	5.64 $\pm$ 0.16	52.14 $\pm$ 1.25	33.61 $\pm$ 1.15
FAGCN	87.21 $\pm$ 0.68	42.00 $\pm$ 0.64	54.62 $\pm$ 1.33	36.51 $\pm$ 2.99	37.33 $\pm$ 1.12	62.81 $\pm$ 13.20	5.47 $\pm$ 0.10	52.05 $\pm$ 0.82	23.41 $\pm$ 2.00
Transformer	88.44 $\pm$ 0.36	39.24 $\pm$ 0.68	57.03 $\pm$ 0.42	41.09 $\pm$ 2.15	39.64 $\pm$ 0.96	76.79 $\pm$ 7.10	10.71 $\pm$ 1.28	52.74 $\pm$ 0.54	37.32 $\pm$ 1.53
NodeFormer	83.94 $\pm$ 0.84	40.47 $\pm$ 2.33	53.29 $\pm$ 1.73	37.42 $\pm$ 3.58	39.40 $\pm$ 1.28	62.68 $\pm$ 3.99	13.71 $\pm$ 0.06	51.34 $\pm$ 0.74	14.54 $\pm$ 4.72
<i>Best observed configurations by adaptation rule</i>									
Full fine-tuning	88.99 $\pm$ 0.57	42.45 $\pm$ 1.23	<u>58.65</u> $\pm$ 2.34	42.66 $\pm$ 2.37	42.30 $\pm$ 1.70	69.37 $\pm$ 3.03	5.26 $\pm$ 0.07	53.56 $\pm$ 1.56	42.46 $\pm$ 0.31
All-in-One	88.80 $\pm$ 0.41	45.06 $\pm$ 0.14	54.83 $\pm$ 0.32	43.34 $\pm$ 0.86	39.95 $\pm$ 0.43	<u>74.13</u> $\pm$ 5.46	11.78 $\pm$ 0.57	60.96 $\pm$ 1.64	37.50 $\pm$ 0.55
EdgePrompt	83.10 $\pm$ 1.76	46.89 $\pm$ 0.59	58.84 $\pm$ 3.95	42.56 $\pm$ 1.66	42.10 $\pm$ 0.96	66.56 $\pm$ 12.92	9.22 $\pm$ 0.33	53.98 $\pm$ 0.63	33.69 $\pm$ 3.79
EdgePrompt+	85.07 $\pm$ 1.50	44.96 $\pm$ 1.34	57.27 $\pm$ 2.83	47.67 $\pm$ 1.34	42.15 $\pm$ 1.08	72.01 $\pm$ 4.86	9.26 $\pm$ 0.32	54.54 $\pm$ 3.25	39.31 $\pm$ 2.50
GPF	89.35 $\pm$ 0.46	46.68 $\pm$ 0.80	58.22 $\pm$ 1.30	41.95 $\pm$ 1.55	41.36 $\pm$ 1.30	72.60 $\pm$ 3.42	33.24 $\pm$ 1.56	53.79 $\pm$ 2.33	34.22 $\pm$ 4.13
GPF+	89.23 $\pm$ 0.78	43.02 $\pm$ 0.27	55.63 $\pm$ 0.73	42.86 $\pm$ 1.64	41.01 $\pm$ 1.30	69.06 $\pm$ 10.62	32.62 $\pm$ 0.72	53.72 $\pm$ 0.91	35.53 $\pm$ 1.48
GPPT	88.21 $\pm$ 0.67	36.06 $\pm$ 0.60	53.84 $\pm$ 3.53	40.53 $\pm$ 1.70	40.25 $\pm$ 0.64	66.83 $\pm$ 9.07	43.58 $\pm$ 9.14	50.82 $\pm$ 0.62	31.34 $\pm$ 1.48
GraphPrompt	87.81 $\pm$ 0.51	24.46 $\pm$ 0.65	<u>58.65</u> $\pm$ 3.19	40.51 $\pm$ 1.98	32.76 $\pm$ 4.18	72.98 $\pm$ 6.05	9.73 $\pm$ 0.32	47.17 $\pm$ 1.26	38.30 $\pm$ 0.52
GraphPrompt+	87.78 $\pm$ 0.95	22.58 $\pm$ 0.48	57.49 $\pm$ 2.09	41.47 $\pm$ 1.55	32.13 $\pm$ 4.58	57.29 $\pm$ 14.44	8.10 $\pm$ 4.09	46.52 $\pm$ 1.64	38.11 $\pm$ 0.40
ProNoG	87.40 $\pm$ 0.47	41.35 $\pm$ 0.68	56.44 $\pm$ 0.53	37.97 $\pm$ 1.24	37.56 $\pm$ 1.46	65.97 $\pm$ 10.09	4.06 $\pm$ 0.07	45.69 $\pm$ 1.34	31.31 $\pm$ 0.75
<i>Graph mixture-of-experts models</i>									
AnyGraph	84.00 $\pm$ 1.90	50.24 $\pm$ 1.76	83.81 $\pm$ 1.78	40.40 $\pm$ 2.89	36.63 $\pm$ 2.21	57.20 $\pm$ 8.08	11.92 $\pm$ 0.73	53.15 $\pm$ 0.65	33.23 $\pm$ 1.60
GMoE	85.76 $\pm$ 2.14	25.68 $\pm$ 0.57	54.83 $\pm$ 0.77	38.18 $\pm$ 3.77	33.09 $\pm$ 1.69	58.28 $\pm$ 10.40	113.74 $\pm$ 2.41	55.51 $\pm$ 2.71	34.65 $\pm$ 3.08
Mowst	87.19 $\pm$ 0.77	38.97 $\pm$ 0.64	55.32 $\pm$ 0.46	42.68 $\pm$ 1.89	35.24 $\pm$ 0.93	64.21 $\pm$ 9.67	103.16 $\pm$ 1.53	52.23 $\pm$ 2.52	28.32 $\pm$ 6.74
GraphMoRE	85.56 $\pm$ 0.72	33.85 $\pm$ 0.92	54.10 $\pm$ 0.40	38.94 $\pm$ 1.90	35.06 $\pm$ 0.70	57.19 $\pm$ 9.03	22.93 $\pm$ 39.40	52.78 $\pm$ 0.47	29.49 $\pm$ 0.44
GMoPE	89.66 $\pm$ 0.57	50.26 $\pm$ 1.03	81.22 $\pm$ 1.62	48.11 $\pm$ 1.91	<u>44.52</u> $\pm$ 1.73	75.91 $\pm$ 6.10	5.04 $\pm$ 0.16	59.85 $\pm$ 0.56	<u>44.63</u> $\pm$ 1.52
<i>Matched-pool routing and frozen-expert mixtures</i>									
SAGMM-PE	89.43 $\pm$ 0.47	49.72 $\pm$ 0.80	82.80 $\pm$ 1.23	48.54 $\pm$ 1.44	44.35 $\pm$ 1.17	<u>76.80</u> $\pm$ 5.20	4.99 $\pm$ 0.11	60.10 $\pm$ 1.46	43.62 $\pm$ 1.13
MetaGL-U	89.03 $\pm$ 0.45	48.06 $\pm$ 0.84	80.43 $\pm$ 1.40	47.48 $\pm$ 1.76	42.33 $\pm$ 1.16	73.36 $\pm$ 4.79	5.14 $\pm$ 0.12	59.84 $\pm$ 1.50	42.51 $\pm$ 1.05
ROUTERGFM-G	89.71 $\pm$ 0.40	<u>50.79</u> $\pm$ 0.81	83.31 $\pm$ 1.26	49.23 $\pm$ 1.53	44.05 $\pm$ 1.45	76.33 $\pm$ 4.65	4.91 $\pm$ 0.12	60.99 $\pm$ 1.22	44.16 $\pm$ 1.26
ROUTERGFM	94.89 $\pm$ 0.34	57.60 $\pm$ 0.60	92.04 $\pm$ 0.75	62.18 $\pm$ 1.20	59.14 $\pm$ 1.90	79.81 $\pm$ 4.55	<u>4.42</u> $\pm$ 0.43	62.64 $\pm$ 1.13	46.18 $\pm$ 1.12

D.2 FIXED-TEAM PREDICTION INTEGRATION

Table 4 gives the measurements in Figure 4 and adds local-winner agreement. With predictors fixed, uniform weighting, global weighting, and simplex stacking give average losses 0.247, 0.239, and 0.235, whereas local transfer reaches 0.211. Its advantage over Local-MLP is 0.015 in average loss and 0.033 in worst-cell loss. Removing centering or shuffling correspondence increases loss to 0.228 or 0.244, respectively, and lowers winner agreement from 67.3% to 55.1% or 42.7%. These observations support transferring structured local evidence rather than treating all historical evaluations as equally useful. The controls alter different aspects of the procedure, so their gaps are not additive component contributions.

Table 4: Five-shot fixed-team diagnosis with identical checkpoints, adapted heads, and expert predictions. Risks are point estimates; agreement is a percentage and is not classification accuracy. The comparison does not alone establish matched historical supervision for learned integration rules.

Integration on the same team	Brier risk ↓	Worst-cell risk ↓	Winner agreement (%) ↑
Uniform	0.247	0.402	39.2
Global risk weights	0.239	0.388	43.6
Simplex stacking	0.235	0.381	45.6
Local-MLP	0.226	0.354	58.8
Local residual transfer	0.211	0.321	67.3
No residual centering	0.228	0.368	55.1
Shuffled context-residual pairs	0.244	0.407	42.7

D.3 TASK-SCOPED MIXTURES AND EXPERT MERGING

Tables 5–8 compare methods within their evaluated task scopes. On Chameleon, ROUTERGFM exceeds Node-MoE by 1.14 and 13.17 accuracy points at five and 100 shots, respectively. On DBLP, its AUC exceeds Link-MoE by 6.32 and 7.28 points in the displayed blocks; Cornell’s corresponding margins are 1.08 and 1.39 points. On MNIST, ROUTERGFM exceeds GMoPE by 1.62 and 1.55 accuracy points. These comparisons retain method-specific expert inventories; the fixed-team study provides the more direct test of prediction integration.

Table 5: Node-level 5-shot accuracy comparisons (percent, higher is better). KDEM/PPEM use compatible expert architectures; META-DES uses the matched graph-context representation. Best/second-best means in each column are bold/underlined.

Method	Photo	Arxiv	Airports	Chameleon
Node-MoE	80.32 $\pm$ 0.93	29.33 $\pm$ 0.93	40.22 $\pm$ 2.40	38.39 $\pm$ 2.26
META-DES	81.46 $\pm$ 1.00	32.20 $\pm$ 0.93	41.10 $\pm$ 2.88	37.12 $\pm$ 2.56
SAGMM-PE	<u>82.10</u> $\pm$ 0.92	33.15 $\pm$ 0.88	41.35 $\pm$ 2.51	37.05 $\pm$ 2.74
KDEM	81.82 $\pm$ 1.04	33.32 $\pm$ 1.01	41.37 $\pm$ 2.86	36.65 $\pm$ 2.66
PPEM	81.37 $\pm$ 1.00	32.84 $\pm$ 0.87	40.84 $\pm$ 2.87	36.18 $\pm$ 2.58
ROUTERGFM	83.74 $\pm$ 0.89	42.92 $\pm$ 1.55	47.09 $\pm$ 1.87	39.53 $\pm$ 2.68

Table 6: Node-level 100-shot accuracy comparisons (percent, higher is better). KDEM/PPEM use compatible expert architectures; META-DES uses the matched graph-context representation. Best/second-best means in each column are bold/underlined.

Method	Photo	Arxiv	Airports	Chameleon
Node-MoE	88.82 $\pm$ 0.43	44.94 $\pm$ 0.94	47.80 $\pm$ 1.42	<u>45.97</u> $\pm$ 1.44
META-DES	89.50 $\pm$ 0.40	49.70 $\pm$ 0.87	<u>49.18</u> $\pm$ 1.60	44.35 $\pm$ 1.24
SAGMM-PE	89.43 $\pm$ 0.47	49.72 $\pm$ 0.80	48.54 $\pm$ 1.44	44.35 $\pm$ 1.17
KDEM	89.03 $\pm$ 0.41	<u>50.71</u> $\pm$ 0.94	47.75 $\pm$ 1.67	43.88 $\pm$ 1.44
PPEM	88.74 $\pm$ 0.49	50.13 $\pm$ 0.78	47.24 $\pm$ 1.73	43.40 $\pm$ 1.21
ROUTERGFM	94.89 $\pm$ 0.34	57.60 $\pm$ 0.60	62.18 $\pm$ 1.20	59.14 $\pm$ 1.90

Table 7: Link-prediction ROC-AUC comparisons by the support block in Table 1 (percent, higher is better). The two blocks share the edge-split convention in Appendix B.3; shared baseline results are repeated for comparison and do not represent additional runs. Task-specific predictors retain their declared eligible inventories; bold and underline identify the best and second-best means within each block.

Method	5-shot block		100-shot block	
	DBLP	Cornell	DBLP	Cornell
Link-MoE	84.76 $\pm$ 1.35	78.42 $\pm$ 5.75	84.76 $\pm$ 1.35	78.42 $\pm$ 5.75
SAGMM-PE	82.80 $\pm$ 1.23	76.80 $\pm$ 5.20	82.80 $\pm$ 1.23	76.80 $\pm$ 5.20
KDEM	84.15 $\pm$ 1.47	75.82 $\pm$ 5.62	84.15 $\pm$ 1.47	75.82 $\pm$ 5.62
PPEM	83.96 $\pm$ 1.34	75.28 $\pm$ 5.17	83.96 $\pm$ 1.34	75.28 $\pm$ 5.17
ROUTERGFM	91.08 $\pm$ 0.45	79.50 $\pm$ 4.48	92.04 $\pm$ 0.75	79.81 $\pm$ 4.55

Table 8: Graph-classification accuracy on MNIST at both support budgets (percent, higher is better). Best/second-best means in each column are bold/underlined.

Method	5-shot	100-shot
META-DES	30.28 $\pm$ 1.80	43.77 $\pm$ 1.13
GMoPE	<u>30.66</u> $\pm$ 2.31	<u>44.63</u> $\pm$ 1.52
SAGMM-PE	30.11 $\pm$ 1.91	43.62 $\pm$ 1.13
ROUTERGFM	32.28 $\pm$ 1.84	46.18 $\pm$ 1.12

D.4 SELECTION, INSERTION, AND SOURCE CALIBRATION

Table 9 reports both best-expert inclusion and shortlist regret. At five shots, Model Spider and MetaGL have the same 72% hit@5 but different regrets (0.029 and 0.032); ROUTERGFM improves both criteria to 84% and 0.019. At 100 shots, ROUTERGFM retains the strongest values, 88% and 0.013. These diagnostics concern the globally best individual expert, not coverage of every local specialist.

Table 9: Global expert-selection diagnostics with $K = 5$ on node and single-label graph classification. Hit@5 is a percentage (higher is better), and regret@5 uses Brier risk (lower is better). All methods use the same eligible inventory, while target-execution costs of transferability estimators are not included in these selection-quality scores.

Method	5-shot		100-shot	
	Hit@5	Regret@5	Hit@5	Regret@5
Metadata MLP	64.0	0.045	68.0	0.036
Nearest application	60.0	0.053	64.0	0.044
MetaGL	72.0	0.032	76.0	0.025
LogME	68.0	0.036	80.0	0.023
Model Spider	72.0	0.029	80.0	0.022
ROUTERGFM	84.0	0.019	88.0	0.013

Table 10 separates new applications, new configurations, unseen architectures, and joint novelty while keeping router parameters fixed. Relative to MetaGL+metadata, ROUTERGFM reduces risk by 0.021, 0.031, 0.030, and 0.033, respectively. Nevertheless, its risk is 0.034 higher for unseen architectures than for new configurations built from observed factors. Source calibration reduces joint-novelty loss by 0.036 with four applications and by a further 0.003 with eight (Table 11). Residual-estimation error falls from 0.086 without calibration to 0.046 with four applications and 0.041 with eight. The calibration results distinguish additional evidence about an expert from metadata-only insertion.

Table 10: Inductive insertion diagnosis using Brier risk (lower is better). MetaGL+metadata is an explicitly extended baseline. Architecture-family holdouts group checkpoints differing only by seed.

Frozen-router setting	MetaGL+metadata	ROUTERGFM
New application; known experts	0.232	0.211
New expert; seen metadata factors	0.256	0.225
New architecture family	0.289	0.259
New application and expert	0.296	0.263

Table 11: Calibration of a newly inserted expert on separate source applications. Router parameters remain fixed; no target query labels are used. Calibration runs count toward the experiment budget.

Source calibration applications	Risk ↓	Mean residual error ↓
0	0.263	0.086
1	0.247	0.070
2	0.235	0.057
4	0.227	0.046
8	0.224	0.041

D.5 COVERAGE, SPECIALIZATION, AND EVIDENCE RELIABILITY

Table 12 follows nested teams under one global ranking. Moving from one to five experts lowers loss by 0.058 and raises winner coverage by 50.2 points. Beyond eight experts, coverage continues to increase while the observed prediction loss remains near its minimum. The loss-reduction column is a difference of mixture losses, not an estimate of the omission term Γ_K .

Table 12: Shortlist size, prediction risk, and local-winner coverage. Risk reduction is the absolute difference from the singleton mixture, calculated within this table. Expert contributions count predictors in the mixture, not wall-clock executions.

Team size K	Brier risk ↓	Risk reduction	Winner coverage (%) ↑	Expert contributions
1	0.269	0.000	40.4	1
2	0.232	0.037	65.9	2
3	0.218	0.051	78.2	3
5	0.211	0.058	90.6	5
8	0.209	0.060	96.1	8
16	0.212	0.057	98.8	16

With half the archived cells removed, source-validated shrinkage retains an advantage over global weighting, but removing a context family or reversing residual relationships gives higher loss (Table 13). Shrinkage reduces the corresponding excess losses relative to full correction without eliminating them. This agrees with the role of residual shift in Theorem 3: a nearby context need not carry transferable expert behavior.

Table 13: Coverage and failure analysis using Brier risk. Source-selected shrinkage may still fail under an unanticipated target shift; it is not an oracle that always recovers the global baseline.

Available context evidence	Global only	Local, $\rho = 1$	Source-validated ρ
Matched coverage	0.239	0.211	0.211
Half of context cells missing	0.239	0.235	0.227
Entire context family missing	0.239	0.281	0.242
Residual relation reversed	0.239	0.334	0.251

Table 14 and Figure 7 compare global and local weighting under the reported specialization conditions. Local transfer increases loss by 0.004 in the low condition but decreases it by 0.028 and 0.053 in the medium and high conditions. The conditions can differ in both expert behavior and estimation difficulty, so these contrasts are descriptive rather than causal effects of one graph property. The analytical example in Appendix A.2 separately isolates the direction of transferred residuals.

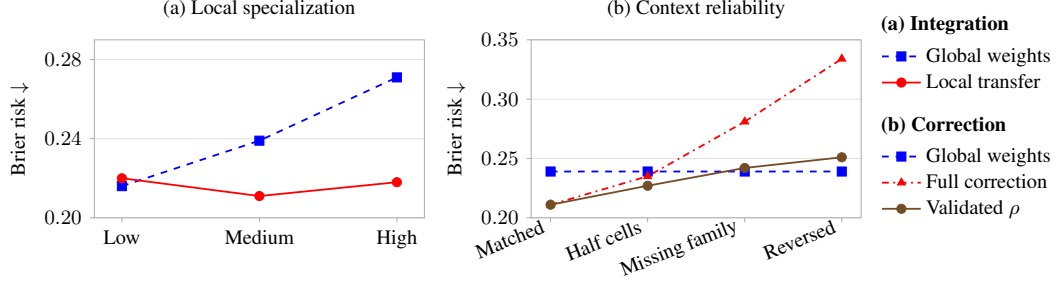


Figure 7: Local-prediction limits under (a) weak specialization and (b) unreliable context. Only global and local prediction risks are compared in (a); full correction in (b) sets $\rho = 1$, and ρ is selected on source validation applications (Tables 14 and 13).

Table 14: Prediction risks within the reported specialization conditions at a fixed support budget. The final column is local minus global risk, not an oracle gap or a paired uncertainty estimate.

Specialization condition	Global only	Local transfer	Risk difference
Low	0.216	0.220	+0.004
Medium	0.239	0.211	-0.028
High	0.271	0.218	-0.053

D.6 DISTRIBUTION-SHIFT COMPARISONS

Table 15 shows complementary strengths across shift conditions. ROUTERGFM gives the lowest feature-shift risk (0.232), whereas GeoMoE is better under structural shift (0.231 versus 0.247) and mixed shift (0.263 versus 0.269). Thus historical local integration is not uniformly preferable to a shift-oriented model. The comparison uses each method’s own inventory and adaptation procedure and does not establish that inserting a standalone shift specialist into the pool would reproduce its performance.

Table 15: Distribution-shift comparisons with task-appropriate graph experts, reporting Brier risk. Methods retain task-appropriate expert inventories and training budgets; lower is better.

Shift condition	GraphMETRO	OGMM	GeoMoE	ROUTERGFM
Feature shift	0.245	<u>0.237</u>	0.238	0.232
Structural shift	<u>0.239</u>	0.251	0.231	0.247
Mixed unseen shift	0.277	<u>0.266</u>	0.263	0.269

E CONFIGURATION-LEVEL RESULTS

Tables 16 and 3 summarize seven architectures and ten adaptation strategies on nine datasets; the following tables retain their configuration-level results. Each adaptation summary takes the best observed test mean over the completed architecture-objective grid and retains that configuration’s standard deviation. Figure 1 ranks the best configurations containing each factor, so its cells summarize attainable choices rather than isolated factor effects. Photo’s leading architecture changes from FAGCN (81.27%) to GAT (89.35%) across budgets, while Chameleon’s changes from Transformer (35.39%) to H2GCN (42.30%). The task dependence also appears in graph prediction: ProNoG obtains QM7b MAE 4.06 ± 0.07 with Transformer/InfoGraph at 100 shots, whereas All-in-One gives the best ToxCast adaptation mean, 60.96 ± 1.64 ROC-AUC, with tied GCN/EdgePred and GCN/InfoGraph configurations. These results motivate selecting among different pretrained choices, while the matched-pool and fixed-team experiments assess routing and integration more directly.

Table 16: Five-shot single-model configuration summary on nine datasets. Each adaptation entry is the best observed test mean over the architecture-objective grid, with the standard deviation of the same configuration. Accuracy and ROC-AUC are percentages; QM7b uses MAE. Bold and underline identify the best and second distinct means; LP conventions are specified in Appendix B.3.

Models	Homophily			Heterophily			Graph		
	Photo NC / Acc. $\uparrow$	ogbn-arxiv NC / Acc. $\uparrow$	DBLP LP / AUC $\uparrow$	Airports NC / Acc. $\uparrow$	Chameleon NC / Acc. $\uparrow$	Cornell LP / AUC $\uparrow$	QM7b GR / MAE $\downarrow$	ToxCast GC / AUC $\uparrow$	MNIST GC / Acc. $\uparrow$
<i>Supervised graph models</i>									
GCN	77.08 $\pm$ 1.52	22.86 $\pm$ 1.72	54.94 $\pm$ 0.45	32.51 $\pm$ 3.89	28.79 $\pm$ 4.71	58.29 $\pm$ 8.87	5.59 $\pm$ 0.28	48.76 $\pm$ 0.98	21.08 $\pm$ 0.72
GAT	72.23 $\pm$ 2.53	19.79 $\pm$ 1.41	55.01 $\pm$ 1.13	28.80 $\pm$ 3.49	27.74 $\pm$ 3.74	63.72 $\pm$ 4.61	21.56 $\pm$ 1.26	48.58 $\pm$ 1.53	22.77 $\pm$ 2.40
GIN	71.84 $\pm$ 1.52	17.86 $\pm$ 1.31	55.91 $\pm$ 0.82	35.18 $\pm$ 3.08	29.01 $\pm$ 3.20	61.91 $\pm$ 4.48	41.65 $\pm$ 5.39	45.97 $\pm$ 2.36	26.79 $\pm$ 2.20
H2GCN	76.77 $\pm$ 1.94	20.25 $\pm$ 1.34	55.33 $\pm$ 0.78	25.50 $\pm$ 3.47	29.80 $\pm$ 3.80	70.13 $\pm$ 5.92	6.14 $\pm$ 0.17	48.83 $\pm$ 1.31	23.83 $\pm$ 2.38
FAGCN	75.89 $\pm$ 1.79	23.60 $\pm$ 0.84	53.81 $\pm$ 0.86	28.02 $\pm$ 2.26	31.36 $\pm$ 5.31	62.37 $\pm$ 7.41	6.16 $\pm$ 0.43	48.22 $\pm$ 0.80	21.89 $\pm$ 1.35
Transformer	76.19 $\pm$ 3.34	22.89 $\pm$ 1.68	56.95 $\pm$ 0.52	33.98 $\pm$ 3.06	28.41 $\pm$ 3.29	72.42 $\pm$ 8.03	24.64 $\pm$ 37.88	48.73 $\pm$ 0.70	22.79 $\pm$ 2.93
NodeFormer	79.04 $\pm$ 2.30	4.43 $\pm$ 1.98	54.20 $\pm$ 4.08	33.47 $\pm$ 3.84	28.06 $\pm$ 6.79	58.75 $\pm$ 9.62	52.70 $\pm$ 47.67	50.51 $\pm$ 0.75	14.64 $\pm$ 3.46
<i>Best observed configurations by adaptation rule</i>									
Full fine-tuning	80.32 $\pm$ 1.36	29.02 $\pm$ 1.19	59.91 $\pm$ 3.28	34.63 $\pm$ 2.56	32.67 $\pm$ 3.82	63.58 $\pm$ 9.74	5.60 $\pm$ 0.69	50.88 $\pm$ 1.32	28.22 $\pm$ 1.25
All-in-One	79.45 $\pm$ 1.29	30.31 $\pm$ 0.26	54.40 $\pm$ 0.54	37.64 $\pm$ 4.45	33.61 $\pm$ 1.78	74.63 $\pm$ 14.46	35.77 $\pm$ 1.65	57.51 $\pm$ 3.09	26.89 $\pm$ 1.30
EdgePrompt	71.53 $\pm$ 5.19	31.38 $\pm$ 0.88	57.38 $\pm$ 1.50	38.17 $\pm$ 6.37	34.71 $\pm$ 3.01	61.08 $\pm$ 11.77	36.00 $\pm$ 1.98	52.39 $\pm$ 1.93	24.48 $\pm$ 2.13
EdgePrompt+	70.42 $\pm$ 3.71	29.51 $\pm$ 1.33	57.16 $\pm$ 3.77	38.80 $\pm$ 5.58	35.39 $\pm$ 2.10	61.81 $\pm$ 16.74	32.62 $\pm$ 1.56	52.46 $\pm$ 2.22	28.54 $\pm$ 2.58
GPF	80.88 $\pm$ 4.46	31.20 $\pm$ 0.91	56.91 $\pm$ 0.93	38.67 $\pm$ 4.85	34.95 $\pm$ 2.84	65.54 $\pm$ 8.29	33.95 $\pm$ 2.13	52.60 $\pm$ 1.47	26.67 $\pm$ 2.78
GPF+	81.27 $\pm$ 3.42	26.40 $\pm$ 1.26	56.28 $\pm$ 0.49	37.62 $\pm$ 4.03	34.40 $\pm$ 2.46	65.32 $\pm$ 10.05	34.99 $\pm$ 1.92	52.35 $\pm$ 1.69	25.53 $\pm$ 5.97
GPPT	78.46 $\pm$ 1.94	28.39 $\pm$ 4.30	55.35 $\pm$ 2.88	35.03 $\pm$ 2.05	33.72 $\pm$ 2.27	60.29 $\pm$ 6.19	41.86 $\pm$ 1.66	51.09 $\pm$ 1.67	22.35 $\pm$ 2.61
GraphPrompt	77.56 $\pm$ 1.63	14.67 $\pm$ 2.23	54.83 $\pm$ 0.55	35.40 $\pm$ 4.13	27.83 $\pm$ 6.74	66.76 $\pm$ 7.43	42.99 $\pm$ 2.25	47.53 $\pm$ 1.92	28.16 $\pm$ 2.70
GraphPrompt+	77.22 $\pm$ 2.58	13.59 $\pm$ 1.88	55.20 $\pm$ 0.69	35.61 $\pm$ 4.09	26.15 $\pm$ 6.21	61.90 $\pm$ 8.10	25.02 $\pm$ 4.18	44.65 $\pm$ 2.66	24.61 $\pm$ 1.79
ProNoG	80.56 $\pm$ 3.53	27.10 $\pm$ 3.22	57.03 $\pm$ 0.70	32.03 $\pm$ 3.68	30.57 $\pm$ 3.07	70.00 $\pm$ 5.18	8.40 $\pm$ 3.46	48.06 $\pm$ 0.75	27.28 $\pm$ 0.44

Table 17: 5-shot configuration results for GCN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	76.29±1.77	18.33±8.05	54.08±0.32	32.70±3.78	28.80±4.97	48.80±7.63	69.86±58.95	48.84±0.86	22.72±1.93
All-in-One	75.48±2.56	10.60±2.23	52.22±0.75	37.56 ±2.58	28.04±5.05	53.25±4.92	106.36±0.49	55.14 ±2.79	12.29±3.90
EdgePrompt	64.04±4.16	24.04±3.29	53.99±0.53	34.12±4.93	32.38 ±8.04	56.37 ±8.07	109.57±0.17	50.89±1.49	19.54±6.46
EdgePrompt+	66.05±5.86	20.00±2.91	53.75±0.56	34.70±6.09	31.88±3.00	53.56±9.97	109.25±0.20	51.68±0.76	24.42±1.50
GPF	76.70±2.18	23.36±2.41	54.11±0.53	31.08±5.22	31.53±1.05	49.52±10.32	110.98±0.26	51.94±0.80	20.42±4.38
GPF+	77.15 ±2.55	20.41±1.81	53.68±0.49	32.50±4.48	31.30±2.12	49.32±8.56	110.93±0.28	50.31±2.41	23.03±1.96
GPPT	72.48±3.39	17.51±5.27	50.86±1.52	31.64±4.36	27.50±4.80	52.22±8.24	108.52±0.77	49.90±0.87	18.43±1.52
GraphPrompt	68.39±3.65	11.08±2.18	53.42±0.42	31.81±1.43	25.00±5.49	52.51±4.12	99.99±0.20	43.07±2.99	24.64 ±0.77
GraphPrompt+	45.13±8.54	12.63±2.55	54.35 ±0.43	<u>55.21</u> ±3.95	22.14±2.37	<u>54.76</u> ±9.26	83.94±8.25	42.83±2.77	20.04±1.49
ProNoG	72.48±4.64	24.66 ±3.36	<u>54.12</u> ±1.01	28.36±2.87	26.79±2.43	52.50±5.90	10.13 ±4.09	43.23±2.32	23.38±1.01
Pretraining: ContextPred									
Full fine-tuning	76.95±4.53	19.71±10.06	53.87 ±0.30	30.91±5.89	29.33±4.53	44.88±6.25	5.78 ±0.61	49.65±1.42	22.53±2.85
All-in-One	72.52±1.87	6.62±1.03	52.49±0.27	32.55±2.42	29.86±4.80	53.77±10.17	109.54±0.70	50.08±0.56	18.29±4.34
EdgePrompt	68.26±7.93	29.24±2.51	53.27±0.43	34.43±4.80	31.16±2.51	49.87±10.24	110.51±0.10	<u>50.51</u> ±0.68	21.66±0.83
EdgePrompt+	67.70±5.29	25.95±1.70	52.90±0.45	35.95 ±6.14	31.92 ±2.94	52.98±11.53	110.27±0.17	50.67 ±1.65	25.37±1.21
GPF	78.37 ±3.80	29.75 ±2.29	52.80±0.66	34.12±4.71	31.58±2.61	44.44±8.82	111.08±0.15	50.11±0.98	20.94±3.57
GPF+	78.31±2.82	26.40±1.26	52.71±0.64	34.87±3.87	31.49±2.13	45.33±9.70	111.02±0.12	50.48±1.38	24.54±1.77
GPPT	76.88±2.48	16.81±7.21	50.67±1.29	30.29±4.69	30.13±1.57	52.71±9.85	109.06±0.79	50.36±0.87	20.12±0.63
GraphPrompt	74.80±4.84	10.51±1.28	52.72±0.59	29.42±4.90	23.09±4.43	56.58 ±9.25	101.18±0.50	47.18±3.69	25.64±1.73
GraphPrompt+	64.50±8.04	8.33±3.20	53.00±0.71	33.28±2.52	22.41±3.57	59.26 ±10.70	87.03±8.01	42.97±3.00	19.62±1.35
ProNoG	76.78±1.88	22.72±3.23	<u>53.29</u> ±0.91	29.98±4.45	27.53±2.75	55.84±8.02	<u>12.16</u> ±1.99	45.57±1.87	26.01 ±1.25
Pretraining: DGI									
Full fine-tuning	77.37±2.31	21.77±1.58	54.03 ±0.16	31.79±4.06	29.53±5.36	49.63±11.54	5.78 ±0.50	46.52±2.48	19.93±5.94
All-in-One	74.79±1.49	17.58±2.20	52.47±0.60	33.68 ±5.48	30.21±4.22	53.42±4.30	104.46±1.05	46.69±1.00	16.50±4.01
EdgePrompt	68.90±2.33	22.87±3.45	53.67±0.79	32.29±5.15	34.14 ±3.98	53.17±9.72	108.74±0.38	49.07±0.36	20.05±5.04
EdgePrompt+	66.76±5.24	22.18±1.65	53.34±0.36	33.42±4.82	33.37 ±3.34	51.93±6.26	108.44±0.13	<u>49.33</u> ±0.71	24.37 ±2.94
GPF	77.84 ±1.94	22.71±1.00	53.06±0.36	26.75±2.42	32.00±3.58	50.83±12.43	110.30±1.18	48.60±0.73	20.76±4.66
GPF+	77.04±2.08	21.67±1.75	53.07±0.35	28.79±4.09	33.29±3.64	53.17±8.78	109.29±0.10	48.71±0.55	23.98±1.89
GPPT	44.63±4.05	17.01±4.40	49.60±0.59	26.34±6.34	21.37±4.67	53.23±12.62	104.90±2.04	50.32 ±1.57	11.27±1.68
GraphPrompt	74.47±3.97	9.96±2.27	52.85±1.18	31.38±1.49	23.99±4.54	55.92 ±10.73	87.45±0.76	42.85±3.06	24.23±1.98
GraphPrompt+	70.06±9.35	10.39±2.27	53.15±0.69	30.82±2.30	25.30±6.51	61.90 ±8.10	63.87±9.30	43.07±2.83	21.13±0.50
ProNoG	74.13±5.42	24.55 ±3.32	52.33±0.61	27.06±3.29	28.69±2.84	53.34±3.67	<u>17.14</u> ±6.97	45.35±1.40	23.17±0.85
Pretraining: EdgePred									
Full fine-tuning	76.49 ±1.88	15.55±9.12	53.79 ±0.28	30.19±4.24	29.41±5.13	43.88±5.64	5.60 ±0.69	46.66±2.30	19.93±5.94
All-in-One	24.39±2.10	3.06±1.32	50.90±1.31	34.50±3.67	29.80±4.51	55.46 ±7.28	114.39±0.75	46.67±2.19	13.97±2.99
EdgePrompt	45.40±10.29	10.80±2.40	53.29±0.18	35.15 ±3.98	24.31±4.56	49.66±7.58	110.51±0.10	50.93 ±1.20	19.66±4.50
EdgePrompt+	63.60±5.16	23.01 ±1.67	53.46±0.52	35.06±7.36	30.33±6.92	50.64±10.10	110.27±0.17	50.48±1.19	<u>24.27</u> ±1.00
GPF	38.96±5.94	9.03±2.04	51.33±0.92	30.87±6.17	30.01±1.66	41.76±9.06	116.65±0.45	50.74±1.03	13.18±7.45
GPF+	63.61±5.21	12.52±2.47	53.34±0.25	29.74±4.87	31.17 ±1.89	41.00±8.14	111.02±0.12	49.20±1.65	22.35±1.67
GPPT	16.72±3.99	3.28±1.24	51.71±1.97	33.93±3.29	26.71±2.54	54.44±7.88	109.06±0.79	50.08±0.22	20.80±0.83
GraphPrompt	23.01±1.45	4.02±0.66	51.17±0.56	30.17±2.65	24.19±3.24	53.35±8.36	113.85±1.87	45.72±1.79	23.57±1.01
GraphPrompt+	45.89±9.69	2.74±0.92	53.22±0.62	32.21±2.46	25.13±5.83	55.23±8.87	87.03±8.01	42.84±2.71	19.16±1.09
ProNoG	20.78±3.10	5.09±0.55	52.76±0.30	32.03±3.68	27.14±2.07	51.33±8.06	<u>12.16</u> ±1.99	42.49±2.71	25.82 ±0.90
Pretraining: GraphCL									
Full fine-tuning	77.31±3.64	12.17±8.16	54.35 ±0.07	32.32±3.37	28.84±4.94	43.85±5.70	34.35±13.23	46.43±1.29	21.22±5.40
All-in-One	71.01±3.11	6.54±2.18	52.37±0.25	34.94±3.67	28.13±4.35	51.36±5.57	114.31±0.68	54.89 ±2.61	20.03±2.76
EdgePrompt	68.43±6.51	22.10±1.67	53.31±0.63	34.79±4.61	30.81±2.31	46.99±5.81	110.34±0.27	47.87±1.82	21.55±1.56
EdgePrompt+	69.36±5.13	23.01±1.67	53.11±1.00	35.20 ±4.70	30.58±2.56	47.69±14.84	110.00±0.14	47.83±2.01	25.16 ±1.88
GPF	77.81±1.38	23.22 ±1.42	53.22±0.50	26.68±3.30	31.51±2.08	43.89±9.18	116.14±0.13	48.26±2.52	22.68±1.16
GPF+	77.92 ±1.33	22.32±1.53	53.50±0.37	27.57±4.50	31.77 ±2.64	44.41±7.81	116.07±0.13	48.11±2.30	24.28±1.36
GPPT	66.71±1.56	12.27±4.16	50.22±0.27	27.83±3.46	26.23±2.35	52.83±5.75	115.34±0.13	49.16±0.95	19.10±1.83
GraphPrompt	69.52±3.67	10.76±1.27	53.32±0.48	29.85±2.76	22.32±3.47	49.83±5.28	113.67±0.12	<u>44.76</u> ±2.37	24.24±1.96
GraphPrompt+	74.01±4.90	11.00±1.44	<u>53.76</u> ±0.61	32.24±1.34	24.59±6.20	59.00 ±10.48	97.09±12.95	42.63±3.21	19.38±2.37
ProNoG	67.47±1.59	18.54±1.36	53.74±0.82	28.85±2.59	26.85±3.00	<u>53.95</u> ±8.12	9.68 ±2.42	45.93±1.66	24.26±1.22
Pretraining: InfoGraph									
Full fine-tuning	77.58 ±1.74	21.94 ±2.16	53.81 ±0.40	32.26±3.46	28.74±5.10	46.37±9.52	5.67 ±0.67	48.87±0.85	19.93±5.94
All-in-One	46.06±2.64	3.06±1.32	51.81±0.99	34.50±3.67	28.46±4.82	56.03±7.48	114.39±0.75	57.51 ±3.09	13.97±2.99
EdgePrompt	61.78±5.70	11.40±2.20	52.34±0.99	35.15 ±3.98	31.40 ±4.85	51.07±9.04	112.15±0.06	50.93 ±1.20	19.66±4.50
EdgePrompt+	64.39±7.89	14.63±0.98	53.04±0.37	34.15±5.76	30.87±3.67	50.02±11.48	111.88±0.09	50.48±1.19	24.27 ±1.00
GPF	70.38±4.68	10.97±1.53	52.52±0.39	27.98±4.51	28.32±1.57	46.75±9.86	116.30±0.05	50.78±1.08	20.23±2.85
GPF+	77.51±3.02	12.17±1.22	52.25±0.30	27.56±3.92	30.99±1.47	46.28±9.74	116.29±0.06	50.51±0.95	23.88±0.92
GPPT	48.07±3.59	4.35±0.51	<u>50.57</u> ±1.29	29.11±7.75	29.79±3.05	52.29±8.77	114.89±0.39	50.08±0.22	20.80±0.83
GraphPrompt	44.42±2.94	5.32±0.64	52.40±0.72	30.09±2.17	25.09±5.51	53.28±7.46	112.52±0.06	45.68±1.85	23.25±0.91
GraphPrompt+	63.79±6.28	3.44±1.34	52.51±0.62	32.00±2.15	23.62±5.58	58.26 ±10.18	96.34±12.56	42.57±2.68	20.94±0.34
ProNoG	52.61±2.75	8.87±0.69	53.09±0.86	28.51±3.59	28.77±2.68	55.38±6.67	<u>12.53</u> ±3.09	43.89±2.14	22.98±1.18

Table 18: 5-shot configuration results for GAT across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	GR / MAE ↓	GC / AUC ↑	GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	76.45 $\pm$ 4.21	26.60 $\pm$ 1.53	54.11 $\pm$ 0.47	28.82 $\pm$ 2.61	28.80 $\pm$ 4.19	47.98 $\pm$ 5.27	33.13 $\pm$ 23.94	50.63 $\pm$ 0.71	22.25 $\pm$ 2.22
All-in-One	76.71 $\pm$ 4.34	15.03 $\pm$ 1.60	51.95 $\pm$ 0.92	31.76 $\pm$ 1.57	24.43 $\pm$ 2.91	56.06 $\pm$ 5.81	108.70 $\pm$ 0.06	55.98 $\pm$ 2.50	11.20 $\pm$ 1.64
EdgePrompt	59.57 $\pm$ 18.75	28.41 $\pm$ 1.25	53.31 $\pm$ 0.75	22.32 $\pm$ 7.46	26.92 $\pm$ 2.53	53.96 $\pm$ 11.29	111.97 $\pm$ 0.21	50.19 $\pm$ 0.36	13.45 $\pm$ 5.17
EdgePrompt+	69.07 $\pm$ 4.88	23.53 $\pm$ 2.21	53.56 $\pm$ 0.36	31.13 $\pm$ 4.31	29.67 $\pm$ 4.11	56.49 $\pm$ 13.38	111.97 $\pm$ 0.23	50.19 $\pm$ 0.98	20.39 $\pm$ 4.07
GPF	77.65 $\pm$ 2.16	28.27 $\pm$ 1.51	53.43 $\pm$ 0.65	26.27 $\pm$ 1.44	27.77 $\pm$ 3.12	50.08 $\pm$ 8.40	113.13 $\pm$ 0.23	50.48 $\pm$ 0.18	12.96 $\pm$ 4.15
GPF+	78.20 $\pm$ 1.15	23.43 $\pm$ 1.69	53.36 $\pm$ 0.56	23.79 $\pm$ 4.49	31.17 $\pm$ 3.29	48.51 $\pm$ 7.95	113.12 $\pm$ 0.18	50.83 $\pm$ 0.40	13.56 $\pm$ 5.60
GPPT	75.91 $\pm$ 6.79	17.36 $\pm$ 1.76	51.94 $\pm$ 1.60	28.80 $\pm$ 3.98	27.81 $\pm$ 4.60	51.43 $\pm$ 12.84	110.13 $\pm$ 0.91	50.27 $\pm$ 0.78	19.38 $\pm$ 2.29
GraphPrompt	71.79 $\pm$ 3.34	11.72 $\pm$ 1.01	53.48 $\pm$ 0.55	34.03 $\pm$ 3.30	24.96 $\pm$ 4.88	54.70 $\pm$ 2.31	111.51 $\pm$ 5.82	42.70 $\pm$ 3.00	27.46 $\pm$ 1.41
GraphPrompt+	61.42 $\pm$ 5.15	12.22 $\pm$ 1.30	53.63 $\pm$ 1.01	35.18 $\pm$ 3.94	22.32 $\pm$ 2.43	47.97 $\pm$ 3.91	99.66 $\pm$ 6.87	42.94 $\pm$ 2.51	24.61 $\pm$ 1.79
ProNoG	76.55 $\pm$ 5.00	22.80 $\pm$ 2.43	54.63 $\pm$ 0.47	26.63 $\pm$ 1.60	27.59 $\pm$ 3.21	50.94 $\pm$ 4.46	13.74 $\pm$ 3.36	42.28 $\pm$ 2.62	<u>25.92</u> $\pm$ 1.27
Pretraining: ContextPred									
Full fine-tuning	75.50 $\pm$ 3.20	29.02 $\pm$ 1.19	54.21 $\pm$ 0.44	27.32 $\pm$ 2.20	30.29 $\pm$ 4.70	55.84 $\pm$ 10.35	20.35 $\pm$ 0.75	49.58 $\pm$ 0.93	25.26 $\pm$ 2.54
All-in-One	69.36 $\pm$ 3.59	20.00 $\pm$ 0.92	51.56 $\pm$ 0.51	28.07 $\pm$ 3.82	25.12 $\pm$ 3.13	52.38 $\pm$ 5.15	115.82 $\pm$ 0.08	46.74 $\pm$ 1.31	22.51 $\pm$ 2.40
EdgePrompt	68.58 $\pm$ 5.23	31.38 $\pm$ 0.88	52.99 $\pm$ 0.98	36.99 $\pm$ 2.41	29.56 $\pm$ 1.60	54.03 $\pm$ 13.24	115.57 $\pm$ 0.10	49.34 $\pm$ 0.76	22.20 $\pm$ 5.26
EdgePrompt+	68.42 $\pm$ 6.28	26.30 $\pm$ 0.63	52.85 $\pm$ 0.46	37.45 $\pm$ 3.03	31.07 $\pm$ 3.28	49.57 $\pm$ 15.02	115.38 $\pm$ 0.07	49.97 $\pm$ 1.33	28.54 $\pm$ 2.58
GPF	73.93 $\pm$ 3.95	<u>31.20</u> $\pm$ 0.91	52.91 $\pm$ 1.00	26.87 $\pm$ 2.87	28.84 $\pm$ 2.22	50.49 $\pm$ 12.92	116.56 $\pm$ 0.02	49.58 $\pm$ 1.46	26.67 $\pm$ 2.78
GPF+	74.48 $\pm$ 3.02	20.10 $\pm$ 11.29	53.02 $\pm$ 0.47	27.15 $\pm$ 2.61	28.68 $\pm$ 3.74	49.68 $\pm$ 8.84	116.62 $\pm$ 0.13	50.05 $\pm$ 0.84	25.53 $\pm$ 5.97
GPPT	67.34 $\pm$ 2.66	26.80 $\pm$ 5.03	50.70 $\pm$ 1.28	31.59 $\pm$ 2.72	28.22 $\pm$ 1.48	44.38 $\pm$ 12.54	116.27 $\pm$ 0.17	50.04 $\pm$ 1.16	22.35 $\pm$ 2.61
GraphPrompt	70.24 $\pm$ 1.84	12.25 $\pm$ 1.09	53.10 $\pm$ 0.66	28.17 $\pm$ 3.75	23.91 $\pm$ 4.30	60.99 $\pm$ 10.52	114.58 $\pm$ 0.15	42.89 $\pm$ 2.64	27.43 $\pm$ 2.21
GraphPrompt+	69.02 $\pm$ 2.53	11.13 $\pm$ 1.57	53.01 $\pm$ 0.71	34.50 $\pm$ 3.81	22.45 $\pm$ 2.24	47.05 $\pm$ 4.02	113.57 $\pm$ 1.09	42.73 $\pm$ 2.77	21.15 $\pm$ 2.82
ProNoG	71.75 $\pm$ 2.34	25.39 $\pm$ 2.42	55.41 $\pm$ 0.50	26.60 $\pm$ 1.27	28.07 $\pm$ 2.66	51.89 $\pm$ 11.92	13.32 $\pm$ 2.16	45.16 $\pm$ 1.85	25.59 $\pm$ 1.69
Pretraining: DGI									
Full fine-tuning	75.55 $\pm$ 2.05	22.73 $\pm$ 1.25	<u>54.05</u> $\pm$ 0.26	28.26 $\pm$ 1.93	30.47 $\pm$ 5.14	55.72 $\pm$ 9.44	31.11 $\pm$ 7.76	49.48 $\pm$ 1.01	26.08 $\pm$ 3.15
All-in-One	59.23 $\pm$ 5.81	20.51 $\pm$ 0.88	52.26 $\pm$ 0.75	28.21 $\pm$ 3.58	33.61 $\pm$ 1.78	<u>54.33</u> $\pm$ 11.78	111.70 $\pm$ 0.12	50.06 $\pm$ 0.90	19.87 $\pm$ 0.92
EdgePrompt	48.10 $\pm$ 2.80	22.15 $\pm$ 1.22	53.06 $\pm$ 0.63	28.14 $\pm$ 4.12	34.58 $\pm$ 3.45	42.71 $\pm$ 10.24	110.52 $\pm$ 0.09	51.36 $\pm$ 1.59	18.63 $\pm$ 4.05
EdgePrompt+	61.19 $\pm$ 5.38	23.09 $\pm$ 0.74	53.10 $\pm$ 1.04	32.82 $\pm$ 3.27	31.20 $\pm$ 3.88	51.40 $\pm$ 7.99	110.40 $\pm$ 0.08	50.78 $\pm$ 1.56	21.00 $\pm$ 4.07
GPF	53.91 $\pm$ 4.12	23.02 $\pm$ 2.28	52.88 $\pm$ 0.56	25.61 $\pm$ 1.07	33.61 $\pm$ 2.05	45.73 $\pm$ 12.23	115.50 $\pm$ 0.05	51.01 $\pm$ 0.93	19.65 $\pm$ 4.71
GPF+	57.67 $\pm$ 3.47	19.22 $\pm$ 3.35	52.71 $\pm$ 0.64	26.55 $\pm$ 2.57	32.09 $\pm$ 4.75	44.58 $\pm$ 11.80	115.67 $\pm$ 0.37	50.05 $\pm$ 0.74	20.66 $\pm$ 3.07
GPPT	43.14 $\pm$ 7.78	13.53 $\pm$ 6.11	51.33 $\pm$ 0.69	30.96 $\pm$ 5.73	29.13 $\pm$ 5.80	46.48 $\pm$ 9.55	113.86 $\pm$ 0.42	50.02 $\pm$ 1.16	16.56 $\pm$ 2.90
GraphPrompt	49.69 $\pm$ 8.13	8.60 $\pm$ 1.75	53.11 $\pm$ 0.52	29.16 $\pm$ 1.95	23.26 $\pm$ 3.83	47.25 $\pm$ 10.71	111.87 $\pm$ 0.12	43.14 $\pm$ 3.19	26.91 $\pm$ 2.06
GraphPrompt+	53.07 $\pm$ 7.97	10.22 $\pm$ 2.53	53.14 $\pm$ 0.52	32.85 $\pm$ 6.66	23.77 $\pm$ 5.27	50.42 $\pm$ 6.69	110.87 $\pm$ 0.63	42.95 $\pm$ 2.57	21.61 $\pm$ 2.25
ProNoG	44.42 $\pm$ 8.12	24.46 $\pm$ 2.79	54.48 $\pm$ 0.45	27.37 $\pm$ 2.32	30.47 $\pm$ 3.51	44.08 $\pm$ 10.84	14.04 $\pm$ 2.33	42.77 $\pm$ 2.60	20.81 $\pm$ 1.66
Pretraining: EdgePred									
Full fine-tuning	67.75 $\pm$ 6.64	22.89 $\pm$ 1.70	54.04 $\pm$ 0.61	26.82 $\pm$ 2.17	27.53 $\pm$ 4.45	53.19 $\pm$ 9.87	20.65 $\pm$ 2.88	46.57 $\pm$ 1.73	26.68 $\pm$ 1.81
All-in-One	26.01 $\pm$ 1.79	5.04 $\pm$ 0.61	50.84 $\pm$ 0.58	30.60 $\pm$ 4.98	25.99 $\pm$ 1.53	54.60 $\pm$ 9.31	115.82 $\pm$ 0.08	45.16 $\pm$ 2.05	19.44 $\pm$ 5.10
EdgePrompt	28.00 $\pm$ 2.49	10.40 $\pm$ 1.99	51.54 $\pm$ 0.53	24.17 $\pm$ 2.28	31.74 $\pm$ 2.67	58.67 $\pm$ 7.16	115.57 $\pm$ 0.10	51.35 $\pm$ 2.27	16.52 $\pm$ 5.74
EdgePrompt+	56.86 $\pm$ 7.73	<u>14.64</u> $\pm$ 1.21	53.16 $\pm$ 0.35	29.76 $\pm$ 3.65	<u>30.56</u> $\pm$ 5.36	54.60 $\pm$ 6.77	115.38 $\pm$ 0.07	51.43 $\pm$ 1.68	25.23 $\pm$ 1.19
GPF	20.24 $\pm$ 6.16	9.75 $\pm$ 1.08	51.92 $\pm$ 1.06	25.56 $\pm$ 1.10	28.29 $\pm$ 2.93	54.72 $\pm$ 7.45	116.56 $\pm$ 0.02	52.60 $\pm$ 1.47	21.70 $\pm$ 1.36
GPF+	32.61 $\pm$ 5.46	11.37 $\pm$ 1.52	52.81 $\pm$ 0.58	25.11 $\pm$ 0.04	29.25 $\pm$ 2.39	54.67 $\pm$ 6.61	116.56 $\pm$ 0.02	52.35 $\pm$ 1.69	22.34 $\pm$ 1.97
GPPT	12.16 $\pm$ 5.96	2.40 $\pm$ 0.68	49.79 $\pm$ 0.43	29.81 $\pm$ 3.20	25.19 $\pm$ 2.98	49.22 $\pm$ 9.36	116.27 $\pm$ 0.17	49.81 $\pm$ 0.71	15.33 $\pm$ 2.38
GraphPrompt	16.96 $\pm$ 2.58	3.23 $\pm$ 0.36	50.70 $\pm$ 0.57	31.21 $\pm$ 2.04	24.05 $\pm$ 3.56	54.05 $\pm$ 8.07	114.58 $\pm$ 0.15	42.60 $\pm$ 2.84	25.24 $\pm$ 1.54
GraphPrompt+	36.22 $\pm$ 6.18	3.78 $\pm$ 0.88	52.65 $\pm$ 0.54	33.81 $\pm$ 3.34	25.25 $\pm$ 2.49	54.28 $\pm$ 8.29	113.57 $\pm$ 1.09	42.67 $\pm$ 2.91	21.79 $\pm$ 2.74
ProNoG	14.28 $\pm$ 3.62	3.97 $\pm$ 0.68	<u>53.70</u> $\pm$ 0.60	25.03 $\pm$ 7.24	27.31 $\pm$ 1.86	60.26 $\pm$ 5.04	13.32 $\pm$ 2.16	42.74 $\pm$ 2.75	24.50 $\pm$ 1.51
Pretraining: GraphCL									
Full fine-tuning	73.68 $\pm$ 3.84	23.28 $\pm$ 1.58	53.88 $\pm$ 0.42	27.37 $\pm$ 2.31	30.08 $\pm$ 3.03	57.21 $\pm$ 5.98	45.35 $\pm$ 7.04	48.46 $\pm$ 1.10	17.91 $\pm$ 1.95
All-in-One	65.18 $\pm$ 2.67	18.55 $\pm$ 1.29	51.59 $\pm$ 0.92	28.60 $\pm$ 2.60	26.60 $\pm$ 3.80	51.95 $\pm$ 11.05	112.93 $\pm$ 0.05	48.35 $\pm$ 1.59	14.08 $\pm$ 2.59
EdgePrompt	69.60 $\pm$ 6.19	<u>22.63</u> $\pm$ 2.97	53.45 $\pm$ 0.44	35.04 $\pm$ 2.69	30.32 $\pm$ 2.41	<u>56.72</u> $\pm$ 5.14	111.21 $\pm$ 0.25	49.04 $\pm$ 0.67	13.52 $\pm$ 2.39
EdgePrompt+	69.97 $\pm$ 5.37	22.11 $\pm$ 0.96	52.90 $\pm$ 0.52	36.65 $\pm$ 3.56	31.18 $\pm$ 2.73	49.33 $\pm$ 5.14	111.25 $\pm$ 0.22	48.95 $\pm$ 0.98	19.32 $\pm$ 3.93
GPF	75.10 $\pm$ 2.02	22.41 $\pm$ 4.01	53.19 $\pm$ 0.72	30.36 $\pm$ 4.26	30.07 $\pm$ 2.56	54.72 $\pm$ 5.39	116.65 $\pm$ 0.08	49.00 $\pm$ 1.12	12.25 $\pm$ 2.02
GPF+	76.61 $\pm$ 2.27	20.52 $\pm$ 3.33	53.52 $\pm$ 0.45	27.98 $\pm$ 2.13	30.46 $\pm$ 2.48	55.43 $\pm$ 4.06	116.61 $\pm$ 0.15	48.93 $\pm$ 1.27	11.40 $\pm$ 1.25
GPPT	67.83 $\pm$ 2.14	15.37 $\pm$ 5.46	50.56 $\pm$ 1.08	28.29 $\pm$ 6.71	25.80 $\pm$ 2.21	55.58 $\pm$ 9.04	114.86 $\pm$ 0.30	49.41 $\pm$ 0.95	12.20 $\pm$ 0.76
GraphPrompt	66.17 $\pm$ 2.60	11.25 $\pm$ 1.53	53.55 $\pm$ 0.65	28.68 $\pm$ 4.84	20.92 $\pm$ 3.46	52.83 $\pm$ 4.78	112.67 $\pm$ 0.09	45.60 $\pm$ 2.64	<u>19.94</u> $\pm$ 2.15
GraphPrompt+	71.98 $\pm$ 2.32	10.50 $\pm$ 1.42	53.67 $\pm$ 0.41	31.23 $\pm$ 3.99	26.15 $\pm$ 6.21	51.52 $\pm$ 6.37	110.84 $\pm$ 0.41	42.86 $\pm$ 3.15	17.73 $\pm$ 3.20
ProNoG	64.90 $\pm$ 1.41	21.93 $\pm$ 3.18	53.46 $\pm$ 1.21	28.56 $\pm$ 3.75	26.60 $\pm$ 2.46	56.51 $\pm$ 10.77	10.44 $\pm$ 1.66	46.82 $\pm$ 2.01	14.82 $\pm$ 1.22
Pretraining: InfoGraph									
Full fine-tuning	73.73 $\pm$ 1.98	23.14 $\pm$ 1.91	54.61 $\pm$ 0.28	29.11 $\pm$ 1.87	28.78 $\pm$ 3.53	53.85 $\pm$ 11.02	43.11 $\pm$ 7.43	49.19 $\pm$ 1.05	21.96 $\pm$ 4.74
All-in-One	29.75 $\pm$ 2.30	8.88 $\pm$ 1.52	50.83 $\pm$ 0.79	28.19 $\pm$ 2.67	29.90 $\pm$ 3.86	50.48 $\pm$ 8.18	114.58 $\pm$ 0.09	52.84 $\pm$ 1.32	11.97 $\pm$ 1.41
EdgePrompt	51.93 $\pm$ 7.22	7.77 $\pm$ 2.56	52.87 $\pm$ 0.53	27.01 $\pm$ 4.91	32.35 $\pm$ 2.44	56.05 $\pm$ 7.15	114.32 $\pm$ 0.30	49.69 $\pm$ 1.31	14.23 $\pm$ 2.64
EdgePrompt+	66.75 $\pm$ 5.36	<u>13.69</u> $\pm$ 1.65	53.47 $\pm$ 0.44	33.76 $\pm$ 2.98	31.82 $\pm$ 3.56	52.87 $\pm$ 10.30	114.35 $\pm$ 0.20	494	

Table 19: 5-shot configuration results for GIN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	75.84 ± 1.45	27.54 ± 0.71	53.49 ± 0.84	33.64 ± 2.66	29.67 ± 2.38	54.68 ± 4.60	41.38 ± 2.62	47.99 ± 2.53	28.22 ± 1.25
All-in-One	72.98 ± 3.70	10.30 ± 2.10	52.15 ± 0.11	32.14 ± 2.67	27.98 ± 4.92	54.79 ± 4.78	59.26 ± 1.43	53.97 ± 4.02	19.82 ± 5.25
EdgePrompt	60.45 ± 4.78	14.13 ± 1.32	54.31 ± 0.77	33.21 ± 3.23	24.47 ± 4.28	46.11 ± 6.56	95.57 ± 1.49	48.47 ± 1.23	22.92 ± 0.97
EdgePrompt+	54.49 ± 18.55	13.60 ± 1.56	54.95 ± 1.10	31.90 ± 3.94	24.99 ± 4.52	51.25 ± 10.73	93.85 ± 1.33	48.18 ± 0.52	25.28 ± 0.98
GPF	68.74 ± 4.70	16.97 ± 3.68	52.33 ± 0.33	35.85 ± 4.58	23.99 ± 4.10	46.11 ± 9.75	97.29 ± 1.49	49.54 ± 1.94	24.95 ± 2.23
GPF+	72.10 ± 3.12	18.47 ± 2.32	52.57 ± 0.68	35.78 ± 4.30	26.54 ± 4.25	46.61 ± 8.09	97.07 ± 0.69	49.39 ± 1.64	24.14 ± 2.88
GPPT	51.52 ± 10.79	18.60 ± 6.61	51.76 ± 1.42	32.44 ± 4.46	23.68 ± 2.82	51.00 ± 7.33	75.19 ± 5.35	50.05 ± 0.65	17.49 ± 2.39
GraphPrompt	66.87 ± 4.31	13.58 ± 1.97	53.29 ± 0.81	34.99 ± 3.65	24.58 ± 4.56	51.11 ± 4.41	67.37 ± 33.79	44.11 ± 3.34	23.70 ± 1.15
GraphPrompt+	44.22 ± 7.38	8.90 ± 1.83	53.78 ± 0.46	35.33 ± 4.17	22.23 ± 2.29	49.63 ± 7.23	41.73 ± 1.57	42.73 ± 2.79	21.36 ± 1.07
ProNoG	74.40 ± 5.06	25.51 ± 4.74	55.17 ± 0.79	30.77 ± 4.48	25.79 ± 1.81	55.38 ± 8.61	55.81 ± 41.58	42.81 ± 2.67	24.03 ± 1.18
Pretraining: ContextPred									
Full fine-tuning	71.93 ± 2.94	22.44 ± 1.70	56.71 ± 0.52	33.91 ± 3.31	29.80 ± 3.09	58.44 ± 9.09	42.98 ± 3.78	47.05 ± 1.07	27.14 ± 1.92
All-in-One	67.78 ± 2.00	12.93 ± 2.18	54.40 ± 0.54	32.60 ± 4.09	27.80 ± 4.15	59.01 ± 5.36	35.77 ± 1.65	54.01 ± 3.01	21.17 ± 7.56
EdgePrompt	60.50 ± 1.28	4.97 ± 0.66	56.95 ± 0.26	30.34 ± 3.27	26.71 ± 4.45	57.12 ± 5.31	42.33 ± 2.04	52.39 ± 1.93	24.27 ± 3.04
EdgePrompt+	56.71 ± 3.51	6.96 ± 2.10	56.70 ± 0.31	33.21 ± 1.82	26.95 ± 5.39	55.48 ± 6.81	32.62 ± 1.56	52.46 ± 2.22	25.09 ± 2.26
GPF	69.72 ± 2.51	10.82 ± 1.93	56.34 ± 0.75	36.07 ± 5.10	25.65 ± 4.60	55.89 ± 7.05	33.95 ± 2.13	51.05 ± 2.16	21.21 ± 6.25
GPF+	70.03 ± 2.72	10.21 ± 4.18	56.28 ± 0.49	36.21 ± 4.91	30.37 ± 5.85	54.08 ± 8.60	34.99 ± 1.92	49.50 ± 2.52	21.76 ± 6.75
GPPT	70.08 ± 3.75	3.65 ± 1.65	50.49 ± 0.48	32.55 ± 1.63	24.77 ± 4.70	60.29 ± 6.19	41.86 ± 1.66	50.78 ± 1.16	22.13 ± 2.10
GraphPrompt	71.31 ± 4.37	7.11 ± 0.84	54.83 ± 0.55	31.95 ± 6.42	24.05 ± 4.04	66.76 ± 7.43	42.99 ± 2.25	44.49 ± 2.07	26.52 ± 2.68
GraphPrompt+	51.63 ± 13.25	3.80 ± 1.93	55.20 ± 0.69	35.11 ± 4.03	22.28 ± 2.17	59.57 ± 6.54	43.85 ± 1.33	43.21 ± 2.30	21.79 ± 1.00
ProNoG	73.91 ± 3.66	13.65 ± 3.07	55.78 ± 0.50	27.56 ± 6.16	23.25 ± 2.73	54.57 ± 8.97	25.85 ± 11.80	46.28 ± 1.06	24.79 ± 1.25
Pretraining: DGI									
Full fine-tuning	75.87 ± 5.11	17.91 ± 2.13	53.79 ± 0.73	34.34 ± 3.18	31.27 ± 4.41	53.34 ± 7.67	41.31 ± 4.51	46.90 ± 2.46	22.33 ± 2.04
All-in-One	70.25 ± 3.60	5.25 ± 3.19	51.43 ± 1.38	32.36 ± 3.03	29.57 ± 4.17	47.63 ± 8.56	63.35 ± 1.50	53.68 ± 1.61	16.72 ± 8.07
EdgePrompt	58.65 ± 6.71	20.95 ± 3.05	53.68 ± 0.71	26.75 ± 8.63	33.85 ± 4.08	55.72 ± 12.21	98.26 ± 0.65	48.71 ± 2.53	18.33 ± 4.27
EdgePrompt+	56.97 ± 6.86	21.73 ± 1.20	53.85 ± 0.71	32.91 ± 6.22	33.06 ± 5.12	53.95 ± 9.18	96.91 ± 1.40	48.96 ± 2.26	22.46 ± 1.77
GPF	72.78 ± 6.35	13.56 ± 2.81	52.78 ± 0.73	32.38 ± 5.59	31.16 ± 3.44	52.82 ± 4.51	99.56 ± 1.22	49.83 ± 1.96	16.19 ± 6.13
GPF+	72.34 ± 2.20	19.39 ± 5.46	52.72 ± 0.57	31.50 ± 4.01	32.59 ± 3.91	51.56 ± 4.47	98.82 ± 0.97	47.86 ± 0.99	19.94 ± 2.30
GPPT	23.66 ± 11.54	5.03 ± 2.04	50.50 ± 1.48	29.28 ± 3.57	20.02 ± 7.35	43.26 ± 13.81	79.70 ± 5.45	49.54 ± 0.85	13.11 ± 0.96
GraphPrompt	52.81 ± 5.43	11.82 ± 2.50	53.76 ± 0.64	30.92 ± 3.38	21.04 ± 3.31	53.91 ± 9.79	46.57 ± 2.37	44.05 ± 2.01	19.64 ± 0.87
GraphPrompt+	44.14 ± 16.99	5.03 ± 1.13	53.01 ± 0.53	34.50 ± 5.75	23.05 ± 1.92	54.22 ± 8.02	44.83 ± 1.98	43.22 ± 2.79	21.84 ± 2.06
ProNoG	67.63 ± 6.50	21.94 ± 4.95	56.57 ± 0.52	27.79 ± 2.31	28.23 ± 4.19	50.84 ± 14.59	16.58 ± 9.57	43.71 ± 2.65	15.43 ± 0.51
Pretraining: EdgePred									
Full fine-tuning	65.44 ± 4.00	7.37 ± 1.30	52.35 ± 0.75	33.69 ± 2.13	29.02 ± 3.19	48.56 ± 10.42	44.29 ± 6.18	45.32 ± 2.05	27.24 ± 1.80
All-in-One	22.49 ± 1.91	5.09 ± 1.17	50.86 ± 0.68	28.68 ± 2.54	25.38 ± 3.24	51.25 ± 12.17	36.08 ± 2.00	53.91 ± 5.96	24.76 ± 3.33
EdgePrompt	50.07 ± 10.53	3.95 ± 1.43	52.55 ± 0.93	33.73 ± 5.49	27.05 ± 5.05	60.69 ± 10.24	36.00 ± 1.98	45.07 ± 1.91	23.28 ± 2.63
EdgePrompt+	48.11 ± 14.93	4.01 ± 0.56	53.40 ± 1.14	31.86 ± 3.03	26.86 ± 3.26	48.10 ± 9.29	35.07 ± 1.59	45.22 ± 1.82	25.63 ± 0.81
GPF	37.56 ± 3.75	5.28 ± 1.98	50.52 ± 1.21	35.44 ± 4.50	26.97 ± 5.07	43.16 ± 14.77	36.77 ± 3.24	44.92 ± 2.22	19.51 ± 5.84
GPF+	48.47 ± 3.53	3.61 ± 0.73	52.28 ± 0.52	32.38 ± 1.99	30.37 ± 2.21	45.14 ± 16.25	36.74 ± 1.65	45.45 ± 2.17	23.45 ± 2.98
GPPT	13.46 ± 9.04	3.21 ± 4.28	50.05 ± 0.84	33.27 ± 2.95	26.81 ± 3.06	44.99 ± 11.44	51.24 ± 26.55	51.09 ± 1.67	18.33 ± 0.66
GraphPrompt	16.21 ± 2.27	2.42 ± 0.70	50.86 ± 0.48	33.09 ± 2.40	22.78 ± 3.42	60.10 ± 13.21	44.88 ± 2.01	42.66 ± 3.06	22.41 ± 2.16
GraphPrompt+	25.92 ± 7.09	2.60 ± 0.73	51.54 ± 0.49	31.76 ± 3.72	23.99 ± 5.01	50.40 ± 6.48	44.96 ± 1.99	42.23 ± 2.11	23.55 ± 2.48
ProNoG	15.35 ± 2.77	2.62 ± 0.46	54.81 ± 0.54	32.02 ± 3.34	24.81 ± 1.91	59.69 ± 7.11	25.85 ± 11.80	42.60 ± 2.71	20.59 ± 1.70
Pretraining: GraphCL									
Full fine-tuning	61.30 ± 6.00	17.52 ± 1.40	52.90 ± 0.61	34.63 ± 2.56	28.93 ± 3.53	51.61 ± 6.09	39.88 ± 4.84	46.74 ± 1.51	26.51 ± 2.06
All-in-One	50.02 ± 2.39	19.00 ± 0.64	51.97 ± 0.93	29.74 ± 2.07	27.20 ± 2.66	46.75 ± 10.58	36.18 ± 2.03	52.42 ± 3.35	22.39 ± 7.37
EdgePrompt	59.67 ± 3.13	17.05 ± 1.24	52.97 ± 1.10	32.92 ± 5.26	28.97 ± 6.29	55.83 ± 10.57	41.99 ± 2.30	47.65 ± 1.36	23.24 ± 2.97
EdgePrompt+	57.82 ± 2.83	18.19 ± 0.61	53.83 ± 1.18	33.66 ± 4.51	27.24 ± 6.12	50.89 ± 11.12	40.58 ± 2.80	47.82 ± 1.28	26.48 ± 1.39
GPF	57.34 ± 1.43	15.84 ± 1.86	52.50 ± 0.72	31.38 ± 6.56	28.57 ± 5.38	42.85 ± 9.13	42.43 ± 4.18	49.03 ± 1.77	20.33 ± 8.16
GPF+	56.19 ± 3.49	18.27 ± 1.39	52.91 ± 0.63	29.30 ± 3.77	28.93 ± 4.05	45.02 ± 6.30	41.83 ± 2.97	48.99 ± 1.35	24.06 ± 2.79
GPPT	14.03 ± 5.34	11.17 ± 5.25	50.53 ± 0.65	32.29 ± 2.86	20.01 ± 1.90	50.99 ± 10.94	156.03 ± 224.12	49.57 ± 0.96	22.09 ± 2.18
GraphPrompt	51.11 ± 3.64	9.03 ± 1.51	53.21 ± 0.72	30.70 ± 4.33	22.29 ± 3.02	47.37 ± 15.04	44.86 ± 2.04	44.34 ± 2.54	27.42 ± 2.72
GraphPrompt+	61.25 ± 7.89	7.76 ± 1.63	53.31 ± 0.63	34.51 ± 4.83	22.18 ± 3.58	56.29 ± 10.48	44.97 ± 1.96	42.84 ± 2.96	21.71 ± 0.78
ProNoG	56.27 ± 2.78	18.07 ± 2.34	57.03 ± 0.70	30.91 ± 3.78	25.99 ± 3.98	49.34 ± 10.99	51.79 ± 37.24	46.77 ± 1.18	26.85 ± 2.38
Pretraining: InfoGraph									
Full fine-tuning	71.65 ± 1.78	14.77 ± 2.36	54.11 ± 0.26	33.95 ± 2.52	30.04 ± 3.00	53.06 ± 7.30	36.17 ± 3.05	46.36 ± 1.76	25.50 ± 1.84
All-in-One	59.24 ± 0.96	2.69 ± 1.56	51.58 ± 1.00	29.91 ± 3.49	27.38 ± 2.70	53.70 ± 6.86	110.13 ± 1.01	53.90 ± 3.17	17.47 ± 5.38
EdgePrompt	60.48 ± 4.41	2.80 ± 0.50	53.98 ± 1.04	32.36 ± 4.58	24.31 ± 4.56	54.15 ± 8.82	111.52 ± 0.24	46.28 ± 1.43	22.23 ± 4.34
EdgePrompt+	62.64 ± 4.87	5.02 ± 1.11	53.78 ± 0.91	32.48 ± 3.89	23.93 ± 3.76	52.93 ± 9.53	111.29 ± 0.04	47.28 ± 1.38	23.77 ± 2.18
GPF	71.86 ± 3.63	5.92 ± 1.62	54.08 ± 0.78	32					

Table 20: 5-shot configuration results for H2GCN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	76.95±1.74	28.60±1.45	54.29±0.30	25.03±3.16	29.72±4.74	57.04±12.34	11.87±0.34	50.05±1.75	26.79±1.28
All-in-One	74.69±4.22	27.26±3.14	53.81±0.31	35.76±3.93	27.90±3.52	61.24±9.64	105.06±0.65	51.40±1.11	22.09±2.93
EdgePrompt	70.42±2.20	31.25±3.26	54.04±0.45	25.97±2.18	31.11±4.13	61.08±11.77	109.00±0.26	50.15±2.33	21.01±6.28
EdgePrompt+	70.21±4.30	26.47±2.80	53.15±0.66	28.21±5.17	30.43±3.80	61.81±16.74	108.06±0.35	49.19±1.20	24.62±1.38
GPF	75.03±2.21	29.19±2.77	54.05±0.45	25.16±0.78	30.22±2.39	63.97±9.70	110.75±0.21	51.68±2.39	21.15±1.67
GPF+	75.02±3.15	21.08±7.62	53.37±0.52	26.27±1.48	31.40±2.90	65.32±10.05	110.76±0.23	51.74±1.21	20.86±2.22
GPPT	66.56±4.83	21.11±3.46	50.02±0.54	28.14±8.33	31.23±3.25	53.51±12.08	108.56±1.22	50.31±0.70	18.05±3.91
GraphPrompt	67.31±3.97	12.05±1.32	53.23±0.45	34.77±3.24	25.05±4.76	64.30±11.87	99.27±0.82	42.60±2.64	21.80±2.24
GraphPrompt+	54.23±3.17	10.61±1.62	54.21±0.65	35.16±3.93	21.82±3.27	53.38±10.17	58.94±18.31	42.93±2.83	21.22±1.33
ProNoG	71.54±2.25	26.98±5.03	53.97±0.83	26.80±2.67	28.52±4.12	70.00±5.18	16.30±9.25	43.10±2.70	22.05±3.33
Pretraining: ContextPred									
Full fine-tuning	76.93±3.61	26.60±1.98	54.28±0.27	24.80±3.29	30.56±3.48	58.07±11.74	6.43±0.91	48.90±0.85	26.71±1.96
All-in-One	73.98±3.61	22.26±2.03	53.45±0.61	35.97±5.00	27.26±3.07	56.54±9.81	98.75±0.82	47.06±1.08	26.11±0.61
EdgePrompt	67.72±5.34	28.81±2.52	53.49±0.62	25.61±0.30	29.81±2.77	51.77±7.50	108.77±0.35	51.02±0.71	24.48±2.13
EdgePrompt+	68.54±3.95	24.62±0.81	53.57±0.40	25.28±0.49	31.99±2.80	55.26±11.52	108.79±0.24	50.96±0.65	24.64±2.30
GPF	76.79±3.86	24.07±12.47	53.53±0.62	25.32±0.72	30.32±2.23	54.15±9.56	108.79±0.25	50.47±1.05	19.13±3.87
GPF+	77.06±3.04	23.31±6.14	53.93±0.44	25.61±1.78	30.57±1.83	55.55±10.41	109.13±0.67	50.72±0.54	16.28±4.95
GPPT	76.04±2.42	28.39±4.30	50.50±1.09	33.83±2.78	31.47±1.85	50.23±8.46	104.39±1.99	50.07±0.31	22.23±0.83
GraphPrompt	76.03±3.39	13.80±2.05	53.76±0.50	34.07±5.20	26.79±6.10	57.40±8.20	97.03±13.10	42.98±2.58	25.51±0.99
GraphPrompt+	72.54±5.04	9.72±2.51	53.67±0.53	35.11±3.94	22.63±2.33	49.66±4.51	69.12±14.13	42.45±2.98	20.22±2.24
ProNoG	78.92±2.23	25.87±4.28	53.35±0.76	26.26±1.45	28.42±3.31	57.37±5.76	18.82±10.39	43.47±2.58	24.14±0.44
Pretraining: DGI									
Full fine-tuning	76.60±2.59	23.31±1.26	53.95±0.07	27.21±2.81	30.88±3.63	57.55±10.73	9.08±0.82	48.86±1.42	25.71±1.50
All-in-One	74.30±2.54	18.66±1.88	54.34±1.00	32.97±4.26	28.71±3.43	55.15±14.41	103.84±0.28	48.74±2.07	24.50±1.56
EdgePrompt	69.31±4.89	22.77±2.28	53.02±0.61	26.02±2.19	30.68±2.23	59.79±9.21	88.47±2.13	50.26±2.01	23.45±0.91
EdgePrompt+	70.42±3.71	22.84±1.35	53.58±0.56	26.29±2.60	30.68±2.44	58.86±9.70	84.45±1.98	51.35±1.85	21.31±3.45
GPF	77.74±3.28	19.38±7.94	53.38±0.69	26.14±5.63	30.56±1.50	54.78±13.57	111.54±0.21	51.53±1.77	17.47±5.33
GPF+	77.37±2.95	17.59±8.06	53.39±0.47	26.19±1.78	30.37±2.98	54.91±12.02	111.87±0.75	51.00±1.14	16.13±6.05
GPPT	19.86±6.40	16.07±5.11	50.08±0.19	30.29±1.06	31.27±2.16	49.99±7.17	106.25±1.30	49.78±0.48	13.85±4.18
GraphPrompt	70.68±4.03	9.35±1.79	52.48±0.63	31.76±4.07	25.99±4.58	48.62±9.70	98.98±3.47	43.06±2.72	24.12±1.08
GraphPrompt+	62.01±6.59	6.19±1.11	52.91±0.34	31.95±1.70	22.08±2.45	51.64±10.04	48.79±6.48	42.24±3.01	21.37±2.03
ProNoG	72.75±6.90	19.85±3.14	53.22±0.86	27.61±2.59	28.88±3.78	62.86±6.78	42.28±31.30	42.87±2.53	24.04±1.32
Pretraining: EdgePred									
Full fine-tuning	74.34±1.94	19.69±1.24	54.24±0.52	25.81±2.70	30.27±4.09	63.58±9.74	6.43±0.91	48.07±1.01	25.65±1.02
All-in-One	50.76±5.97	2.44±0.88	51.46±0.52	37.64±4.45	26.48±4.83	57.79±11.07	98.75±0.82	49.54±1.25	24.56±1.98
EdgePrompt	50.52±13.18	6.20±2.33	52.53±0.64	25.06±0.16	29.11±2.65	59.54±11.58	108.77±0.35	50.21±1.88	23.30±1.56
EdgePrompt+	65.80±5.49	7.33±2.69	52.58±0.16	25.32±0.18	30.23±5.37	58.31±9.29	108.79±0.24	48.83±0.28	23.51±1.73
GPF	40.44±9.38	4.01±1.54	51.85±0.68	25.06±0.13	28.44±3.08	65.54±8.29	108.79±0.25	48.96±0.68	20.65±5.20
GPF+	56.78±8.79	6.25±0.74	52.62±0.85	25.73±0.72	27.91±3.65	61.75±9.93	110.49±3.71	48.92±0.83	23.79±0.49
GPPT	14.48±5.09	2.87±1.70	49.82±0.68	32.65±2.61	22.99±2.74	56.33±8.84	104.39±1.99	50.04±0.17	19.00±1.21
GraphPrompt	19.38±3.43	4.37±0.94	50.36±0.49	34.38±3.48	24.33±2.95	55.37±13.01	87.42±0.46	46.19±2.86	23.64±1.43
GraphPrompt+	43.80±7.04	2.72±0.92	52.10±0.17	35.28±3.84	24.33±3.79	52.39±10.09	69.12±14.13	42.81±2.83	19.22±2.76
ProNoG	17.81±2.60	7.42±0.55	52.44±0.63	27.16±2.49	22.88±2.94	59.62±6.56	18.82±10.39	44.19±2.23	21.88±1.23
Pretraining: GraphCL									
Full fine-tuning	77.02±3.00	23.28±0.62	54.12±0.27	26.53±2.73	28.62±4.32	60.12±12.54	6.97±0.68	49.17±1.42	24.54±1.88
All-in-One	67.14±2.05	22.55±2.42	53.40±0.37	32.39±5.84	27.91±1.61	54.73±11.40	89.82±1.25	49.53±2.70	23.03±0.89
EdgePrompt	67.51±5.31	26.35±1.29	52.92±0.69	28.15±3.73	30.67±2.49	50.70±6.55	106.09±0.47	50.07±0.98	21.51±2.11
EdgePrompt+	70.21±4.96	23.13±1.21	53.25±0.47	34.00±2.81	31.18±2.25	51.01±9.41	106.10±0.34	49.48±0.96	23.74±0.57
GPF	76.89±1.93	24.59±5.13	53.21±0.39	25.98±2.87	30.36±2.45	52.90±10.67	106.14±0.29	49.26±1.68	20.30±4.40
GPF+	76.70±1.71	19.80±6.62	53.63±0.27	26.63±2.40	30.67±2.01	56.87±11.51	106.73±1.15	49.18±1.24	19.77±4.29
GPPT	66.93±1.84	12.12±2.86	50.24±0.90	30.10±3.92	30.12±1.86	53.57±11.03	99.91±3.12	49.43±0.74	18.52±2.44
GraphPrompt	66.58±3.57	10.70±1.53	53.72±0.29	27.81±4.87	26.29±4.81	52.15±9.84	80.25±17.73	44.54±2.51	23.38±1.59
GraphPrompt+	71.09±2.71	11.22±1.76	53.90±0.32	33.28±3.73	26.11±4.36	50.38±4.84	55.93±22.12	42.45±2.91	20.48±3.20
ProNoG	63.55±2.31	18.30±1.73	53.39±1.24	26.58±3.38	29.22±2.55	58.88±15.89	48.49±49.84	46.24±1.69	21.94±1.22
Pretraining: InfoGraph									
Full fine-tuning	76.71±2.53	26.42±1.27	54.32±0.51	25.33±2.99	29.80±4.54	63.28±9.70	7.39±1.17	50.88±1.32	26.40±0.95
All-in-One	62.42±1.40	19.45±1.02	53.21±0.54	35.90±4.66	29.15±4.47	58.73±11.21	111.67±0.14	52.29±2.72	21.94±1.14
EdgePrompt	67.53±4.96	28.10±2.17	52.90±0.55	25.01±0.28	31.12±2.93	51.26±5.74	115.17±0.17	52.18±0.92	20.72±1.89
EdgePrompt+	69.73±2.62	25.73±0.75	53.38±0.20	27.42±2.96	30.96±4.46	61.01±7.53	115.13±0.17	52.18±1.33	24.42±1.45
GPF	74.30±2.74	26.51±1.72	52.97±0.59	25.95±1.66	31.51±2.59	51.20±14.11	115.51±0.18	50.76±1.68	19.16±6.14
GPF+	76.29±2.48	20.26±7.50	53.54±0.44	27.03±1.88	32.29±2.74	61.19±10.51	115.55±0.13	51.90±0.93	22.29±3.74
GPPT	65.30±1.47	13.92±2.28	50.20±1.41	27.47±3.06	30.75±3.24	50.61±9.93	113.01±0.73	50.04±0.39	16.55±3.39
GraphPrompt	62.45±2.89	8.99±1.38	53.33±0.75	31.06±4.79	25.96±4.87	57.41±9.91	111.03±0.11	44.91±2.45	24.84±1.60
GraphPrompt+	73.92±4.48	6.84±1.28	53.87±0.62	34.31±4.36	23.77±2.59	51.77±3.82	88.02±15.79	42.75±2.96	21.86±1.96
ProNoG	68.41±0.66	23.13±2.41	54.91±0.73	27.35±0.24	28.06±1.73	52.93±12.51	8.40±3.46	44.52±1.94	22.28±1.06

Table 21: 5-shot configuration results for FAGCN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	76.57 $\pm$ 1.07	27.53 $\pm$ 1.09	54.17 $\pm$ 0.25	29.28 $\pm$ 1.77	32.49 $\pm$ 2.97	54.17 $\pm$ 8.06	6.38 $\pm$ 0.63	48.50 $\pm$ 0.86	23.80 $\pm$ 1.65
All-in-One	76.84 $\pm$ 2.32	29.69 $\pm$ 0.97	53.96 $\pm$ 0.40	31.95 $\pm$ 2.81	28.35 $\pm$ 2.32	52.20 $\pm$ 11.40	78.06 $\pm$ 0.55	52.13 $\pm$ 3.57	22.29 $\pm$ 2.47
EdgePrompt	68.55 $\pm$ 6.14	24.37 $\pm$ 3.15	53.83 $\pm$ 0.53	32.32 $\pm$ 4.39	24.89 $\pm$ 5.02	53.26 $\pm$ 9.50	91.73 $\pm$ 0.60	49.31 $\pm$ 1.50	23.98 $\pm$ 1.23
EdgePrompt+	69.09 $\pm$ 3.76	24.14 $\pm$ 3.95	54.13 $\pm$ 0.41	33.68 $\pm$ 5.39	25.95 $\pm$ 4.61	49.04 $\pm$ 9.61	90.65 $\pm$ 0.71	48.96 $\pm$ 0.71	25.06 $\pm$ 1.03
GPF	76.56 $\pm$ 1.75	28.82 $\pm$ 1.63	54.07 $\pm$ 0.50	31.74 $\pm$ 5.16	24.97 $\pm$ 1.88	50.57 $\pm$ 7.60	101.41 $\pm$ 1.86	49.93 $\pm$ 0.77	20.61 $\pm$ 4.74
GPF+	77.63 $\pm$ 2.50	20.60 $\pm$ 9.03	53.89 $\pm$ 0.38	29.42 $\pm$ 2.91	25.68 $\pm$ 2.08	51.34 $\pm$ 7.27	101.28 $\pm$ 1.60	49.88 $\pm$ 0.56	22.04 $\pm$ 1.67
GPPT	43.81 $\pm$ 10.53	19.63 $\pm$ 6.20	51.04 $\pm$ 1.47	34.26 $\pm$ 1.91	22.85 $\pm$ 4.18	51.36 $\pm$ 11.47	86.29 $\pm$ 7.90	48.98 $\pm$ 0.81	16.77 $\pm$ 2.96
GraphPrompt	64.18 $\pm$ 4.01	12.05 $\pm$ 2.24	53.71 $\pm$ 0.29	35.16 $\pm$ 3.93	23.24 $\pm$ 1.95	53.10 $\pm$ 7.35	68.05 $\pm$ 33.95	42.70 $\pm$ 2.86	24.27 $\pm$ 1.54
GraphPrompt+	50.86 $\pm$ 3.52	12.87 $\pm$ 1.71	54.19 $\pm$ 0.50	35.16 $\pm$ 3.92	22.01 $\pm$ 2.46	44.58 $\pm$ 4.10	29.22 $\pm$ 5.33	42.83 $\pm$ 2.80	21.41 $\pm$ 4.02
ProNoG	73.18 $\pm$ 3.37	25.75 $\pm$ 4.22	54.05 $\pm$ 0.47	26.84 $\pm$ 2.45	26.55 $\pm$ 2.31	49.29 $\pm$ 5.14	20.23 $\pm$ 2.90	42.93 $\pm$ 2.83	24.26 $\pm$ 1.18
Pretraining: ContextPred									
Full fine-tuning	79.93 $\pm$ 3.11	23.77 $\pm$ 1.46	54.55 $\pm$ 0.54	28.79 $\pm$ 2.12	31.05 $\pm$ 3.59	53.13 $\pm$ 7.01	6.24 $\pm$ 0.38	48.08 $\pm$ 0.93	25.37 $\pm$ 0.93
All-in-One	74.81 $\pm$ 4.43	18.47 $\pm$ 1.98	53.28 $\pm$ 0.73	34.21 $\pm$ 2.45	28.80 $\pm$ 1.38	55.33 $\pm$ 10.88	100.97 $\pm$ 1.51	42.72 $\pm$ 2.96	22.50 $\pm$ 1.88
EdgePrompt	71.53 $\pm$ 5.19	18.42 $\pm$ 3.79	53.71 $\pm$ 0.60	34.19 $\pm$ 5.67	28.23 $\pm$ 3.63	52.65 $\pm$ 4.44	107.05 $\pm$ 0.53	50.06 $\pm$ 1.39	21.97 $\pm$ 2.51
EdgePrompt+	69.81 $\pm$ 4.66	19.51 $\pm$ 1.32	54.15 $\pm$ 0.30	35.09 $\pm$ 4.99	28.23 $\pm$ 6.21	53.65 $\pm$ 7.04	107.40 $\pm$ 0.68	50.03 $\pm$ 1.34	24.41 $\pm$ 1.62
GPF	80.88 $\pm$ 4.46	19.54 $\pm$ 1.13	53.64 $\pm$ 0.44	32.96 $\pm$ 3.11	28.57 $\pm$ 3.46	48.04 $\pm$ 6.75	108.22 $\pm$ 0.99	49.64 $\pm$ 1.34	21.03 $\pm$ 3.01
GPF+	81.27 $\pm$ 3.42	15.34 $\pm$ 5.89	53.70 $\pm$ 0.35	33.15 $\pm$ 3.44	30.08 $\pm$ 3.31	48.59 $\pm$ 8.18	108.64 $\pm$ 1.87	49.23 $\pm$ 0.59	20.73 $\pm$ 2.55
GPPT	74.46 $\pm$ 2.92	14.06 $\pm$ 2.03	49.56 $\pm$ 1.06	32.21 $\pm$ 3.70	28.37 $\pm$ 3.19	50.92 $\pm$ 11.64	98.75 $\pm$ 4.23	50.38 $\pm$ 1.20	18.77 $\pm$ 2.34
GraphPrompt	75.76 $\pm$ 5.33	10.47 $\pm$ 2.00	53.59 $\pm$ 0.46	31.74 $\pm$ 2.92	25.15 $\pm$ 5.97	55.42 $\pm$ 2.97	73.11 $\pm$ 1.46	43.52 $\pm$ 2.82	24.69 $\pm$ 1.42
GraphPrompt+	75.67 $\pm$ 3.84	10.15 $\pm$ 2.14	54.17 $\pm$ 0.48	34.51 $\pm$ 3.32	22.42 $\pm$ 2.53	53.14 $\pm$ 7.50	63.28 $\pm$ 14.33	42.71 $\pm$ 3.14	21.74 $\pm$ 3.89
ProNoG	80.56 $\pm$ 3.53	14.69 $\pm$ 3.24	53.43 $\pm$ 1.07	28.36 $\pm$ 3.21	27.52 $\pm$ 2.45	53.70 $\pm$ 7.34	15.27 $\pm$ 5.49	43.66 $\pm$ 2.80	24.87 $\pm$ 1.44
Pretraining: DGI									
Full fine-tuning	76.40 $\pm$ 4.64	27.18 $\pm$ 1.88	53.75 $\pm$ 0.55	30.80 $\pm$ 4.58	32.67 $\pm$ 3.82	57.43 $\pm$ 12.84	7.68 $\pm$ 1.86	47.71 $\pm$ 0.90	23.36 $\pm$ 1.42
All-in-One	74.84 $\pm$ 5.49	25.28 $\pm$ 0.70	53.39 $\pm$ 0.51	31.91 $\pm$ 0.85	32.30 $\pm$ 3.66	54.60 $\pm$ 8.16	105.82 $\pm$ 1.53	45.37 $\pm$ 1.75	21.34 $\pm$ 3.44
EdgePrompt	63.47 $\pm$ 5.05	29.90 $\pm$ 1.82	53.50 $\pm$ 0.49	28.89 $\pm$ 7.49	34.07 $\pm$ 2.93	52.58 $\pm$ 9.43	108.19 $\pm$ 0.14	50.79 $\pm$ 0.41	21.04 $\pm$ 1.77
EdgePrompt+	67.43 $\pm$ 6.78	29.51 $\pm$ 1.33	53.69 $\pm$ 0.46	29.62 $\pm$ 6.57	34.22 $\pm$ 1.42	54.89 $\pm$ 9.76	108.04 $\pm$ 0.62	50.99 $\pm$ 0.56	26.23 $\pm$ 0.64
GPF	74.44 $\pm$ 5.93	28.92 $\pm$ 1.11	53.12 $\pm$ 0.53	27.42 $\pm$ 3.18	31.59 $\pm$ 2.34	59.74 $\pm$ 6.53	110.40 $\pm$ 0.21	48.67 $\pm$ 1.10	20.57 $\pm$ 3.10
GPF+	76.10 $\pm$ 4.66	22.97 $\pm$ 10.48	53.32 $\pm$ 0.48	27.16 $\pm$ 6.98	31.33 $\pm$ 0.90	53.20 $\pm$ 11.71	110.65 $\pm$ 0.65	48.94 $\pm$ 1.17	20.76 $\pm$ 6.73
GPPT	11.25 $\pm$ 0.23	22.77 $\pm$ 4.69	49.97 $\pm$ 1.60	28.58 $\pm$ 6.06	30.37 $\pm$ 2.31	53.38 $\pm$ 14.55	106.89 $\pm$ 1.71	49.97 $\pm$ 1.46	19.67 $\pm$ 2.16
GraphPrompt	66.59 $\pm$ 9.67	8.03 $\pm$ 1.80	52.98 $\pm$ 0.58	29.54 $\pm$ 4.02	23.05 $\pm$ 4.63	50.67 $\pm$ 5.92	95.45 $\pm$ 0.56	43.42 $\pm$ 2.63	26.79 $\pm$ 0.57
GraphPrompt+	68.25 $\pm$ 7.13	9.44 $\pm$ 2.03	53.63 $\pm$ 0.77	29.50 $\pm$ 2.12	24.31 $\pm$ 3.65	45.20 $\pm$ 6.98	68.06 $\pm$ 17.05	43.00 $\pm$ 2.87	19.39 $\pm$ 2.78
ProNoG	68.56 $\pm$ 9.04	26.96 $\pm$ 3.12	51.96 $\pm$ 0.59	28.46 $\pm$ 2.32	28.42 $\pm$ 3.04	65.18 $\pm$ 11.23	21.26 $\pm$ 16.10	46.09 $\pm$ 1.41	27.28 $\pm$ 0.44
Pretraining: EdgePred									
Full fine-tuning	75.45 $\pm$ 1.19	21.32 $\pm$ 0.96	54.13 $\pm$ 0.31	31.25 $\pm$ 3.14	30.53 $\pm$ 2.47	48.59 $\pm$ 8.32	6.24 $\pm$ 0.38	47.05 $\pm$ 1.36	23.96 $\pm$ 1.59
All-in-One	33.44 $\pm$ 2.90	5.33 $\pm$ 0.69	51.71 $\pm$ 0.43	35.27 $\pm$ 2.17	30.91 $\pm$ 3.50	53.13 $\pm$ 6.94	100.97 $\pm$ 1.51	46.92 $\pm$ 1.24	20.40 $\pm$ 1.79
EdgePrompt	19.05 $\pm$ 4.67	4.35 $\pm$ 1.15	51.77 $\pm$ 0.64	32.26 $\pm$ 4.28	25.28 $\pm$ 7.45	49.20 $\pm$ 7.50	107.05 $\pm$ 0.53	47.95 $\pm$ 0.97	21.27 $\pm$ 1.72
EdgePrompt+	50.13 $\pm$ 8.34	5.36 $\pm$ 1.15	52.55 $\pm$ 0.39	29.13 $\pm$ 8.46	29.40 $\pm$ 5.61	48.32 $\pm$ 8.55	107.40 $\pm$ 0.68	48.00 $\pm$ 0.67	24.24 $\pm$ 2.68
GPF	36.76 $\pm$ 6.51	4.74 $\pm$ 1.14	51.63 $\pm$ 0.59	29.76 $\pm$ 4.86	30.31 $\pm$ 2.98	49.42 $\pm$ 4.78	108.22 $\pm$ 0.99	48.78 $\pm$ 0.62	22.65 $\pm$ 1.52
GPF+	67.73 $\pm$ 7.31	10.89 $\pm$ 2.25	53.62 $\pm$ 0.28	27.54 $\pm$ 3.16	26.94 $\pm$ 5.15	48.23 $\pm$ 4.38	107.94 $\pm$ 0.49	48.53 $\pm$ 0.73	23.16 $\pm$ 1.25
GPPT	35.35 $\pm$ 3.15	2.95 $\pm$ 0.65	50.41 $\pm$ 0.60	30.89 $\pm$ 2.04	27.69 $\pm$ 5.88	52.50 $\pm$ 10.27	98.75 $\pm$ 4.23	50.33 $\pm$ 0.14	18.65 $\pm$ 2.04
GraphPrompt	29.45 $\pm$ 3.09	3.73 $\pm$ 0.19	49.90 $\pm$ 0.88	53.13 $\pm$ 3.55	21.17 $\pm$ 4.40	52.35 $\pm$ 5.24	73.11 $\pm$ 1.46	44.83 $\pm$ 1.66	26.49 $\pm$ 0.91
GraphPrompt+	55.11 $\pm$ 3.57	3.87 $\pm$ 0.29	52.15 $\pm$ 0.68	33.69 $\pm$ 3.97	24.29 $\pm$ 4.71	48.68 $\pm$ 4.07	63.28 $\pm$ 14.33	42.84 $\pm$ 2.93	18.82 $\pm$ 2.52
ProNoG	32.42 $\pm$ 2.28	4.43 $\pm$ 0.52	51.15 $\pm$ 0.63	26.02 $\pm$ 4.40	28.77 $\pm$ 3.31	52.86 $\pm$ 6.46	15.27 $\pm$ 5.49	43.62 $\pm$ 2.57	23.98 $\pm$ 1.51
Pretraining: GraphCL									
Full fine-tuning	77.25 $\pm$ 2.30	27.87 $\pm$ 1.15	53.87 $\pm$ 0.42	29.28 $\pm$ 1.74	31.18 $\pm$ 2.91	53.60 $\pm$ 10.29	6.16 $\pm$ 0.53	48.64 $\pm$ 1.48	20.66 $\pm$ 1.13
All-in-One	77.65 $\pm$ 1.73	30.31 $\pm$ 0.26	53.55 $\pm$ 0.26	29.08 $\pm$ 4.11	28.32 $\pm$ 2.25	54.44 $\pm$ 8.45	94.09 $\pm$ 2.07	55.07 $\pm$ 2.91	20.89 $\pm$ 2.23
EdgePrompt	64.00 $\pm$ 4.96	28.59 $\pm$ 2.48	53.64 $\pm$ 0.38	36.07 $\pm$ 3.43	26.87 $\pm$ 2.14	48.68 $\pm$ 6.27	102.40 $\pm$ 0.28	49.65 $\pm$ 1.42	21.33 $\pm$ 1.84
EdgePrompt+	64.65 $\pm$ 5.19	26.89 $\pm$ 3.38	53.78 $\pm$ 0.41	34.63 $\pm$ 2.84	27.49 $\pm$ 2.13	52.21 $\pm$ 6.24	102.23 $\pm$ 0.39	49.41 $\pm$ 1.43	21.27 $\pm$ 1.69
GPF	77.65 $\pm$ 2.36	30.09 $\pm$ 2.36	53.65 $\pm$ 0.37	24.89 $\pm$ 4.75	27.74 $\pm$ 1.48	51.49 $\pm$ 5.08	103.01 $\pm$ 0.81	49.92 $\pm$ 1.66	19.44 $\pm$ 1.55
GPF+	78.47 $\pm$ 1.45	19.74 $\pm$ 8.23	53.42 $\pm$ 0.75	30.79 $\pm$ 2.85	28.94 $\pm$ 1.84	52.08 $\pm$ 7.56	103.99 $\pm$ 2.55	49.94 $\pm$ 1.13	18.25 $\pm$ 3.66
GPPT	78.46 $\pm$ 1.94	15.22 $\pm$ 0.90	51.63 $\pm$ 0.76	33.09 $\pm$ 1.71	27.95 $\pm$ 4.86	50.14 $\pm$ 10.08	91.04 $\pm$ 6.00	49.11 $\pm$ 0.87	13.02 $\pm$ 0.74
GraphPrompt	76.46 $\pm$ 2.77	14.67 $\pm$ 2.23	53.89 $\pm$ 0.29	35.08 $\pm$ 2.93	24.96 $\pm$ 6.86	55.70 $\pm$ 5.60	55.96 $\pm$ 2.14	45.30 $\pm$ 2.25	21.00 $\pm$ 2.00
GraphPrompt+	76.99 $\pm$ 3.02	13.59 $\pm$ 1.88	53.97 $\pm$ 0.45	34.89 $\pm$ 3.12	23.21 $\pm$ 2.92	49.98 $\pm$ 4.64	44.72 $\pm$ 15.83	43.12 $\pm$ 2.63	17.13 $\pm$ 2.85
ProNoG	76.99 $\pm$ 1.61	27.10 $\pm$ 3.22	53.73 $\pm$ 0.82	28.27 $\pm$ 6.22	26.99 $\pm$ 3.13	50.69 $\pm$ 8.88	15.91 $\pm$ 5.14	45.79 $\pm$ 1.84	23.58 $\pm$ 1.66
Pretraining: InfoGraph									
Full fine-tuning	77.04 $\pm$ 1.41	26.12 $\pm$ 1.78	54.30 $\pm$ 0.35	30.36 $\pm$ 3.79	31.76 $\pm$ 5.33	54.73 $\pm$ 8.20	6.37 $\pm$ 0.73	48.71 $\pm$ 1.11	24.08 $\pm$ 2.35
All-in-One	75.84 $\pm$ 2.48	25.05 $\pm$ 0.82	53.44 $\pm$ 0.47	37.42 $\pm$ 4.48	30.75 $\pm$ 2.70	52.29 $\pm$ 6.78	112.06 $\pm$ 0.25	56.80 $\pm$ 2.85	16.49 $\pm$ 4.67
EdgePrompt	58.86 $\pm$ 3.31	20.63 $\pm$ 2.43	53.61 $\pm$ 0.49	26.73 $\pm$ 2.26	34.17 $\pm$ 3.41	52.68 $\pm$ 7.97	112.23 $\pm$ 0.14	48.68 $\pm$ 0.56	21.46 $\pm$ 1.82
EdgePrompt+	63.23 $\pm$ 4.78	18.86 $\pm$ 1.45	53.92 $\pm$ 0.41	32.48 $\pm$ 4.99	33.45 $\pm$ 3.62	47.92 $\pm$ 6.98	112.11 $\pm$ 0.07	48.71 $\pm$ 0.36	24.64 $\pm$ 2.63
GPF	76.20 $\pm$ 3.34	24.28 $\pm$ 2.10	53.61 $\pm$ 0.45	28.60 $\pm$ 5.29	32.47 $\pm$ 3.25	49.68 $\pm$ 5.26	113.86 $\pm$ 0.24	49.95 $\pm$ 1.97	21.85 $\pm$ 1.81
GPF+	76.64 $\pm$ 3.75	18.01 $\pm$ 6.75	53.80 $\pm$ 0.40	26					

Table 22: 5-shot configuration results for Transformer across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	80.32±1.36	26.98±1.49	54.84±0.50	33.38±5.47	29.33±3.77	56.19±11.19	15.17±1.60	49.88±1.33	26.40±2.51
All-in-One	79.45±1.29	30.08±0.92	53.37±0.30	35.90±3.94	30.60±2.74	56.58±11.18	94.59±5.02	48.01±1.50	23.51±2.22
EdgePrompt	67.44±4.94	25.31±2.59	52.98±0.56	34.14±3.27	30.44±3.37	42.83±7.86	107.44±0.28	49.03±0.40	18.02±3.87
EdgePrompt+	68.39±6.42	23.73±2.37	52.78±0.35	34.62±3.34	30.84±3.79	50.08±6.86	106.96±0.40	49.13±0.81	22.55±2.33
GPF	80.81±2.33	23.54±1.59	52.87±0.51	38.10±3.36	30.66±3.36	46.71±4.21	111.34±0.29	49.08±0.60	17.53±4.34
GPF+	80.27±1.70	22.42±1.52	52.63±0.61	37.62±4.03	32.08±2.20	45.89±5.52	111.29±0.26	49.08±0.54	18.88±2.99
GPPT	73.02±10.93	18.80±6.48	50.54±1.66	29.98±4.56	32.50±3.56	52.65±11.68	108.23±1.59	48.95±1.55	12.01±0.75
GraphPrompt	72.76±1.47	12.56±1.72	52.89±0.39	35.40±4.13	27.83±6.74	48.51±3.37	103.93±4.21	44.30±3.65	21.84±1.26
GraphPrompt+	64.06±7.51	11.78±1.38	52.49±0.62	55.23±4.04	22.18±2.35	47.74±7.83	45.22±35.40	42.47±2.86	22.78±1.27
ProNoG	78.39±4.68	25.13±4.00	55.60±0.53	29.23±5.98	28.34±2.53	55.28±9.25	14.45±8.52	42.41±3.26	24.62±1.32
Pretraining: ContextPred									
Full fine-tuning	76.70±3.54	22.98±1.64	55.17±0.79	33.45±3.28	30.42±2.94	59.50±10.37	8.79±2.00	48.76±0.76	23.86±1.43
All-in-One	76.09±1.11	21.20±2.31	52.65±0.83	30.19±5.26	31.89±2.10	56.22±12.80	96.94±0.16	42.49±3.37	22.52±1.15
EdgePrompt	69.11±7.34	17.68±1.30	57.38±1.50	38.17±6.37	30.65±4.74	51.44±14.74	107.31±0.19	49.35±1.10	22.15±2.17
EdgePrompt+	69.56±3.97	15.45±2.48	55.21±1.29	38.80±5.58	30.52±3.75	56.88±13.31	107.35±0.13	49.65±0.61	21.31±2.18
GPF	78.85±0.63	19.80±1.72	56.91±0.93	37.88±5.08	29.92±3.43	55.86±12.45	107.31±0.37	49.99±1.16	20.15±2.74
GPF+	78.85±1.18	17.20±2.43	54.36±0.60	37.32±4.45	29.92±3.43	52.18±11.82	107.69±0.93	49.32±1.23	18.28±3.16
GPPT	76.17±1.93	12.01±4.72	50.87±1.74	31.85±2.14	30.18±2.08	53.75±10.93	101.58±3.74	49.52±1.08	15.12±1.21
GraphPrompt	77.56±1.63	9.40±1.64	51.99±0.86	28.48±5.32	25.63±6.32	57.41±11.69	77.86±1.06	42.64±2.90	20.15±1.27
GraphPrompt+	74.01±3.90	7.32±1.84	51.72±0.64	34.36±2.53	23.94±3.79	55.21±8.57	62.37±10.61	42.52±2.92	18.39±2.83
ProNoG	79.42±1.95	15.43±2.76	56.29±0.53	28.80±2.17	28.40±3.03	61.66±11.25	13.12±3.81	42.84±2.52	19.46±1.62
Pretraining: DGI									
Full fine-tuning	79.02±1.72	24.57±1.20	55.53±1.00	32.65±3.28	31.19±4.24	52.63±12.21	6.16±0.27	49.94±0.87	26.24±1.13
All-in-One	76.22±2.52	22.67±0.68	52.82±0.81	30.99±4.69	32.46±2.01	55.54±9.25	47.88±5.38	52.54±1.16	20.60±4.42
EdgePrompt	71.30±4.02	24.68±0.75	55.81±1.29	33.44±6.06	34.71±3.01	51.76±7.21	97.14±0.73	51.14±1.95	17.19±3.65
EdgePrompt+	70.41±2.58	23.22±0.87	54.51±0.88	32.55±7.02	35.39±2.10	58.59±8.43	97.02±0.73	51.57±2.50	21.39±3.42
GPF	78.12±2.32	24.02±0.93	55.78±2.16	34.41±2.47	34.95±2.84	47.75±10.83	100.32±1.07	51.39±1.16	20.19±1.58
GPF+	79.11±0.33	19.77±9.61	54.42±0.65	33.45±3.48	34.40±2.46	48.81±8.16	101.06±2.04	51.24±0.45	18.93±6.12
GPPT	44.42±11.79	14.15±6.94	49.85±0.57	31.25±2.40	29.29±2.94	49.54±6.47	80.94±4.37	49.19±1.42	18.80±2.40
GraphPrompt	72.20±7.04	7.58±0.94	52.55±0.93	30.34±2.69	23.83±4.18	58.25±13.10	44.47±1.01	42.95±2.87	23.86±1.42
GraphPrompt+	68.32±2.99	9.45±0.85	52.95±1.35	29.01±4.05	23.24±5.61	56.27±6.62	25.02±4.18	42.81±2.33	22.42±1.66
ProNoG	74.11±5.64	25.12±2.70	56.61±0.42	24.41±3.22	30.12±1.69	54.54±8.81	21.97±7.51	42.98±3.09	25.52±1.03
Pretraining: EdgePred									
Full fine-tuning	64.42±22.74	22.87±1.19	54.25±0.64	33.06±3.33	29.31±3.03	57.72±12.84	7.27±1.37	48.89±0.89	23.93±2.15
All-in-One	30.46±2.61	7.12±0.97	51.78±0.99	32.91±5.06	29.33±3.21	54.80±9.00	96.94±0.16	48.18±1.55	21.65±2.70
EdgePrompt	38.53±6.09	5.09±2.43	50.95±0.49	38.12±3.98	28.58±1.44	51.96±10.20	109.83±3.84	50.37±0.55	20.85±2.86
EdgePrompt+	53.82±10.12	6.39±1.57	50.99±0.87	38.63±5.32	29.32±1.99	55.84±10.01	107.35±0.13	50.14±0.80	23.98±1.85
GPF	35.39±3.48	6.80±2.49	51.21±0.53	37.71±4.32	28.98±2.22	53.60±9.89	107.31±0.37	50.36±1.17	19.86±3.59
GPF+	54.48±2.35	11.65±2.02	52.23±0.67	37.16±3.62	29.08±3.68	54.96±10.10	107.30±0.12	50.54±0.79	21.39±3.50
GPPT	33.90±1.11	2.28±0.48	49.95±0.32	32.44±1.81	28.58±2.55	49.32±14.50	101.58±3.74	50.00±0.65	17.99±2.28
GraphPrompt	25.28±1.29	3.99±0.80	49.43±0.61	33.13±4.11	25.09±5.28	51.99±11.19	77.86±1.06	44.69±1.91	25.76±0.93
GraphPrompt+	41.25±7.00	3.77±0.96	49.76±0.94	34.65±4.09	25.70±3.95	51.49±9.32	62.37±10.61	42.42±3.02	20.10±2.98
ProNoG	29.26±4.06	5.19±0.63	54.29±0.78	29.27±4.21	28.31±1.85	59.08±10.42	13.12±3.81	43.88±2.41	23.26±1.38
Pretraining: GraphCL									
Full fine-tuning	75.10±3.59	25.07±1.05	54.67±0.72	33.61±4.20	28.50±2.88	58.10±7.87	7.50±2.25	48.50±0.91	25.75±0.60
All-in-One	70.74±2.21	24.10±1.55	53.71±0.40	32.24±4.27	29.44±1.65	55.64±9.73	91.21±0.21	52.43±1.16	26.89±1.30
EdgePrompt	68.03±4.96	26.03±0.47	53.52±0.74	35.73±5.43	30.92±2.56	51.31±7.76	109.62±0.12	48.25±2.05	22.68±1.81
EdgePrompt+	68.38±4.80	24.90±1.39	52.76±1.11	34.09±4.47	30.75±2.55	52.20±7.35	109.56±0.09	48.56±1.23	23.75±1.52
GPF	73.69±3.19	27.73±1.12	51.73±0.83	38.67±4.85	29.46±2.19	55.48±4.92	109.83±0.14	48.06±1.96	21.01±2.81
GPF+	73.61±1.84	22.42±5.75	52.20±0.60	37.32±4.72	29.46±2.19	54.24±9.96	109.82±0.10	48.31±1.19	20.88±1.88
GPPT	62.52±3.17	18.26±2.34	50.37±1.42	33.62±2.87	28.75±3.37	51.05±12.57	106.96±2.55	49.72±0.76	19.20±2.37
GraphPrompt	67.13±1.39	12.78±1.43	53.06±0.48	29.03±2.40	23.12±4.22	53.82±6.30	92.81±0.64	43.95±2.66	28.16±2.70
GraphPrompt+	77.22±2.58	11.86±1.45	53.04±0.60	34.58±4.48	23.98±4.25	50.65±4.48	50.38±18.61	42.67±2.79	20.28±3.22
ProNoG	65.97±1.90	20.70±1.48	57.01±0.60	31.21±4.15	27.83±3.28	52.22±9.46	18.78±24.14	46.45±1.94	25.18±1.66
Pretraining: InfoGraph									
Full fine-tuning	76.99±3.75	25.09±1.34	55.39±0.82	33.20±3.90	29.88±3.57	57.19±9.37	6.38±0.50	50.22±0.76	21.62±2.50
All-in-One	63.63±6.80	22.26±2.23	52.66±0.73	32.21±2.72	30.64±2.61	56.43±10.90	105.34±0.15	57.03±2.42	11.84±1.71
EdgePrompt	62.26±5.73	18.30±2.00	55.84±1.79	36.85±5.34	31.63±1.77	45.48±6.15	114.05±0.18	51.00±1.34	21.66±2.89
EdgePrompt+	69.15±6.24	18.53±1.83	54.48±1.05	37.83±5.02	32.85±1.21	51.83±13.17	113.68±0.25	51.07±1.17	21.76±2.65
GPF	75.90±2.80	19.33±1.15	55.47±1.00	37.95±3.33	31.00±3.17	53.86±6.31	115.28±0.12	50.46±1.55	13.99±4.67
GPF+	76.62±3.88	19.17±1.27	52.68±0.91	36.91±2.83	32.82±1.01	52.37±6.54	115.23±0.08	49.59±1.00	16.85±9.47
GPPT	62.64±0.94	10.61±2.03	49.89±0.38	32.62±1.12	33.72±2.27	50.80±9.56	111.89±0.94	50.02±0.23	13.40±1.97
GraphPrompt	57.86±2.41	7.12±0.95	52.74±0.91	31.30±1.56	25.22±4.90	59.67±10.75	111.78±2.61	43.75±2.33	18.84±0.74
GraphPrompt+	68.66±5.4	8.67±2.42	53.50±0.99	35.35±4.36	25.22±4.90	50.82±9.93	81.15±21.78	43.16±1.38	22.24±3.28
ProNoG	63.12±1.25	19.94±1.67	56.68±0.54	29.13±0.29	30.57±0.37	59.87±6.97	8.83±3.31	43.81±2.48	14.77±0.66

Table 23: 5-shot configuration results for NodeFormer across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	GR / MAE ↓	GC / AUC ↑	GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	69.96±5.09	5.62±2.50	59.91±3.28	24.92±0.19	18.91±2.48	49.89±15.96	50.29±32.58	50.21±0.32	13.81±2.50
All-in-One	68.58±2.85	15.36±2.28	51.80±3.86	31.15±4.74	28.83±5.16	74.63±14.46	103.55±0.62	44.77±2.44	11.29±0.60
EdgePrompt	65.36±6.62	4.96±2.55	55.29±3.99	24.94±0.04	23.40±2.92	44.04±10.54	113.00±0.80	50.71±0.24	16.70±2.93
EdgePrompt+	64.60±5.09	4.65±2.13	57.16±3.77	24.96±0.00	23.06±1.91	59.60±17.39	113.03±0.87	50.66±0.31	15.78±4.58
GPF	68.06±3.49	4.25±2.83	52.43±3.15	24.15±1.51	20.91±2.15	53.44±17.96	113.07±0.44	50.74±0.30	10.21±0.67
GPF+	68.38±2.94	3.17±2.24	54.44±1.83	25.23±1.12	20.76±2.83	51.53±18.98	113.00±0.70	50.81±0.30	12.69±4.84
GPPT	70.32±5.29	4.07±1.17	50.85±4.52	35.03±2.05	28.00±4.54	60.08±21.39	108.25±0.61	50.20±0.60	14.13±3.49
GraphPrompt	62.81±6.46	5.53±1.06	52.18±4.51	35.13±3.97	23.66±4.01	53.44±10.44	104.92±0.68	42.79±2.70	25.07±2.40
GraphPrompt+	45.99±12.24	3.29±0.90	53.43±3.57	35.27±3.92	22.25±2.23	53.72±16.38	82.24±4.45	43.23±3.09	23.12±2.64
ProNoG	72.53±5.52	25.31±3.82	53.51±0.99	30.92±5.63	29.02±3.76	58.32±18.03	8.78±3.85	42.93±2.69	24.46±1.36
Pretraining: ContextPred									
Full fine-tuning	70.83±4.17	15.57±8.04	59.55±3.81	24.91±0.13	25.33±4.16	49.66±10.58	51.23±32.45	50.38±0.46	16.33±4.41
All-in-One	78.90±2.98	2.82±2.59	52.32±0.20	30.32±1.11	28.89±2.88	52.49±8.79	109.95±0.31	51.21±2.13	19.19±1.58
EdgePrompt	69.29±8.48	3.96±3.56	54.07±1.56	25.06±0.07	27.42±3.71	57.27±15.86	113.38±1.08	49.29±0.37	14.22±3.89
EdgePrompt+	68.84±8.03	6.72±4.88	54.55±5.50	25.13±0.22	26.83±3.98	50.58±8.78	113.34±1.16	49.34±0.21	14.21±4.08
GPF	77.56±3.66	6.47±3.68	50.78±2.54	24.99±0.13	22.67±2.07	54.03±11.41	113.85±1.04	49.91±0.23	13.73±2.78
GPF+	78.83±2.29	8.08±4.54	52.85±2.68	25.32±1.13	28.24±4.44	53.20±11.82	113.85±1.05	49.97±0.24	13.74±4.29
GPPT	74.95±3.46	3.34±3.59	55.35±2.88	33.20±2.03	29.38±3.84	48.78±10.45	110.43±1.34	49.98±1.31	19.56±1.23
GraphPrompt	76.85±1.65	3.07±0.96	54.38±3.72	35.04±3.54	24.15±3.96	43.45±3.06	109.38±2.70	43.15±2.63	20.59±2.11
GraphPrompt+	75.09±3.53	4.55±1.40	54.96±5.03	35.37±4.17	24.58±3.75	50.06±4.41	99.49±0.45	43.00±2.77	19.44±3.45
ProNoG	75.50±0.58	13.45±0.90	56.41±1.14	27.56±7.81	27.11±1.51	43.39±5.76	15.12±1.72	42.77±2.29	21.59±0.96
Pretraining: DGI									
Full fine-tuning	54.81±14.81	15.44±4.76	52.30±2.43	24.96±0.12	22.67±2.71	52.70±25.45	50.85±32.68	49.77±0.14	9.83±0.57
All-in-One	71.79±7.63	20.54±1.66	50.85±4.36	29.88±1.24	32.24±2.93	58.95±31.37	107.01±0.13	45.05±2.32	9.97±0.42
EdgePrompt	36.92±6.69	5.38±4.62	52.53±2.16	24.97±0.07	23.72±3.32	53.20±10.45	111.54±0.07	50.79±0.38	9.69±0.35
EdgePrompt+	45.97±13.89	5.79±4.68	53.12±2.11	25.01±0.05	23.69±3.23	52.89±29.47	111.55±0.12	50.53±0.92	9.85±0.00
GPF	23.76±12.05	3.58±3.14	51.68±2.73	24.87±0.55	21.33±2.86	43.19±27.00	111.54±0.07	51.32±0.78	9.60±0.47
GPF+	27.73±13.66	9.00±4.71	52.38±2.72	26.73±4.41	25.25±1.92	54.93±26.49	111.54±0.07	51.22±0.65	9.85±0.11
GPPT	68.07±7.73	7.83±4.43	50.38±1.79	34.75±2.29	30.75±2.27	49.16±29.09	107.80±0.07	49.92±1.09	10.54±0.42
GraphPrompt	61.40±10.07	3.62±1.12	49.13±2.46	35.08±3.91	22.02±2.26	49.61±17.32	105.05±0.24	42.73±2.79	14.96±2.83
GraphPrompt+	70.16±6.55	8.71±2.04	51.72±2.76	35.61±4.09	23.16±2.90	53.56±4.39	94.72±2.95	42.79±2.75	15.55±3.24
ProNoG	59.36±3.49	12.64±3.35	56.01±1.35	24.32±4.51	25.84±3.90	56.28±13.46	15.99±2.06	42.92±2.99	14.16±1.07
Pretraining: EdgePred									
Full fine-tuning	39.37±12.26	18.04±2.53	56.63±4.11	24.94±0.93	23.16±5.01	42.81±16.12	55.63±46.90	49.38±0.86	10.49±0.74
All-in-One	17.97±4.74	4.69±2.49	52.50±4.59	26.94±2.93	25.40±4.16	41.52±20.37	109.88±0.24	54.68±2.01	19.99±2.09
EdgePrompt	8.18±2.31	3.22±2.80	50.84±1.36	26.00±0.86	20.68±0.71	51.34±10.44	112.87±0.11	51.85±0.69	9.86±0.58
EdgePrompt+	12.20±3.81	1.56±0.94	51.08±0.80	26.15±1.33	20.69±1.04	45.61±11.27	112.82±0.13	51.99±0.93	9.28±0.51
GPF	14.43±2.21	1.72±1.85	49.98±0.69	24.67±0.48	22.58±3.10	49.18±16.53	113.38±0.09	52.42±1.42	11.90±4.20
GPF+	17.12±6.05	3.32±4.78	50.66±3.05	24.39±1.10	25.19±3.49	47.32±11.69	113.37±0.09	51.92±1.48	9.41±0.48
GPPT	16.69±7.41	3.59±4.42	50.59±1.81	32.89±2.79	27.05±4.52	49.99±11.95	109.84±0.07	50.35±0.74	9.92±0.00
GraphPrompt	20.10±2.87	2.85±0.81	51.07±1.76	33.88±2.93	22.71±0.23	47.92±8.45	108.16±0.13	43.23±2.23	20.29±1.88
GraphPrompt+	19.36±4.74	3.01±1.15	53.47±1.13	30.85±5.46	23.05±2.65	54.79±8.91	99.49±0.45	43.15±2.86	15.55±3.30
ProNoG	18.77±3.05	3.80±0.58	56.98±1.20	28.27±2.78	25.19±2.25	57.91±10.31	15.04±1.74	43.41±2.57	18.81±1.87
Pretraining: GraphCL									
Full fine-tuning	70.22±6.65	9.73±5.77	52.24±2.07	24.97±0.04	28.06±3.46	52.15±17.67	49.08±33.56	49.21±1.39	9.87±0.03
All-in-One	62.77±3.02	20.05±1.52	50.74±2.19	26.05±1.72	29.98±3.96	61.74±22.39	111.31±0.06	52.89±1.86	9.38±0.34
EdgePrompt	44.88±7.81	9.49±0.90	51.95±2.93	25.37±0.92	27.92±3.93	53.41±10.88	114.98±0.03	49.77±1.10	9.57±0.47
EdgePrompt+	50.24±7.04	10.32±1.66	50.65±3.11	24.96±0.00	28.22±4.54	52.98±12.98	114.94±0.06	49.79±1.11	9.52±0.42
GPF	58.84±3.88	11.64±3.24	52.01±1.57	25.09±2.04	23.88±4.95	53.34±16.81	115.11±0.03	49.46±1.00	10.07±0.17
GPF+	59.90±4.42	11.95±3.55	48.19±4.77	25.45±1.04	27.50±4.81	51.03±13.40	115.11±0.03	49.81±1.21	10.07±0.17
GPPT	64.03±2.66	10.11±1.49	51.32±1.38	35.01±2.04	28.99±4.09	49.37±21.86	111.79±0.04	49.71±0.66	9.94±0.05
GraphPrompt	59.55±2.85	7.01±1.33	49.42±2.12	35.20±3.95	24.25±4.02	52.58±12.91	111.29±0.26	47.38±1.93	15.06±3.38
GraphPrompt+	67.64±4.19	3.58±0.94	51.25±2.98	35.28±3.95	22.06±2.43	49.96±7.70	100.15±0.50	44.65±2.66	14.76±3.13
ProNoG	66.09±2.18	19.12±1.71	53.10±0.56	30.31±6.74	27.06±2.60	48.37±10.58	14.77±1.79	48.06±0.75	13.89±1.77
Pretraining: InfoGraph									
Full fine-tuning	61.85±9.16	2.67±1.13	52.36±3.63	24.99±0.14	24.09±6.43	45.07±14.86	52.46±16.50	50.52±0.65	10.03±0.41
All-in-One	64.34±7.35	3.24±2.54	52.29±0.62	32.09±5.38	25.75±3.46	60.29±11.79	110.44±0.13	55.36±2.45	9.67±0.55
EdgePrompt	48.11±10.27	6.54±5.76	50.98±0.62	24.99±0.30	21.79±2.93	47.13±15.34	113.99±0.07	50.20±1.18	9.40±0.46
EdgePrompt+	49.63±8.45	6.01±6.12	50.97±0.62	25.06±0.29	21.94±3.62	50.84±11.13	114.04±0.03	50.13±1.21	9.75±0.38
GPF	60.87±4.20	2.98±3.40	52.40±2.72	25.08±0.23	21.41±2.40	47.09±13.77	114.34±0.02	50.30±0.87	9.74±0.38
GPF+	65.42±5.01	3.02±3.28	53.87±2.31	23.79±1.98	23.13±2.71	49.70±12.50	114.34±0.02	50.89±1.75	10.20±1.18
GPPT	55.51±8.95	3.84±3.86	49.71±0.39	34.36±5.54	25.69±2.16	52.15±13.21	111.00±0.07	49.98±0.87	9.75±0.38
GraphPrompt	62.75±3.93	2.49±0.97	52.19±2.42	35.30±3.96	22.91±2.39	56.27±5.75	110.14±0.22	43.44±2.72	15.01±3.06
GraphPrompt+	59.42±6.99	2.39±0.97	53.52±1.77	35.28±3.88	22.50±1.47	54.29±9.44	100.83±1.11	42.82±2.68	15.42±3.08
ProNoG	48.39±1.19	13.38±2.21	51.35±1.32	29.42±6.46	22.37±3.22	47.50±8.84	14.70±1.65	44.34±2.16	14.83±3.13

Table 24: 100-shot configuration results for GCN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	87.75±0.63	37.84±0.46	55.10±0.29	40.03±3.77	33.79±1.16	45.35±6.30	40.11±48.15	50.34±1.30	25.61±8.79
All-in-One	87.92±0.57	32.11±3.41	51.26±0.47	39.92±0.51	36.78±1.18	52.01±6.25	88.08±2.16	57.51±1.51	22.97±0.81
EdgePrompt	82.32±0.87	44.05±0.73	54.08±0.28	39.34±0.94	39.68±1.17	55.17±6.38	80.98±0.22	52.32±0.63	25.20±3.19
EdgePrompt+	84.17±1.17	39.40±0.73	53.87±0.35	45.52±2.13	38.22±1.25	52.94±12.37	80.43±0.86	52.47±0.64	28.78±1.41
GPF	89.11±0.33	43.64±0.61	54.12±0.32	39.62±1.91	39.49±0.97	46.63±10.31	104.80±0.57	52.07±0.95	24.05±3.06
GPF+	87.36±0.43	41.38±0.54	53.85±0.26	41.80±2.24	37.88±1.32	49.11±7.94	104.42±0.43	52.53±0.34	27.87±0.53
GPPT	88.21±0.67	25.28±2.97	51.38±1.38	33.06±5.94	33.13±4.04	48.69±9.40	113.64±1.78	50.02±1.92	19.18±1.16
GraphPrompt	81.10±0.31	19.58±0.96	53.54±0.46	38.66±0.99	28.75±0.51	48.13±13.08	20.46±0.37	44.45±1.41	30.47±1.12
GraphPrompt+	85.25±1.30	19.29±0.81	54.12±0.30	37.92±1.09	26.13±1.10	48.06±8.81	18.33±1.33	42.67±1.16	25.51±2.42
ProNoG	86.26±0.30	39.08±0.51	54.62±0.82	32.99±1.92	31.21±1.30	48.70±9.25	7.92±2.02	42.85±1.41	26.59±0.42
Pretraining: ContextPred									
Full fine-tuning	87.57±0.50	37.75±0.48	54.94±0.29	40.71±1.70	35.00±0.86	49.46±12.30	43.34±45.36	52.18±0.84	24.33±7.46
All-in-One	86.01±0.51	16.42±0.70	52.00±0.45	39.97±0.64	37.05±0.79	52.68±7.88	98.92±2.70	50.83±1.41	28.81±0.55
EdgePrompt	81.43±1.78	43.08±0.65	53.31±0.66	40.48±1.39	38.78±1.83	57.36±5.48	93.07±0.43	52.21±0.70	28.11±3.09
EdgePrompt+	83.66±2.43	39.22±1.09	53.62±0.33	46.94±2.55	37.63±1.30	50.54±6.03	92.62±0.56	52.30±1.56	30.73±1.11
GPF	88.33±0.42	43.20±0.94	52.89±0.44	41.59±1.07	38.59±0.94	45.04±11.15	105.54±0.41	52.02±0.71	28.88±0.90
GPF+	86.60±0.41	41.93±0.76	53.44±0.49	41.32±1.78	37.55±0.77	46.54±5.98	105.23±0.35	51.89±1.14	30.46±0.75
GPPT	86.83±0.57	29.30±3.73	51.09±0.78	35.70±1.01	36.26±1.88	55.22±9.88	113.94±1.87	50.25±0.61	20.76±1.20
GraphPrompt	86.12±0.32	20.01±0.62	52.48±0.53	38.10±2.25	29.29±1.50	50.75±6.57	20.60±0.50	47.17±1.26	29.99±0.36
GraphPrompt+	85.62±0.53	18.90±2.23	52.94±0.68	37.95±1.71	28.29±2.27	56.93±7.07	19.70±0.19	44.58±1.21	27.20±1.88
ProNoG	85.76±0.70	36.49±0.47	53.49±0.61	35.16±1.26	30.47±2.28	53.60±11.50	10.31±1.91	44.75±1.25	26.28±0.71
Pretraining: DGI									
Full fine-tuning	87.67±0.60	33.24±0.45	55.08±0.63	42.28±1.40	35.40±1.16	44.49±8.67	5.36±0.20	44.94±2.04	29.26±1.05
All-in-One	86.54±0.39	31.59±2.09	51.52±0.66	40.23±1.17	38.05±0.92	51.94±12.93	80.60±4.12	53.77±0.86	26.91±2.43
EdgePrompt	80.73±2.98	30.50±2.95	54.45±0.44	38.66±0.84	41.06±0.87	58.48±9.58	86.04±0.88	51.04±0.67	25.23±3.45
EdgePrompt+	83.61±1.70	32.85±1.23	53.87±0.72	45.62±1.61	40.43±0.99	52.82±15.03	85.22±0.73	51.20±0.51	30.93±1.30
GPF	88.30±0.36	30.63±2.93	52.94±0.47	40.78±2.68	40.14±1.14	51.72±10.83	97.76±0.72	51.02±0.57	25.51±2.80
GPF+	86.60±0.53	30.08±2.71	53.75±0.54	40.00±1.42	38.27±0.57	50.73±10.30	97.16±0.69	51.20±0.61	29.29±1.61
GPPT	84.54±0.78	36.01±0.92	49.86±0.16	34.51±3.83	36.68±0.87	50.20±12.22	104.84±8.45	50.18±1.08	23.22±1.43
GraphPrompt	81.67±0.30	19.03±1.38	52.84±0.38	35.27±1.62	28.49±1.41	52.44±12.14	18.38±0.08	44.37±1.01	31.80±0.56
GraphPrompt+	86.25±0.41	16.37±2.01	53.31±0.28	38.33±2.93	29.66±2.28	54.70±9.31	18.58±0.28	44.86±1.20	27.48±2.88
ProNoG	79.23±0.57	37.97±0.35	52.86±0.83	34.66±1.71	31.22±2.70	51.99±8.68	9.63±1.89	45.04±0.72	27.46±0.46
Pretraining: EdgePred									
Full fine-tuning	87.17±0.40	38.18±0.78	54.68±1.16	41.62±1.42	35.39±0.93	42.09±7.79	5.39±0.19	45.94±2.84	29.26±1.05
All-in-One	60.67±0.78	12.22±1.04	50.90±0.52	43.34±0.86	37.52±1.53	48.80±8.89	98.92±2.70	60.96±1.64	24.97±1.76
EdgePrompt	73.82±3.73	28.97±1.33	53.82±1.17	39.67±1.11	37.15±0.73	47.54±5.69	93.07±0.43	52.65±0.89	27.04±2.37
EdgePrompt+	83.56±1.24	32.70±1.34	53.49±0.41	46.96±2.17	38.59±0.79	48.51±7.03	92.62±0.56	52.88±1.01	30.34±1.00
GPF	74.90±2.71	26.77±2.01	51.57±1.31	40.53±1.90	37.48±0.99	40.02±12.53	105.54±0.41	52.29±0.97	23.70±3.66
GPF+	86.29±0.86	29.57±1.91	53.46±0.49	41.57±0.48	38.47±1.46	41.14±6.39	105.23±0.35	50.33±2.36	29.05±0.66
GPPT	32.33±2.73	3.30±0.28	51.21±0.93	35.62±1.76	30.96±1.34	53.76±10.85	113.94±1.87	49.99±0.23	23.22±1.43
GraphPrompt	37.21±1.49	10.73±0.52	50.68±0.62	34.99±1.22	29.27±3.01	44.59±6.71	39.54±42.82	46.17±1.42	31.86±0.39
GraphPrompt+	81.55±1.35	9.64±2.64	53.15±1.40	37.90±2.31	28.46±1.59	46.83±7.11	19.70±0.19	43.33±1.45	23.95±0.66
ProNoG	33.02±0.48	14.54±0.54	52.17±0.86	37.47±1.76	31.23±1.25	50.77±12.47	10.31±1.91	41.53±1.20	26.13±0.66
Pretraining: GraphCL									
Full fine-tuning	87.66±0.70	37.30±0.83	55.23±0.42	39.90±1.91	34.80±1.03	41.43±6.45	5.67±0.11	50.82±1.52	29.76±0.49
All-in-One	86.78±0.55	14.33±1.18	51.72±0.64	42.28±1.31	36.43±0.76	49.27±9.30	111.92±1.17	56.66±1.66	28.01±0.73
EdgePrompt	82.92±0.52	37.89±2.92	53.30±0.32	39.27±0.67	39.07±0.92	50.49±8.49	69.30±0.53	50.91±0.98	28.10±0.32
EdgePrompt+	84.06±2.19	33.56±1.03	53.62±0.33	46.28±2.27	38.31±1.91	46.85±8.94	63.53±1.01	52.15±1.07	28.76±0.79
GPF	88.75±0.50	35.56±1.64	53.25±0.19	41.47±2.44	37.57±1.22	42.24±8.04	114.92±1.24	51.02±1.36	28.01±0.30
GPF+	87.56±0.70	35.36±2.34	53.57±0.31	39.80±1.95	36.78±1.18	46.90±7.83	113.95±0.94	51.22±1.10	29.28±0.44
GPPT	87.82±1.03	29.86±0.64	50.24±0.94	32.84±1.97	37.12±0.79	50.11±12.69	116.83±0.24	50.09±0.28	19.34±0.98
GraphPrompt	84.32±0.50	20.28±0.23	53.15±0.45	36.46±1.48	26.15±1.98	41.40±5.76	102.03±0.35	45.75±2.03	19.10±0.81
GraphPrompt+	87.38±0.60	20.32±0.22	53.75±0.45	37.47±0.66	29.34±1.14	48.54±6.20	26.21±9.61	44.92±1.39	27.46±0.65
ProNoG	86.81±0.28	30.46±0.49	54.45±0.78	36.18±2.12	29.17±1.38	51.84±11.29	5.37±1.14	43.71±1.36	25.76±0.68
Pretraining: InfoGraph									
Full fine-tuning	87.19±0.75	38.25±1.11	54.83±0.34	41.09±1.77	34.59±1.22	44.38±10.86	5.26±0.07	50.42±1.75	29.32±0.57
All-in-One	74.86±0.55	19.65±8.10	51.88±0.78	43.34±0.86	37.91±0.73	50.51±6.17	112.04±1.15	60.96±1.64	24.97±1.76
EdgePrompt	79.45±1.07	36.47±0.57	52.76±0.37	39.67±1.11	39.74±0.82	53.91±3.31	96.44±1.28	52.65±0.89	27.04±2.37
EdgePrompt+	83.60±1.80	36.33±0.10	53.48±0.49	46.96±2.17	38.87±0.70	49.54±6.28	95.50±0.52	52.88±1.01	30.34±1.00
GPF	87.38±0.27	34.92±0.68	52.42±0.22	39.90±1.09	39.09±0.60	42.52±9.00	115.53±0.86	52.54±0.97	25.84±3.26
GPF+	87.50±0.50	33.72±1.15	53.41±0.25	39.87±1.64	37.82±0.41	42.71±6.92	114.90±0.98	52.77±1.00	30.25±0.62
GPPT	73.22±1.01	5.51±0.34	51.17±1.34	35.62±1.76	36.20±0.01	49.54±7.24	116.78±0.27	49.99±0.23	20.04±0.70
GraphPrompt	74.63±0.75	14.20±1.05	52.02±0.34	36.03±0.69	30.07±2.14	49.27±8.11	99.37±0.56	46.60±2.07	31.97±0.46
GraphPrompt+	85.65±0.86	15.07±2.42	52.51±0.23	36.99±1.39	27.80±1.31	47.60±6.75	20.30±1.06	44.15±1.35	30.09±3.23
ProNoG	76.77±0.13	20.33±0.70	53.68±0.63	34.76±1.49	34.39±1.37	50.74±10.10	8.04±1.46	41.86±0.30	27.28±0.43

Table 25: 100-shot configuration results for GAT across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	87.29±0.75	39.88±0.64	54.85±0.26	34.84±1.67	36.11±1.05	51.75±10.96	24.94±6.90	52.87±1.01	38.93±1.24
All-in-One	87.79 ±0.37	34.79±0.99	51.62±0.56	36.86±2.17	32.04±0.92	53.89±9.06	97.19±0.61	57.64 ±0.56	18.14±8.00
EdgePrompt	75.92±5.10	44.89 ±0.53	52.91±0.25	26.48±1.74	34.30±1.54	52.46±6.44	98.86±0.75	50.69±0.59	18.40±3.79
EdgePrompt+	77.56±3.95	40.84±1.07	53.58±0.33	39.47 ±1.10	36.80 ±0.74	57.46 ±3.94	98.30±0.89	52.22±1.35	33.82±3.02
GPF	87.10±1.30	44.84 ±0.61	53.46±0.21	28.81±2.34	34.33±0.68	52.77 ±9.96	109.48±0.43	50.72±0.94	22.65±4.37
GPF+	86.41±1.29	42.64±1.06	53.61±0.19	33.59±0.90	36.70±1.85	51.80±8.26	109.45±0.57	51.12±0.60	23.51±6.44
GPPT	87.74±0.62	24.18±2.39	50.21±1.26	27.67±4.43	33.47±1.92	47.18±8.04	113.77±2.16	49.75±0.71	20.50±2.86
GraphPrompt	83.95±0.76	20.56±0.95	53.50±0.18	38.53±0.86	29.08±1.53	51.11±9.68	24.26±0.52	42.94±1.50	34.40±0.45
GraphPrompt+	84.41±1.07	20.49±1.51	53.54±0.18	37.92±1.09	28.34±2.18	43.40±10.14	23.65±3.69	42.40±1.08	35.46 ±0.90
ProNoG	87.40±0.47	38.38±0.62	55.08 ±0.52	30.20±0.99	30.79±1.87	50.51±6.83	11.32 ±1.69	41.43±1.28	31.31±0.75
Pretraining: ContextPred									
Full fine-tuning	87.76 ±0.76	39.36±0.59	55.27±0.38	34.89±2.30	36.20 ±1.28	57.48±15.44	20.94±1.19	52.56±1.13	35.54±3.52
All-in-One	85.97±0.65	35.07±0.64	50.69±0.46	34.03±1.14	31.04±1.06	53.32±8.03	113.94±0.71	50.40±2.18	35.75±2.58
EdgePrompt	77.43±1.44	43.16±0.77	52.77±0.32	39.39±0.94	32.79±0.80	53.22±14.90	110.30±0.48	51.85±0.82	33.69±3.79
EdgePrompt+	84.31±1.01	39.85±0.67	53.65±0.51	40.00 ±1.06	36.20 ±1.39	50.89±12.96	110.13±0.84	52.73 ±1.14	38.82 ±1.27
GPF	87.22±0.43	43.93 ±0.60	53.15±0.30	30.84±0.96	32.68±0.43	54.77±15.41	116.00±0.49	51.77±1.21	34.22±4.13
GPF+	86.71±1.14	41.15±2.30	53.94±0.24	32.91±1.97	35.82±0.54	52.96±10.66	116.19±0.53	51.98±0.92	35.07±3.84
GPPT	86.06±0.85	34.53±3.68	50.06±0.14	35.06±1.33	31.52±2.23	47.41±7.74	116.95±0.14	49.33±2.95	27.21±0.56
GraphPrompt	82.49±0.82	20.82±0.38	52.85±0.43	37.80±2.05	27.69±1.43	59.50 ±15.78	108.35±0.41	45.01±1.58	34.84±0.34
GraphPrompt+	84.36±1.39	21.57±0.81	53.11±0.40	38.63±1.16	27.62±2.41	49.58±6.54	105.19±1.39	45.41±1.68	35.55±0.84
ProNoG	82.30±1.14	35.78±0.60	55.35 ±0.63	26.63±2.45	29.85±0.80	51.56±14.48	10.54 ±0.90	44.25±1.29	30.97±0.24
Pretraining: DGI									
Full fine-tuning	87.23 ±0.96	36.22±0.54	55.25 ±0.47	32.13±1.86	38.40±0.91	53.75±10.24	27.83±4.37	51.43 ±1.32	38.82±1.41
All-in-One	82.67±0.88	32.65±2.17	52.00±0.46	34.25±6.58	37.93±1.01	55.58±16.84	107.98±0.46	50.05±1.11	22.30±1.17
EdgePrompt	76.24±1.81	35.20±3.42	53.13±0.15	30.53±6.29	38.62±1.16	42.50±16.42	78.37±0.85	51.37±1.60	25.10±6.77
EdgePrompt+	80.97±2.64	36.17±1.10	53.85±0.33	40.71 ±0.46	37.75±2.07	50.77±8.32	71.91±0.24	50.79±1.64	39.20 ±1.01
GPF	82.89±3.11	34.12±4.80	52.74±0.64	27.92±3.64	38.65±1.07	47.37±18.05	113.88±1.09	50.89±0.85	29.24±3.69
GPF+	82.20±0.74	32.78±2.71	53.50±0.16	33.57±2.88	39.10 ±0.94	42.83±12.00	113.91±1.17	50.54±0.25	32.42±2.34
GPPT	69.92±8.95	36.06±0.60	50.11±1.52	32.03±1.50	35.36±2.06	54.03±5.04	116.55±0.47	50.13±0.22	24.92±2.82
GraphPrompt	67.62±0.27	15.49±2.25	52.99±0.58	31.80±0.49	31.11±0.47	52.20±10.85	92.94±0.57	40.37±0.99	38.30±0.52
GraphPrompt+	76.37±2.79	17.59±0.84	53.47±0.56	38.41±3.04	30.73±3.79	52.01±17.13	84.89±2.52	41.02±0.79	38.07±0.60
ProNoG	57.84±1.03	37.86 ±0.51	<u>55.22</u> ±0.66	30.41±1.85	34.99±2.42	55.97 ±10.10	12.79 ±0.49	40.21±0.98	26.93±1.38
Pretraining: EdgePred									
Full fine-tuning	88.27 ±0.27	38.09 ±0.66	54.31 ±0.79	30.33±2.30	37.42±1.19	58.34±3.52	20.94±1.29	49.11±2.53	34.58 ±1.35
All-in-One	55.30±1.92	13.37±1.88	51.24±0.82	33.67±6.96	33.43±2.54	53.86±9.28	113.94±0.71	46.17±1.49	29.10±1.22
EdgePrompt	60.28±7.97	28.00±1.72	51.22±0.46	25.14±0.11	36.20±0.47	56.81±7.86	110.30±0.48	51.07±4.88	22.58±5.98
EdgePrompt+	81.10±0.92	34.76±1.11	54.29±0.40	34.91±3.92	37.70 ±1.11	53.46±6.61	110.13±0.84	54.54 ±3.25	33.69±2.27
GPF	61.18±7.30	26.39±1.01	51.19±0.53	25.29±0.62	35.59±1.34	60.33±6.39	116.00±0.49	53.79±2.33	20.76±3.68
GPF+	84.65±1.99	30.28±3.21	53.11±0.27	25.54±0.70	37.38±0.68	59.57±5.95	116.00±0.54	53.33±2.40	25.64±1.85
GPPT	12.45±1.83	3.06±0.54	50.31±0.49	34.94±1.27	29.06±0.75	62.81 ±13.80	116.95±0.14	50.05±0.48	15.61±2.98
GraphPrompt	23.39±1.55	6.42±0.77	50.07±0.89	35.44±2.23	28.46±2.43	42.78±7.66	108.35±0.41	42.13±2.22	33.91 ±1.38
GraphPrompt+	76.60±2.60	6.57±1.78	52.45±0.32	39.39 ±0.87	27.74±2.29	48.54±7.03	105.19±1.39	41.50±1.01	32.14±2.48
ProNoG	20.47±1.23	7.98±0.70	53.56±0.84	33.11±1.49	30.19±1.61	56.76±6.43	10.54 ±0.90	40.71±0.99	28.99±0.71
Pretraining: GraphCL									
Full fine-tuning	87.54±0.23	37.80 ±0.69	54.92 ±0.40	35.09±1.55	36.85±1.54	53.70±3.36	31.11±1.16	52.46±1.33	20.18±2.09
All-in-One	86.13±0.69	35.35±3.11	51.26±0.45	35.27±3.60	28.33±1.13	54.08±11.38	110.30±0.28	53.21 ±1.84	26.32±0.56
EdgePrompt	83.10±1.76	35.76±1.80	53.54±0.32	38.00±0.43	34.24±0.83	49.58±6.84	84.63±1.06	51.62±0.35	17.07±4.76
EdgePrompt+	84.69±0.93	32.72±1.82	53.74±0.20	41.04 ±0.48	37.52 ±1.46	50.70±5.86	80.95±1.07	52.56 ±1.36	24.82±2.11
GPF	89.35 ±0.46	36.54±1.89	53.70±0.34	34.13±1.28	34.59±0.95	52.68±10.60	116.01±0.49	50.85±0.91	16.77±3.05
GPF+	87.85±0.27	35.08±4.24	53.89±0.33	36.08±0.61	35.49±1.78	52.10±9.50	116.07±0.40	51.03±0.46	12.57±2.67
GPPT	86.58±0.82	31.70±0.91	51.03±1.36	36.81±2.45	34.45±1.41	51.97±9.72	116.76±0.25	49.71±0.73	16.43±2.24
GraphPrompt	84.29±0.23	19.05±0.95	53.55±0.37	33.92±1.07	23.37±0.74	53.08±10.68	102.39±0.84	46.51±0.89	29.92±0.45
GraphPrompt+	85.68±0.31	18.59±0.83	53.67±0.35	39.82 ±1.03	29.21±1.50	54.29 ±10.15	93.03±1.43	45.20±1.17	30.07 ±1.02
ProNoG	84.05±0.82	33.11±0.42	<u>54.51</u> ±0.55	33.57±1.41	29.81±1.40	57.09 ±10.25	5.08 ±0.46	45.21±1.05	24.33±0.28
Pretraining: InfoGraph									
Full fine-tuning	87.63 ±0.75	39.27 ±0.80	55.49 ±0.41	32.76±2.12	38.05±1.19	57.17±7.33	33.41±2.15	53.56±1.56	27.44±3.13
All-in-One	63.43±0.93	24.83±0.98	50.66±0.65	31.39±5.55	36.33±1.53	58.86±10.23	112.10±0.61	56.04 ±0.87	25.18±0.84
EdgePrompt	77.59±3.56	38.69±1.82	53.39±0.33	26.96±4.61	38.54 ±0.97	56.41±9.92	106.33±0.61	52.09±1.09	11.66±2.99
EdgePrompt+	81.29±2.61	36.14±1.22	53.95±0.11	41.24 ±1.35	37.96±1.31	52.18±12.41	105.41±0.47	52.97±1.18	22.00±3.37
GPF	83.13±0.95	38.69±1.29	53.06±0.15	25.16±0.50	36.29±2.37	51.68±6.23	116.09±0.43	52.72±1.05	13.97±3.01
GPF+	85.26±1.22	32.81±1.12	53.71±0.18	26.08±1.83	37.48±1.56	51.77±5.03	116.08±0.52	53.72±0.91	12.89±4.11
GPPT	60.12±1.62	5.17±0.54	49.66±0.27	32.84±5.51	34.77±1.48	50.67±10.81	116.98±0.14	49.96±0.35	20.73±0.80
GraphPrompt	55.93±1.41	14.03±0.47	52.31±0.39	36.20±1.68	31.18±1.45	64.16±6.72	108.50±0.89	46.08±1.45	35.03±0.56
GraphPrompt+	84.01±2.79	14.42±1.19	53.32±0.20	38.23 ±1.63	29.75±2.49	46.42±11.64	87.00±8.59	43.37±1.57	38.11 ±0.50
ProNoG	56.01±2.37	20.15±0.32	<u>54.09</u> ±0.51	30.41±0.98	33.27±0.19	65.97 ±10.09	4.78 ±0.11	41.03±1.03	29.61±0.45

Table 26: 100-shot configuration results for GIN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	87.72 ±0.72	36.95±0.89	53.66±0.53	40.13±2.23	37.79 ±1.27	50.94±11.81	37.03±3.68	50.16±2.01	39.53 ±1.10
All-in-One	86.53±0.28	22.95±1.99	52.20±0.77	38.96±2.29	35.75±1.35	56.46 ±9.03	27.80±0.85	54.87 ±1.37	26.16±4.38
EdgePrompt	79.97±0.73	39.71±1.10	54.28±1.28	38.23±1.39	34.29±2.36	50.37±6.83	41.59±1.71	51.04±0.76	24.75±6.50
EdgePrompt+	82.59±1.76	38.08±1.23	55.11 ±0.64	41.75 ±4.02	34.24±2.22	51.15±6.95	37.08±1.66	52.25 ±1.27	32.91 ±1.24
GPF	87.21±0.45	36.95±1.06	52.65±0.57	39.01±0.68	32.85±2.47	46.06±15.17	53.32±2.87	51.74±1.27	31.93±3.05
GPF+	86.92±0.39	36.51±0.53	53.63±0.71	39.92±1.57	32.56±7.21	48.04±12.89	48.00±2.50	51.19±1.39	31.05±1.47
GPPT	87.54±0.78	27.83±3.16	52.64±1.05	36.41±1.55	29.48±4.95	48.44±2.69	43.58±9.14	50.76±1.39	19.13±0.54
GraphPrompt	85.78±0.94	23.16±0.82	53.20±0.10	38.35±0.97	29.51±1.16	54.70 ±17.13	35.57±4.98	42.18±1.69	31.44±0.37
GraphPrompt+	72.27±14.41	19.27±4.64	53.69±0.53	37.90±1.09	26.08±1.19	40.98±12.32	33.20±1.18	42.69±0.67	29.26±0.05
ProNoG	83.51±1.42	39.73 ±0.74	55.29 ±0.50	37.97±1.24	30.79±1.91	54.44±9.99	17.45 ±9.80	42.00±1.38	25.22±1.25
Pretraining: ContextPred									
Full fine-tuning	87.49 ±0.63	34.07 ±0.58	54.90±0.37	39.95±1.61	38.28±1.17	53.44±14.75	36.24±1.66	52.22±1.59	42.46 ±0.31
All-in-One	83.43±0.35	24.38±0.84	52.51±0.99	40.03±1.27	36.00±1.22	54.65±18.40	36.04±0.94	54.13 ±1.40	30.01±4.39
EdgePrompt	75.17±4.30	18.57±1.81	54.79±0.56	40.03±2.13	35.61±1.50	56.88±16.01	32.97±0.77	53.98±0.63	31.87±1.77
EdgePrompt+	82.79±2.91	20.78±2.28	55.61 ±0.45	44.43 ±2.43	36.07±0.97	54.65±11.79	33.09±1.58	54.12 ±1.14	34.43±7.53
GPF	86.61±2.47	30.94±1.75	54.30±0.72	40.23±1.55	37.36±2.00	51.39±17.29	33.24±1.56	53.71±1.96	31.36±3.79
GPF+	86.00±0.38	32.64±1.79	54.65±0.71	40.00±1.81	38.80 ±1.86	51.49±15.27	32.62±0.72	52.49±2.07	35.53±1.48
GPPT	84.64±0.78	6.21±1.63	50.00±0.31	37.04±3.00	30.41±4.11	47.23±12.75	50.05±21.49	49.46±1.68	27.06±0.80
GraphPrompt	82.94±1.04	9.82±2.23	52.06±0.42	37.67±1.99	28.50±0.60	61.52 ±21.74	45.46±1.72	46.49±1.38	36.24±0.47
GraphPrompt+	74.15±18.49	2.40±0.47	52.73±0.89	37.34±1.14	26.65±1.56	56.15±7.86	45.32±1.51	45.85±1.02	32.80±6.17
ProNoG	83.20±1.04	23.33±2.79	55.03 ±0.55	35.54±2.08	30.26±2.06	48.72±10.73	13.77 ±4.69	44.94±1.15	30.12±0.83
Pretraining: DGI									
Full fine-tuning	88.04 ±1.05	32.41±1.02	54.35±0.35	40.68±2.59	39.48±0.98	57.74±7.73	37.00±2.73	49.74±2.12	37.96 ±0.88
All-in-One	85.15±1.15	8.22±3.80	51.27±0.61	39.75±0.72	35.91±0.91	51.89±2.44	34.42±0.82	53.72 ±1.39	19.49±9.47
EdgePrompt	67.88±5.14	23.17±3.70	54.53±0.49	34.94±3.11	37.23±1.35	54.15±6.59	50.69±2.26	52.54±1.49	25.46±1.63
EdgePrompt+	75.04±5.32	27.14±4.56	54.72 ±0.65	44.86 ±1.14	37.87±2.02	54.29±8.64	45.75±1.68	52.69 ±1.60	31.20±2.25
GPF	86.22±1.09	16.61±2.67	52.85±0.43	39.92±1.09	39.45±0.98	55.17±15.03	60.35±2.85	50.13±2.48	27.56±3.24
GPF+	86.09±1.00	22.90±5.41	53.25±0.64	40.94±1.79	39.92 ±0.99	53.48±8.32	57.08±2.19	48.87±2.35	27.39±2.12
GPPT	39.21±8.62	22.04±6.48	49.88±0.26	38.05±1.73	34.62±0.89	53.01±6.11	49.71±11.97	49.92±0.40	23.53±1.12
GraphPrompt	63.05±4.76	20.77±0.44	53.37±0.40	35.85±4.08	27.33±2.02	51.70±7.36	44.65±1.00	43.47±1.73	32.94±0.39
GraphPrompt+	60.75±23.42	20.01±1.18	52.85±0.45	39.47±2.65	27.64±1.84	45.33±12.94	44.81±1.01	44.30±1.70	30.03±5.33
ProNoG	63.89±1.90	35.26 ±1.11	56.06 ±0.67	31.11±2.48	30.10±4.00	59.64 ±7.23	10.17 ±2.98	41.86±1.09	24.21±0.22
Pretraining: EdgePred									
Full fine-tuning	87.05 ±0.66	27.13 ±1.13	53.60±0.47	40.84±1.79	38.06 ±1.07	54.79±9.97	36.24±1.66	46.44±4.87	40.97 ±1.87
All-in-One	33.07±0.96	9.01±1.38	50.48±0.36	38.99±1.60	36.07±1.43	55.98 ±6.23	36.08±0.99	56.94 ±2.25	30.21±3.44
EdgePrompt	74.63±1.81	12.07±1.29	53.74±0.42	35.85±2.19	33.57±1.59	52.25±12.23	32.97±0.77	46.34±2.08	27.57±3.50
EdgePrompt+	76.44±4.74	16.34±1.85	54.12 ±0.59	47.24 ±0.61	34.60±0.70	52.70±8.82	33.09±1.58	47.18±3.64	34.67 ±4.25
GPF	75.16±2.23	11.49±2.48	51.84±0.75	40.78±1.57	36.22±1.16	51.99±14.74	33.24±1.56	43.80±1.00	30.84±1.86
GPF+	84.47±1.25	18.79±2.08	52.96±0.32	41.06±1.86	37.66±1.54	49.61±14.65	32.62±0.72	45.36±2.73	31.87±3.81
GPPT	19.08±4.82	3.42±3.27	49.97±0.47	39.16±2.42	31.32±1.91	51.45±12.85	50.05±21.49	50.31±0.70	24.42±1.13
GraphPrompt	22.49±0.46	3.14±1.15	51.51±0.17	38.53±2.14	27.30±1.19	57.98 ±16.94	44.51±0.58	44.26±3.45	34.48±0.84
GraphPrompt+	31.21±3.75	5.99±0.72	52.48±0.25	38.48±1.71	26.03±2.07	45.99±11.51	45.32±1.51	43.73±1.86	32.29±7.29
ProNoG	19.04±0.82	2.39±0.65	54.22 ±0.60	37.01±2.52	29.21±2.42	54.11±7.88	13.77 ±4.69	41.62±1.10	29.83±0.56
Pretraining: GraphCL									
Full fine-tuning	87.71 ±0.90	29.55±1.37	53.40±0.39	40.63±1.06	38.26 ±1.80	53.71±5.45	36.20±2.91	51.40±1.93	42.05 ±1.76
All-in-One	79.64±1.80	24.72±2.43	51.78±1.09	36.89±2.60	34.49±1.63	47.37±6.42	36.17±0.99	53.55 ±2.17	29.54±4.84
EdgePrompt	79.21±3.19	30.51±2.89	52.70±0.82	34.99±1.16	33.87±3.47	57.00±5.21	35.72±0.76	49.74±1.89	30.38±4.17
EdgePrompt+	82.11±1.85	30.17±3.44	54.62 ±0.55	45.24 ±2.21	36.16±1.58	58.57 ±4.98	35.92±1.44	49.99±1.54	39.31 ±2.50
GPF	84.98±1.06	23.55±3.29	51.97±0.87	40.96±1.31	36.11±2.06	42.78±7.07	37.19±2.05	51.35±1.72	32.45±0.76
GPF+	84.74±0.75	25.19±2.55	52.55±0.73	42.86±1.64	35.52±1.33	49.92±3.73	35.69±3.48	50.65±1.49	34.80±1.52
GPPT	22.75±2.59	11.63±2.70	50.36±0.79	40.53±1.70	23.46±3.58	51.61±6.71	108.95±94.89	49.99±1.93	31.34±1.48
GraphPrompt	67.70±2.15	15.06±0.61	52.75±0.54	33.04±2.87	22.84±2.15	41.02±6.27	45.73±1.06	44.38±1.75	37.20±0.59
GraphPrompt+	72.14±10.48	11.91±1.51	53.01±0.61	40.41±1.95	25.24±3.13	49.89±9.17	46.29±2.73	45.77±1.53	33.60±6.64
ProNoG	79.05±0.66	32.73 ±0.48	56.33 ±0.40	33.77±2.09	29.49±1.27	49.34±9.66	29.13 ±18.92	44.76±1.60	30.28±0.45
Pretraining: InfoGraph									
Full fine-tuning	88.06 ±0.50	32.98 ±1.06	54.44±0.32	40.51±1.98	38.17 ±1.22	55.46±8.68	34.91±1.55	51.13±2.52	40.43 ±1.47
All-in-One	78.21±0.74	5.03±2.48	51.66±0.63	39.62±0.65	35.35±1.35	52.51±10.70	102.05±3.11	54.77 ±2.75	25.12±5.56
EdgePrompt	76.73±5.34	12.92±1.77	55.09±0.46	39.22±2.11	28.20±3.20	53.82±6.02	91.32±2.50	48.23±1.37	28.49±3.23
EdgePrompt+	83.52±1.13	14.81±1.48	54.67±0.23	47.67 ±1.34	34.06±2.52	51.27±6.60	87.41±1.71	51.30±4.42	33.75 ±1.94
GPF	86.10±0.28	17.03±4.12	54.97±0.30	39.97±1.05	34.76±1.63	50.51±9.08	111.70±0.60	51.88±2.05	27.60±3.90
GPF+	86.72±0.99	24.33 ±2.83	54.61±0.14	40.25±1.49	37.97±2.08	52.15±8.56	111.55±0.64	50.93±1.72	30.36±0.54
GPPT	78.11±0.73	2.14±0.38	50.24±0.72	39.01±1.75	32.03±1.11	51.06±8.39	115.53±1.14	50.12±0.11	23.10±0.46
GraphPrompt	72.92±1.50	4.85±2.07	51.98±0.47	40.10±0.86	26.75±0.35	62.78 ±9.39	81.75±31.52	45.07±1.67	32.19±0.41
GraphPrompt+	68.82±18.27	13.02±2.14	52.98±0.56	41.47±1.55	28.68±1.53	45.49±12.97	43.83±1.06	46.52±1.64	30.17±4.05
ProNoG	71.94±0.52	6.56±1.47	55.20 ±0.63	36.81±1.62	30.43±0.67	55.89 ±7.97	7.59 ±1.21	41.57±1.27	28.84±0.44

Table 27: 100-shot configuration results for H2GCN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	GR / MAE ↓	GC / AUC ↑	GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	88.29±0.49	39.48±0.84	54.95 ±0.37	28.56±4.09	41.68 ±1.48	67.66±7.23	10.69±0.49	51.83±0.66	34.27 ±2.35
All-in-One	88.45±0.32	44.06±0.44	54.27±0.35	40.68 ±1.08	35.27±0.97	74.13 ±5.46	82.90±1.99	52.56±0.78	30.60±0.49
EdgePrompt	80.48±2.47	46.89 ±0.59	54.07±0.23	24.89±0.57	37.95±0.80	66.56±12.92	69.09±0.95	51.46±1.92	27.44±1.37
EdgePrompt+	84.41±0.95	43.01±0.63	53.83±0.31	34.94±3.81	39.43±1.44	72.01±4.86	65.57±0.74	52.38±1.48	<u>31.33</u> ±1.07
GPF	88.86 ±0.39	46.57±0.45	54.12±0.17	25.80±0.64	39.32±0.97	72.60±3.42	106.73±3.79	52.79 ±2.93	26.19±0.97
GPF+	87.34±1.06	39.10±11.87	53.71±0.40	35.90±2.27	40.28±0.62	69.06±10.62	103.72±0.74	53.51 ±0.81	29.26±1.17
GPPT	86.10±3.65	31.99±1.98	50.95±0.96	33.04±4.85	36.03±1.10	57.31±3.33	114.27±3.63	49.74±1.08	20.50±0.62
GraphPrompt	87.16±0.75	22.13±0.76	53.29±0.23	38.35±1.49	32.20±2.57	72.98±6.05	20.21±0.47	44.14±1.68	29.60±0.67
GraphPrompt+	85.09±0.96	20.43±2.02	53.96±0.45	37.92±1.09	28.53±4.49	46.25±9.29	21.74±2.58	43.71±1.07	30.86±0.68
ProNoG	84.21±0.95	41.35±0.68	<u>54.64</u> ±0.62	<u>33.77</u> ±1.48	36.14±1.25	65.61±6.94	6.33 ±1.34	42.06±1.11	27.74±0.64
Pretraining: ContextPred									
Full fine-tuning	88.24±0.45	38.75±0.86	54.60 ±0.43	28.48±3.73	41.20 ±1.18	64.76 ±5.06	6.08 ±0.13	50.70±1.58	38.57 ±0.75
All-in-One	87.97±0.33	38.59±0.53	53.75±0.21	40.94 ±1.27	30.80±1.40	57.34±11.72	59.96±2.64	48.97±0.75	36.97±0.46
EdgePrompt	81.37±1.41	41.53 ±0.42	53.45±0.26	25.49±0.48	37.02±1.07	47.92±9.17	94.78±1.12	51.29±0.96	30.90±3.30
EdgePrompt+	85.07±1.50	39.24±1.04	53.85±0.25	36.00±0.96	38.29±1.03	57.46±7.52	94.64±0.54	51.34 ±0.79	35.94±1.58
GPF	88.34 ±0.49	41.24±0.57	53.51±0.18	26.10±0.27	37.50±0.75	54.24±12.87	96.33±1.58	51.19±0.89	29.14±1.84
GPF+	87.70±0.23	39.98±1.94	<u>54.06</u> ±0.24	34.25±1.60	39.00±1.75	58.55±9.77	97.31±3.00	51.18±0.77	29.17±3.18
GPPT	87.34±0.40	30.60±6.04	50.75±0.90	34.35±2.46	32.86±2.88	50.20±15.81	104.76±4.84	50.19±0.63	24.58±0.61
GraphPrompt	86.93±1.28	21.35±0.63	53.42±0.41	38.03±1.19	29.89±1.15	60.33±13.18	19.16±0.19	42.47±1.62	32.68±1.33
GraphPrompt+	87.46±0.78	20.94±1.21	53.76±0.28	37.92±1.09	26.52±1.50	53.13±15.29	19.32±0.22	41.63±1.02	33.83±2.02
ProNoG	86.58±0.43	37.10±0.95	53.99±0.61	28.23±2.46	31.96±1.25	53.60±10.60	<u>12.67</u> ±6.09	41.36±0.69	28.52±0.54
Pretraining: DGI									
Full fine-tuning	87.49±0.91	40.15 ±1.10	<u>54.60</u> ±0.34	31.67±4.56	42.22 ±0.91	64.30±6.35	6.55 ±0.54	50.96±1.00	37.24 ±2.44
All-in-One	87.10±0.23	34.54±2.12	54.83 ±0.32	41.82 ±1.75	36.54±1.36	57.03±13.97	78.26±0.33	52.78±1.44	37.20±0.41
EdgePrompt	81.04±1.22	39.16±3.25	53.19±0.14	26.86±0.91	39.37±1.11	59.79±14.55	12.77±3.49	53.08±2.46	31.02±2.20
EdgePrompt+	84.47±1.57	39.49±2.43	53.80±0.13	36.68±2.51	37.97±2.05	59.76±11.47	10.63±1.83	53.10 ±1.44	34.65±2.72
GPF	87.74 ±0.36	33.88±4.51	53.31±0.25	27.67±2.52	39.54±1.13	65.76 ±10.54	104.12±0.12	52.54±2.26	30.67±3.13
GPF+	87.43±0.54	31.10±2.82	53.51±0.48	31.62±3.03	38.83±1.12	55.86±11.10	104.55±1.53	53.24 ±0.90	26.25±7.40
GPPT	81.09±1.79	35.07±0.50	50.61±0.83	34.60±0.94	35.94±0.57	48.20±12.63	107.60±7.05	50.08±0.23	27.10±1.50
GraphPrompt	79.26±0.78	19.04±1.06	52.49±0.43	34.18±1.99	28.72±0.61	44.99±10.82	17.76±0.49	42.66±1.81	31.61±1.53
GraphPrompt+	82.90±1.64	17.85±2.15	53.50±0.23	36.89±1.01	25.80±1.12	44.30±10.99	11.19±2.82	42.78±1.21	31.64±4.11
ProNoG	73.72±0.72	37.46±0.44	53.61±0.49	31.11±1.74	37.56±1.46	62.35±10.54	12.54±5.09	40.87±1.05	28.97±0.18
Pretraining: EdgePred									
Full fine-tuning	87.19 ±0.60	38.53 ±0.62	55.46 ±0.41	26.94±0.65	41.17 ±1.10	64.73 ±6.03	6.08 ±0.13	49.32±1.93	34.75 ±3.30
All-in-One	58.70±0.87	16.74±0.90	51.86±0.31	40.43 ±1.94	36.30±1.67	55.41±8.92	59.96±2.64	54.60 ±1.24	33.78±0.56
EdgePrompt	74.09±2.72	24.08±2.17	52.33±0.70	25.29±0.19	36.70±0.96	59.98±15.02	94.78±1.12	52.11±0.93	26.01±1.25
EdgePrompt+	82.98±1.84	33.53±1.08	53.23±0.37	27.27±0.11	39.32±1.09	61.64±4.24	94.64±0.54	50.08±1.08	30.09±0.44
GPF	79.06±3.46	15.55±2.81	52.06±0.28	25.42±0.11	36.96±1.04	56.98±8.34	96.33±1.58	50.20±0.88	25.97±1.24
GPF+	86.87±0.66	25.89±2.50	53.27±0.20	26.46±0.98	37.11±4.98	60.29±3.10	95.95±1.23	49.14±0.67	27.73±0.66
GPPT	24.08±4.45	3.74±2.25	50.12±0.60	35.01±0.93	28.51±0.76	54.20±7.15	104.76±4.84	49.97±0.41	20.78±0.53
GraphPrompt	39.68±1.22	8.43±1.04	50.21±0.24	38.76±0.72	25.68±3.65	48.11±9.00	19.15±0.17	43.81±2.05	<u>34.58</u> ±1.08
GraphPrompt+	79.39±1.94	5.82±1.05	52.55±0.46	37.87±1.36	29.65±5.12	46.37±10.27	19.32±0.22	41.74±1.40	26.99±0.56
ProNoG	32.20±2.11	16.91±0.47	52.11±0.52	30.66±0.79	26.25±0.95	58.80±11.75	<u>12.67</u> ±6.09	41.72±1.09	28.79±0.94
Pretraining: GraphCL									
Full fine-tuning	88.08±1.11	40.12±1.00	55.07 ±0.37	27.32±1.37	41.49 ±1.69	63.35 ±8.73	6.08 ±0.21	51.88±1.28	32.59 ±1.44
All-in-One	87.15±0.39	39.07±0.40	53.45±0.60	42.41 ±0.95	34.54±1.97	56.19±10.62	33.99±3.22	53.20 ±1.20	31.45±0.33
EdgePrompt	80.34±2.21	40.27 ±0.68	53.38±0.64	26.00±0.70	38.23±0.98	56.53±8.98	84.11±1.74	52.50±0.90	27.35±1.98
EdgePrompt+	84.71±0.23	39.58±0.82	53.86±0.16	36.91±0.98	39.53±0.73	58.17±8.17	83.88±0.86	53.14 ±1.50	29.26±1.27
GPF	88.62 ±0.64	38.97±0.58	53.64±0.52	26.20±1.12	37.73±1.14	57.00±9.77	85.70±1.23	51.14±1.25	28.00±2.30
GPF+	87.57±0.37	40.21±0.51	53.74±0.37	34.58±1.07	40.18±1.05	58.24±12.08	85.02±0.58	50.99±0.79	28.41±1.79
GPPT	87.42±0.88	22.06±1.79	50.68±1.01	35.14±2.60	37.15±1.13	51.32±8.58	86.26±7.48	49.73±0.67	21.51±1.10
GraphPrompt	85.54±0.44	19.34±0.48	53.52±0.49	36.58±1.29	27.33±3.67	53.20±12.94	18.98±0.20	45.23±1.60	29.64±0.39
GraphPrompt+	87.78±0.95	18.90±1.01	53.95±0.44	39.37±0.81	30.79±5.44	48.42±12.26	18.97±0.20	45.06±1.66	<u>31.68</u> ±1.89
ProNoG	85.12±1.02	33.32±0.38	<u>54.22</u> ±0.62	32.38±0.68	36.11±1.32	55.10±17.74	<u>16.31</u> ±6.53	44.02±1.72	25.85±0.29
Pretraining: InfoGraph									
Full fine-tuning	87.65±0.50	40.02±1.70	55.45 ±0.49	26.89±0.41	42.30 ±1.70	67.04 ±5.31	5.89±0.11	52.45±1.55	36.74±1.59
All-in-One	86.40±0.92	40.15±0.80	53.56±0.48	41.82 ±0.93	38.17±0.62	53.86±6.51	109.14±0.81	57.42 ±0.84	33.87±0.45
EdgePrompt	80.38±2.76	44.03±0.60	53.05±0.30	25.06±0.22	38.67±1.27	53.60±3.72	109.92±0.68	52.92±0.86	27.59±2.44
EdgePrompt+	83.30±1.42	40.95±1.10	53.59±0.39	27.72±0.65	40.25±1.17	58.31±6.20	108.72±0.50	53.27±1.13	33.92±1.45
GPF	88.51 ±0.71	46.68 ±0.80	53.14±0.22	24.99±0.11	38.74±0.96	52.37±6.36	113.93±1.01	52.60±1.74	27.07±1.68
GPF+	87.61±0.51	39.08±10.70	53.70±0.28	25.87±0.66	41.01±1.30	56.86±6.17	113.94±1.16	53.22±1.56	29.63±0.87
GPPT	84.91±0.46	22.01±2.64	52.17±0.32	36.58±1.06	38.22±0.47	51.00±9.31	116.20±1.24	50.19±0.22	22.81±1.22
GraphPrompt	83.45±0.44	17.37±2.62	52.96±0.16	38.10±1.07	31.62±2.21	60.33±4.51	78.73±0.58	46.11±1.70	36.78 ±0.44
GraphPrompt+	85.81±3.40	12.79±1.10	53.83±0.38	38.53±1.48	27.15±0.90	55.86±11.93	19.08±0.22	45.16±1.42	36.11±2.39
ProNoG	83.50±0.75	37.39±1.21	<u>54.97</u> ±0.63	31.67±1.30	35.32±1.01	53.47±14.79	4.81 ±0.20	42.26±1.17	28.72±0.40

Table 28: 100-shot configuration results for FAGCN across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	GR / MAE ↓	GC / AUC ↑	GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	87.17 $\pm$ 0.90	42.18 $\pm$ 0.76	54.95 $\pm$ 0.73	34.81 $\pm$ 1.45	37.38 $\pm$ 1.39	49.37 $\pm$ 5.70	6.04 $\pm$ 0.41	50.79 $\pm$ 1.61	31.77 $\pm$ 0.89
All-in-One	88.80 $\pm$ 0.41	45.06 $\pm$ 0.14	53.79 $\pm$ 0.26	31.42 $\pm$ 4.50	33.28 $\pm$ 1.12	45.30 $\pm$ 1.07	11.78 $\pm$ 0.57	57.26 $\pm$ 0.83	28.22 $\pm$ 0.83
EdgePrompt	82.09 $\pm$ 0.75	45.63 $\pm$ 0.70	53.86 $\pm$ 0.19	34.33 $\pm$ 6.18	38.30 $\pm$ 1.07	53.39 $\pm$ 9.92	<u>9.22</u> $\pm$ 0.33	51.11 $\pm$ 0.67	28.66 $\pm$ 0.21
EdgePrompt+	83.08 $\pm$ 0.75	44.96 $\pm$ 1.34	54.15 $\pm$ 0.07	43.22 $\pm$ 1.79	39.17 $\pm$ 1.48	53.82 $\pm$ 9.63	9.26 $\pm$ 0.32	51.58 $\pm$ 0.57	31.96 $\pm$ 1.52
GPF	88.88 $\pm$ 0.31	45.22 $\pm$ 0.38	54.00 $\pm$ 0.10	34.35 $\pm$ 4.14	32.70 $\pm$ 1.10	45.76 $\pm$ 4.12	66.49 $\pm$ 3.63	51.63 $\pm$ 0.39	28.05 $\pm$ 0.55
GPF+	88.64 $\pm$ 0.47	42.53 $\pm$ 0.78	53.77 $\pm$ 0.08	31.82 $\pm$ 2.42	36.33 $\pm$ 1.42	50.63 $\pm$ 10.11	63.64 $\pm$ 0.76	51.87 $\pm$ 0.73	27.81 $\pm$ 2.51
GPPT	85.80 $\pm$ 1.54	29.10 $\pm$ 2.32	50.91 $\pm$ 1.20	36.13 $\pm$ 0.48	24.24 $\pm$ 4.35	45.05 $\pm$ 6.67	57.95 $\pm$ 6.47	50.69 $\pm$ 1.57	16.50 $\pm$ 1.45
GraphPrompt	78.80 $\pm$ 1.79	20.28 $\pm$ 0.42	53.62 $\pm$ 0.14	37.92 $\pm$ 1.09	28.68 $\pm$ 1.09	53.25 $\pm$ 8.19	19.60 $\pm$ 0.22	43.45 $\pm$ 1.39	30.47 $\pm$ 0.67
GraphPrompt+	82.03 $\pm$ 2.49	20.22 $\pm$ 0.37	53.86 $\pm$ 0.12	37.92 $\pm$ 1.09	25.80 $\pm$ 0.94	42.78 $\pm$ 11.34	15.45 $\pm$ 1.80	42.21 $\pm$ 0.58	28.13 $\pm$ 2.79
ProNoG	85.20 $\pm$ 1.10	39.14 $\pm$ 0.67	<u>54.75</u> $\pm$ 0.66	30.25 $\pm$ 1.30	27.92 $\pm$ 0.35	49.31 $\pm$ 10.11	17.06 $\pm$ 2.51	41.47 $\pm$ 0.95	24.39 $\pm$ 0.34
Pretraining: ContextPred									
Full fine-tuning	88.47 $\pm$ 0.52	41.85 $\pm$ 0.38	55.17 $\pm$ 0.94	33.67 $\pm$ 1.78	<u>36.36</u> $\pm$ 0.51	48.59 $\pm$ 7.81	5.77 $\pm$ 0.29	48.63 $\pm$ 1.38	27.88 $\pm$ 0.43
All-in-One	87.14 $\pm$ 0.35	35.25 $\pm$ 0.35	53.68 $\pm$ 0.45	39.32 $\pm$ 1.12	34.86 $\pm$ 0.98	54.70 $\pm$ 7.53	67.47 $\pm$ 5.13	51.14 $\pm$ 1.72	29.77 $\pm$ 0.59
EdgePrompt	81.13 $\pm$ 0.43	31.90 $\pm$ 0.87	53.67 $\pm$ 0.39	36.84 $\pm$ 2.59	34.69 $\pm$ 0.87	51.72 $\pm$ 10.31	79.63 $\pm$ 1.00	48.93 $\pm$ 0.61	27.51 $\pm$ 1.01
EdgePrompt+	83.00 $\pm$ 0.79	36.95 $\pm$ 1.39	<u>54.62</u> $\pm$ 0.30	40.38 $\pm$ 1.02	<u>40.72</u> $\pm$ 0.56	52.18 $\pm$ 9.16	79.39 $\pm$ 0.68	48.65 $\pm$ 0.62	31.73 $\pm$ 1.41
GPF	88.39 $\pm$ 0.16	33.09 $\pm$ 0.97	53.47 $\pm$ 0.44	33.29 $\pm$ 2.53	34.19 $\pm$ 1.82	50.54 $\pm$ 7.88	87.34 $\pm$ 2.01	48.59 $\pm$ 0.57	26.12 $\pm$ 0.96
GPF+	88.74 $\pm$ 0.23	31.15 $\pm$ 3.35	53.74 $\pm$ 0.27	34.66 $\pm$ 1.21	35.93 $\pm$ 1.46	54.53 $\pm$ 7.04	90.14 $\pm$ 7.75	48.75 $\pm$ 0.55	25.43 $\pm$ 1.79
GPPT	87.21 $\pm$ 1.06	19.44 $\pm$ 5.38	50.47 $\pm$ 0.52	35.47 $\pm$ 2.88	33.71 $\pm$ 1.02	51.57 $\pm$ 5.55	82.91 $\pm$ 7.15	49.70 $\pm$ 0.92	20.63 $\pm$ 1.02
GraphPrompt	86.61 $\pm$ 0.68	17.26 $\pm$ 0.45	53.26 $\pm$ 0.29	37.37 $\pm$ 1.58	29.95 $\pm$ 1.40	57.38 $\pm$ 15.17	19.45 $\pm$ 0.26	42.25 $\pm$ 1.75	31.79 $\pm$ 0.84
GraphPrompt+	86.93 $\pm$ 0.44	19.05 $\pm$ 1.26	53.62 $\pm$ 0.36	39.75 $\pm$ 1.00	27.91 $\pm$ 1.39	46.59 $\pm$ 8.25	19.61 $\pm$ 0.14	41.76 $\pm$ 0.93	33.77 $\pm$ 1.22
ProNoG	87.31 $\pm$ 0.42	22.83 $\pm$ 0.30	53.95 $\pm$ 0.48	<u>34.66</u> $\pm$ 0.97	30.78 $\pm$ 0.97	51.05 $\pm$ 11.19	<u>12.30</u> $\pm$ 4.31	41.44 $\pm$ 0.94	24.50 $\pm$ 0.56
Pretraining: DGI									
Full fine-tuning	88.29 $\pm$ 0.59	42.23 $\pm$ 0.43	54.28 $\pm$ 0.92	38.30 $\pm$ 3.12	39.47 $\pm$ 1.63	57.88 $\pm$ 7.21	5.86 $\pm$ 0.26	50.90 $\pm$ 1.06	31.93 $\pm$ 2.92
All-in-One	87.65 $\pm$ 0.38	42.12 $\pm$ 0.69	53.66 $\pm$ 0.31	39.39 $\pm$ 1.55	38.92 $\pm$ 1.11	49.66 $\pm$ 4.98	86.50 $\pm$ 6.14	51.08 $\pm$ 0.92	29.29 $\pm$ 2.59
EdgePrompt	79.80 $\pm$ 1.55	44.84 $\pm$ 0.81	53.28 $\pm$ 0.57	29.85 $\pm$ 5.39	37.15 $\pm$ 1.91	60.33 $\pm$ 7.64	55.67 $\pm$ 4.09	51.98 $\pm$ 0.47	28.45 $\pm$ 1.69
EdgePrompt+	81.77 $\pm$ 1.00	44.03 $\pm$ 1.28	53.70 $\pm$ 0.32	44.25 $\pm$ 2.35	38.86 $\pm$ 1.53	54.22 $\pm$ 6.42	57.63 $\pm$ 2.12	52.20 $\pm$ 0.56	33.42 $\pm$ 1.28
GPF	87.22 $\pm$ 0.56	42.94 $\pm$ 0.63	52.50 $\pm$ 0.35	31.75 $\pm$ 3.78	36.66 $\pm$ 0.28	62.62 $\pm$ 12.63	102.30 $\pm$ 0.96	51.29 $\pm$ 0.54	25.89 $\pm$ 2.08
GPF+	<u>88.15</u> $\pm$ 0.27	41.59 $\pm$ 1.53	53.62 $\pm$ 0.19	35.72 $\pm$ 1.66	37.70 $\pm$ 1.16	51.87 $\pm$ 5.91	103.21 $\pm$ 3.04	51.06 $\pm$ 1.13	26.46 $\pm$ 4.42
GPPT	72.62 $\pm$ 2.63	30.83 $\pm$ 5.10	49.87 $\pm$ 0.33	32.71 $\pm$ 5.02	34.88 $\pm$ 1.79	55.36 $\pm$ 8.33	111.23 $\pm$ 2.61	50.52 $\pm$ 2.59	19.92 $\pm$ 0.53
GraphPrompt	72.64 $\pm$ 2.61	16.36 $\pm$ 2.63	52.64 $\pm$ 0.67	30.25 $\pm$ 4.83	32.76 $\pm$ 4.18	54.96 $\pm$ 8.05	19.85 $\pm$ 6.87	43.86 $\pm$ 2.20	31.17 $\pm$ 1.44
GraphPrompt+	84.86 $\pm$ 1.25	22.10 $\pm$ 1.02	53.81 $\pm$ 0.27	33.70 $\pm$ 2.47	30.58 $\pm$ 1.60	41.93 $\pm$ 11.48	14.64 $\pm$ 0.73	45.01 $\pm$ 1.45	27.16 $\pm$ 0.64
ProNoG	70.56 $\pm$ 4.49	40.95 $\pm$ 0.58	52.22 $\pm$ 0.75	28.43 $\pm$ 2.76	32.57 $\pm$ 2.43	65.54 $\pm$ 9.01	<u>8.64</u> $\pm$ 4.27	44.68 $\pm$ 1.12	27.64 $\pm$ 0.43
Pretraining: EdgePred									
Full fine-tuning	87.76 $\pm$ 0.51	42.02 $\pm$ 1.10	54.73 $\pm$ 0.20	33.11 $\pm$ 1.84	38.85 $\pm$ 1.73	51.03 $\pm$ 8.75	5.77 $\pm$ 0.29	48.10 $\pm$ 1.87	22.40 $\pm$ 3.56
All-in-One	57.38 $\pm$ 1.05	24.70 $\pm$ 0.60	52.07 $\pm$ 0.46	38.58 $\pm$ 0.62	36.33 $\pm$ 0.97	49.68 $\pm$ 5.90	67.47 $\pm$ 5.13	49.55 $\pm$ 0.67	25.91 $\pm$ 0.95
EdgePrompt	57.16 $\pm$ 3.33	9.04 $\pm$ 0.89	52.26 $\pm$ 0.50	38.76 $\pm$ 2.05	35.37 $\pm$ 1.74	45.47 $\pm$ 1.93	79.63 $\pm$ 1.00	48.98 $\pm$ 1.16	25.01 $\pm$ 1.07
EdgePrompt+	78.89 $\pm$ 1.58	20.37 $\pm$ 3.07	52.44 $\pm$ 0.66	39.57 $\pm$ 2.15	37.95 $\pm$ 1.13	47.80 $\pm$ 7.80	79.39 $\pm$ 0.68	49.41 $\pm$ 1.05	30.68 $\pm$ 1.22
GPF	75.42 $\pm$ 2.34	9.60 $\pm$ 0.71	51.79 $\pm$ 0.46	28.48 $\pm$ 4.87	34.62 $\pm$ 1.14	47.02 $\pm$ 3.69	87.34 $\pm$ 2.01	51.62 $\pm$ 0.77	26.84 $\pm$ 2.61
GPF+	87.02 $\pm$ 0.43	<u>30.06</u> $\pm$ 1.13	53.72 $\pm$ 0.21	32.23 $\pm$ 2.94	34.80 $\pm$ 3.73	48.99 $\pm$ 5.71	88.48 $\pm$ 4.17	51.43 $\pm$ 1.10	23.94 $\pm$ 3.39
GPPT	55.72 $\pm$ 0.63	3.36 $\pm$ 0.46	50.21 $\pm$ 0.45	32.68 $\pm$ 2.22	32.66 $\pm$ 2.05	52.45 $\pm$ 5.33	82.91 $\pm$ 7.15	49.86 $\pm$ 0.23	18.99 $\pm$ 1.22
GraphPrompt	49.73 $\pm$ 0.74	5.94 $\pm$ 0.27	49.34 $\pm$ 0.76	36.00 $\pm$ 1.44	32.52 $\pm$ 3.59	<u>51.06</u> $\pm$ 11.79	19.27 $\pm$ 0.11	43.25 $\pm$ 1.30	28.58 $\pm$ 0.24
GraphPrompt+	85.04 $\pm$ 1.35	6.99 $\pm$ 0.22	52.15 $\pm$ 0.73	37.95 $\pm$ 1.00	32.13 $\pm$ 4.58	45.37 $\pm$ 9.00	19.61 $\pm$ 0.14	41.54 $\pm$ 1.30	25.36 $\pm$ 0.22
ProNoG	47.14 $\pm$ 1.19	7.91 $\pm$ 0.16	51.72 $\pm$ 0.67	36.03 $\pm$ 1.93	33.10 $\pm$ 0.79	45.69 $\pm$ 8.58	<u>12.30</u> $\pm$ 4.31	41.08 $\pm$ 1.13	26.33 $\pm$ 0.42
Pretraining: GraphCL									
Full fine-tuning	87.07 $\pm$ 0.69	42.45 $\pm$ 1.23	54.14 $\pm$ 0.22	35.77 $\pm$ 1.83	37.94 $\pm$ 1.24	50.08 $\pm$ 6.00	5.96 $\pm$ 0.46	51.77 $\pm$ 1.01	26.28 $\pm$ 1.07
All-in-One	87.05 $\pm$ 0.79	43.51 $\pm$ 0.24	53.81 $\pm$ 0.53	35.01 $\pm$ 2.73	33.92 $\pm$ 0.77	49.92 $\pm$ 6.68	45.75 $\pm$ 6.03	57.09 $\pm$ 0.98	23.97 $\pm$ 1.02
EdgePrompt	81.80 $\pm$ 0.93	44.80 $\pm$ 0.25	53.71 $\pm$ 0.32	42.56 $\pm$ 1.66	34.64 $\pm$ 0.58	<u>52.01</u> $\pm$ 4.84	65.49 $\pm$ 0.48	50.69 $\pm$ 1.73	24.26 $\pm$ 0.22
EdgePrompt+	82.19 $\pm$ 0.57	44.53 $\pm$ 0.28	53.86 $\pm$ 0.33	38.30 $\pm$ 5.84	<u>36.85</u> $\pm$ 1.16	49.20 $\pm$ 4.25	65.38 $\pm$ 0.74	50.53 $\pm$ 1.27	27.79 $\pm$ 0.45
GPF	88.53 $\pm$ 0.59	44.24 $\pm$ 0.61	53.50 $\pm$ 0.34	30.51 $\pm$ 3.63	33.81 $\pm$ 1.03	49.99 $\pm$ 8.19	73.46 $\pm$ 2.61	52.31 $\pm$ 1.69	22.89 $\pm$ 1.13
GPF+	87.09 $\pm$ 1.00	39.67 $\pm$ 7.96	53.72 $\pm$ 0.28	34.56 $\pm$ 1.68	36.36 $\pm$ 0.96	49.49 $\pm$ 8.22	75.81 $\pm$ 10.29	52.36 $\pm$ 1.39	24.84 $\pm$ 0.60
GPPT	88.11 $\pm$ 0.40	18.57 $\pm$ 3.07	50.42 $\pm$ 1.04	34.96 $\pm$ 1.75	33.90 $\pm$ 1.95	50.32 $\pm$ 8.96	64.83 $\pm$ 3.34	50.17 $\pm$ 0.43	17.78 $\pm$ 2.20
GraphPrompt	87.81 $\pm$ 0.51	24.46 $\pm$ 0.65	53.71 $\pm$ 0.37	39.57 $\pm$ 2.13	30.94 $\pm$ 2.50	54.74 $\pm$ 10.17	19.37 $\pm$ 0.52	45.21 $\pm$ 1.85	22.59 $\pm$ 1.12
GraphPrompt+	87.64 $\pm$ 0.41	20.82 $\pm$ 3.29	53.85 $\pm$ 0.27	38.10 $\pm$ 1.42	28.09 $\pm$ 0.76	40.40 $\pm$ 6.26	19.50 $\pm$ 0.06	45.20 $\pm$ 1.51	21.22 $\pm$ 1.71
ProNoG	86.46 $\pm$ 0.19	39.48 $\pm$ 0.68	54.51 $\pm$ 0.44	34.33 $\pm$ 0.84	31.85 $\pm$ 1.64	48.16 $\pm$ 11.98	<u>12.66</u> $\pm$ 4.71	43.07 $\pm$ 1.62	26.45 $\pm$ 0.70
Pretraining: InfoGraph									
Full fine-tuning	87.31 $\pm$ 0.43	42.01 $\pm$ 0.98	54.74 $\pm$ 0.55	36.05 $\pm$ 1.85	38.32 $\pm$ 1.17	52.29 $\pm$ 8.64	6.01 $\pm$ 0.39	50.05 $\pm$ 0.94	28.23 $\pm$ 0.37
All-in-One	87.60 $\pm$ 0.35	41.63 $\pm$ 0.34	53.86 $\pm$ 0.44	42.71 $\pm$ 0.83	39.14 $\pm$ 1.43	52.29 $\pm$ 7.93	109.32 $\pm$ 0.75	60.04 $\pm$ 0.90	27.09 $\pm$ 1.02
EdgePrompt	80.97 $\pm$ 1.78	46.69 $\pm$ 0.24	53.80 $\pm$ 0.34	36.51 $\pm$ 4.42	39.75 $\pm$ 1.04	56.84 $\pm$ 2.66	100.21 $\pm$ 0.69	50.67 $\pm$ 0.68	27.09 $\pm$ 1.92
EdgePrompt+	79.97 $\pm$ 4.18	43.66 $\pm$ 0.74	54.46 $\pm$ 0.14	43.59 $\pm$ 1.29	40.41 $\pm$ 1.26	49.20 $\pm$ 3.81	99.78 $\pm$ 0.62	51.52 $\pm$ 1.33	34.86 $\pm$ 1.15

Table 29: 100-shot configuration results for Transformer across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline rank supplied measurements within each objective and dataset; lower is better for MAE.

Methods	Photo	ogbn-arxiv	DBLP	Airports	Chameleon	Cornell	QM7b	ToxCast	MNIST
	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	NC / Acc. ↑	NC / Acc. ↑	LP / AUC ↑	GR / MAE ↓	GC / AUC ↑	GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	88.99±0.57	40.54±0.95	56.18±0.57	41.42±2.51	39.83±0.81	58.53±9.79	16.45±1.96	51.84±1.75	35.59±1.38
All-in-One	88.32±0.76	43.98±0.27	53.70±0.21	40.33±1.07	38.95±1.06	50.63±9.55	49.04±14.78	52.67±0.65	28.76±1.76
EdgePrompt	63.13±15.86	46.14±0.51	53.99±0.71	39.27±1.29	39.14±0.44	51.22±13.55	44.90±0.59	49.33±1.39	25.07±3.01
EdgePrompt+	57.23±16.89	42.10±0.70	53.05±0.99	40.08±1.09	40.36±0.59	56.98±8.36	43.96±0.47	51.16±1.29	28.83±1.87
GPF	88.28±0.76	45.29±0.56	53.68±0.26	40.10±1.34	35.33±10.37	49.87±11.28	104.48±0.73	48.69±1.48	24.06±1.54
GPF+	89.23±0.78	42.39±0.43	52.62±0.45	39.57±0.84	38.73±0.56	53.15±14.38	104.39±0.69	50.90±1.43	22.89±1.70
GPPT	87.64±0.96	27.97±2.54	50.18±0.87	35.19±4.50	37.84±1.76	47.61±9.73	112.68±3.02	47.64±1.42	16.12±1.10
GraphPrompt	86.04±0.62	22.67±0.93	52.82±0.26	38.63±1.21	30.40±3.69	48.18±11.59	22.57±0.21	41.72±1.28	36.98±0.85
GraphPrompt+	78.17±2.25	22.58±0.48	53.12±0.22	37.95±0.97	26.07±0.98	43.52±12.31	19.59±0.92	41.91±0.80	32.32±2.40
ProNoG	84.25±0.41	40.26±0.56	55.75±0.68	36.63±2.12	34.63±0.79	52.88±11.77	8.00±2.23	40.75±1.09	27.14±0.33
Pretraining: ContextPred									
Full fine-tuning	88.76±0.60	39.71±0.61	56.45±0.20	41.62±1.36	39.76±0.52	68.73±8.00	8.79±1.11	51.96±2.10	34.82±2.38
All-in-One	88.26±0.30	38.85±0.29	53.19±0.34	40.30±2.48	38.06±0.54	52.25±7.66	54.76±0.50	49.17±1.16	33.91±0.86
EdgePrompt	71.63±17.47	38.20±0.35	58.13±0.66	42.35±1.22	38.13±0.75	66.56±8.43	90.15±0.35	51.23±1.14	26.25±3.09
EdgePrompt+	77.51±7.02	32.95±1.22	56.12±0.73	46.00±0.86	39.46±1.39	64.00±11.39	90.27±0.47	51.35±0.98	25.17±1.46
GPF	88.73±0.42	39.67±0.18	57.78±1.26	41.42±0.93	37.68±0.48	61.64±14.48	90.25±1.50	50.99±1.18	28.01±1.69
GPF+	87.72±0.81	35.47±3.00	55.63±0.73	42.08±1.19	38.96±1.24	59.95±11.33	91.63±4.49	50.57±1.09	26.97±2.81
GPPT	87.51±0.90	11.63±3.63	50.56±1.15	38.08±2.43	35.58±1.96	52.44±10.67	90.08±7.13	50.54±0.33	20.40±1.71
GraphPrompt	86.85±0.36	14.89±0.62	52.21±0.52	40.51±1.98	27.93±1.02	64.88±16.03	18.88±0.16	43.06±0.81	28.70±0.68
GraphPrompt+	85.58±1.16	16.44±0.98	52.56±1.15	38.94±1.67	26.91±1.86	57.29±14.44	19.13±0.33	41.85±0.96	29.31±1.15
ProNoG	86.04±0.47	27.37±0.49	55.23±0.30	34.08±2.58	30.37±0.56	60.05±14.08	9.89±1.44	41.06±0.95	23.52±0.71
Pretraining: DGI									
Full fine-tuning	88.82±0.76	37.92±1.44	56.84±0.49	40.73±1.79	40.41±1.75	59.62±5.74	5.89±0.24	52.15±1.01	32.13±4.42
All-in-One	87.22±0.68	33.51±2.23	53.70±0.42	33.82±4.59	39.95±0.43	50.56±7.28	14.89±2.88	54.57±0.46	25.74±2.23
EdgePrompt	71.72±12.57	40.57±1.48	57.25±0.73	40.46±3.83	39.74±0.96	56.46±15.69	19.81±16.43	52.85±1.73	23.42±2.98
EdgePrompt+	80.06±9.29	35.75±3.33	56.23±0.73	45.37±0.92	40.34±0.92	63.21±10.34	11.61±0.57	52.97±0.89	22.56±0.98
GPF	89.00±0.79	33.52±1.55	56.78±0.58	40.25±2.92	40.79±0.38	46.54±17.26	64.01±2.41	52.96±0.59	24.37±1.76
GPF+	87.86±0.48	34.58±2.59	54.62±0.97	39.92±2.32	39.89±2.13	43.59±7.86	66.12±9.76	52.78±0.71	22.20±4.55
GPPT	84.27±2.48	35.37±3.55	50.88±1.32	36.78±1.65	39.43±1.18	49.98±2.29	59.32±17.17	50.43±1.15	22.13±0.83
GraphPrompt	81.06±1.13	12.18±1.53	52.43±0.59	33.04±3.90	27.87±2.56	54.77±14.35	18.81±0.14	43.97±1.20	30.94±0.18
GraphPrompt+	79.18±3.28	17.75±1.26	52.83±0.28	36.48±2.34	29.18±1.88	53.39±14.21	21.00±2.40	44.31±1.80	29.66±1.99
ProNoG	79.58±0.58	38.97±0.43	55.63±0.45	30.94±1.41	31.77±4.53	56.79±11.48	17.96±9.94	41.33±0.95	25.78±1.09
Pretraining: EdgePred									
Full fine-tuning	88.82±0.64	39.87±1.24	55.53±0.29	42.51±0.94	39.01±0.76	58.45±11.47	8.21±1.20	51.89±2.22	34.93±1.10
All-in-One	62.37±0.46	27.31±0.64	52.11±0.67	41.24±1.72	36.70±0.47	53.01±7.12	54.76±0.50	54.21±0.70	31.66±1.16
EdgePrompt	41.14±17.45	24.06±1.08	51.32±0.74	41.97±1.98	35.86±1.02	48.75±13.15	90.15±0.35	51.78±0.93	25.17±1.92
EdgePrompt+	58.79±21.33	24.25±1.51	51.35±1.05	45.62±1.58	36.42±0.64	56.74±13.35	90.27±0.47	51.36±0.43	31.22±1.09
GPF	80.82±1.37	31.81±1.03	51.70±0.60	40.53±1.75	39.03±0.81	51.49±10.52	90.25±1.50	51.43±1.19	26.86±0.68
GPF+	87.28±0.62	33.40±0.94	52.99±0.51	40.03±2.48	37.42±5.17	51.80±7.72	90.05±1.34	51.96±0.71	27.13±0.59
GPPT	55.66±0.55	4.00±2.02	49.80±0.40	38.35±1.91	31.66±0.50	59.23±6.66	90.08±7.13	50.05±0.69	18.69±1.41
GraphPrompt	43.81±0.77	6.92±1.44	50.10±0.26	37.24±0.88	27.56±3.62	61.00±9.34	57.92±53.39	44.36±1.15	31.58±0.27
GraphPrompt+	78.39±4.71	11.74±0.98	50.48±0.51	38.38±1.22	27.82±0.96	42.71±10.78	19.13±0.33	41.58±1.02	30.05±2.11
ProNoG	46.25±0.54	11.93±0.53	54.00±0.93	36.00±2.37	31.96±1.28	64.11±10.62	9.89±1.44	41.85±0.92	27.33±0.61
Pretraining: GraphCL									
Full fine-tuning	88.45±0.38	39.53±0.89	56.32±0.51	41.52±2.31	39.74±0.86	59.57±5.15	6.82±0.58	52.69±0.90	34.33±2.28
All-in-One	87.91±0.11	40.34±0.37	53.92±0.32	41.57±1.82	38.62±0.74	47.73±6.86	37.73±0.51	55.70±1.64	37.50±0.55
EdgePrompt	66.19±16.31	40.00±0.48	54.27±0.55	41.34±2.02	37.92±0.84	44.95±11.71	96.19±0.61	51.88±1.07	28.55±2.73
EdgePrompt+	73.85±15.55	37.35±0.66	53.98±0.96	43.65±1.56	39.79±0.89	52.77±6.45	96.49±0.41	52.11±1.18	34.92±0.97
GPF	88.17±0.27	40.40±1.12	52.37±0.91	40.53±1.31	37.60±0.53	48.87±9.90	100.57±0.86	50.79±1.09	28.53±0.74
GPF+	88.39±1.29	41.04±0.86	52.60±0.31	40.66±1.08	38.55±1.45	52.56±7.49	100.50±0.74	50.55±0.82	31.43±1.65
GPPT	86.96±0.70	25.16±1.31	51.37±0.75	39.29±1.72	38.96±0.95	47.55±7.70	106.20±2.97	49.27±1.26	21.42±1.32
GraphPrompt	84.04±0.69	20.52±1.19	52.98±0.42	40.18±1.50	25.73±2.90	50.13±7.25	9.73±0.32	46.27±1.54	35.07±0.59
GraphPrompt+	87.49±0.50	21.74±1.11	52.97±0.10	39.06±1.37	29.15±5.22	46.52±9.45	12.67±4.78	44.23±2.13	37.69±0.44
ProNoG	83.92±0.67	33.08±0.28	56.44±0.53	36.48±0.97	33.82±1.90	52.58±11.32	5.45±0.65	44.64±1.30	29.42±0.59
Pretraining: InfoGraph									
Full fine-tuning	88.91±0.70	39.06±1.05	56.81±0.40	42.66±2.37	40.42±0.89	69.37±3.03	7.07±0.71	52.75±0.60	22.72±1.88
All-in-One	86.79±0.20	41.34±0.30	53.07±0.36	42.51±1.50	39.46±0.76	55.36±5.50	84.43±0.71	59.71±0.56	16.54±2.59
EdgePrompt	55.51±27.19	43.71±0.62	57.99±1.15	42.51±1.31	42.10±0.96	54.22±7.41	105.37±0.42	52.46±1.10	27.50±1.11
EdgePrompt+	53.28±13.47	39.53±0.84	55.98±1.01	47.01±1.15	42.15±0.88	51.15±8.78	105.65±0.54	52.66±0.49	25.22±3.24
GPF	87.53±0.29	44.70±0.49	58.22±1.30	41.95±1.55	41.36±1.30	53.65±7.44	112.56±1.78	51.81±0.79	12.50±1.49
GPF+	87.49±0.49	42.19±0.75	53.27±0.75	41.52±1.25	40.69±0.97	54.82±8.45	112.54±1.93	52.22±0.81	12.33±1.41
GPPT	82.81±0.48	19.99±0.70	51.53±1.23	40.08±0.97	40.25±0.64	48.53±6.86	115.95±1.02	50.35±0.15	16.58±0.64
GraphPrompt	79.87±0.36	17.83±1.30	53.11±0.46	37.92±2.45	31.09±2.45	63.40±11.07	54.54±0.94	44.73±1.51	27.08±0.39
GraphPrompt+	81.96±0.17	18.11±1.19	54.41±1.14	38.80±1.85	28.54±1.56	53.53±12.51	19.78±5.87	44.35±1.38	29.25±1.46
ProNoG	78.30±1.09	34.27±0.37	55.83±0.30	35.85±1.55	36.92±1.89	58.78±12.56	4.06±0.07	41.80±1.33	20.61±0.81

Table 30: 100-shot configuration results for NodeFormer across six pretraining objectives and ten adaptation methods. Accuracy and ROC-AUC are percentages; QM7b uses MAE. LP partitions and preprocessing conventions follow Appendix B.3. Bold and underline identify the highest and second distinct performance means within each objective and dataset; lower is better for MAE.

Methods	Photo NC / Acc. ↑	ogbn-arxiv NC / Acc. ↑	DBLP LP / AUC ↑	Airports NC / Acc. ↑	Chameleon NC / Acc. ↑	Cornell LP / AUC ↑	QM7b GR / MAE ↓	ToxCast GC / AUC ↑	MNIST GC / Acc. ↑
Pretraining: AttrMasking									
Full fine-tuning	87.59±0.71	6.79±1.02	58.65±2.34	26.41±1.13	30.06±2.05	51.89±12.50	25.18±5.26	51.58±1.18	26.66±2.82
All-in-One	82.10±0.83	29.51±1.61	54.83±4.89	31.29±2.92	35.84±1.49	42.38±23.42	80.11±8.71	44.41±1.42	25.18±0.52
EdgePrompt	73.61±4.89	15.16±7.36	58.84±3.95	29.39±4.03	27.52±1.87	50.51±19.43	106.56±3.14	50.84±0.08	18.66±1.71
EdgePrompt+	73.64±8.11	15.79±7.97	57.27±2.83	32.43±4.54	29.40±3.20	55.81±17.89	105.13±0.20	50.78±0.22	23.42±1.42
GPF	82.62±1.49	9.87±6.66	52.42±1.36	25.29±0.33	25.64±1.52	55.24±9.79	107.78±0.66	50.72±0.19	19.30±3.66
GPF+	82.41±2.16	9.44±5.09	54.40±2.47	27.11±1.64	29.41±3.33	47.68±17.35	107.18±0.51	50.87±0.24	16.10±4.39
GPPT	78.39±3.98	3.40±0.66	50.01±2.00	33.57±3.05	27.05±4.16	60.36±19.46	102.02±1.41	50.82±0.62	15.64±0.86
GraphPrompt	73.09±2.13	10.73±1.05	58.65±3.19	37.92±1.09	26.78±1.71	55.27±11.92	20.70±2.19	40.47±1.21	29.14±0.68
GraphPrompt+	73.94±3.22	5.63±1.63	55.88±2.83	39.52±0.94	30.06±2.30	53.15±17.98	8.10±4.09	40.60±1.32	28.91±2.47
ProNoG	82.84±0.91	36.83±0.67	54.23±0.62	36.76±1.14	30.89±2.68	49.95±11.24	4.94±0.54	40.26±0.90	24.00±0.40
Pretraining: ContextPred									
Full fine-tuning	88.41±1.39	30.46±15.09	58.28±3.73	29.39±3.09	34.23±0.81	49.04±11.34	24.04±5.46	50.89±0.41	21.49±3.51
All-in-One	87.18±1.26	30.91±0.98	52.13±0.09	34.00±1.65	36.99±1.34	56.74±10.53	101.71±0.87	56.20±0.79	31.72±0.68
EdgePrompt	75.11±5.85	4.87±3.68	49.56±4.42	27.70±4.50	29.35±4.06	57.74±7.54	108.43±0.04	49.39±0.05	20.14±1.83
EdgePrompt+	66.25±4.81	9.85±2.94	53.42±4.01	24.53±4.76	30.30±5.44	54.34±10.17	108.34±0.08	49.49±0.15	20.12±4.04
GPF	86.33±1.25	34.48±2.96	51.68±1.54	25.42±1.17	32.65±1.41	55.29±8.68	111.72±2.98	50.67±0.24	17.28±1.68
GPF+	86.88±0.90	33.69±2.21	51.80±2.20	26.10±1.62	32.54±1.55	50.68±3.44	110.37±0.04	50.69±0.35	21.27±4.32
GPPT	83.71±2.54	1.09±0.67	53.84±3.53	33.27±1.77	35.26±1.54	66.83±9.07	107.01±0.04	49.34±1.05	20.11±2.18
GraphPrompt	80.75±1.78	6.93±1.01	56.72±4.09	38.00±1.88	29.06±2.05	41.24±10.11	45.29±0.26	40.56±1.19	26.85±0.30
GraphPrompt+	84.34±1.06	17.18±2.68	57.49±2.09	38.71±1.80	30.88±3.79	50.37±10.14	36.92±0.63	42.00±2.12	28.58±1.21
ProNoG	82.84±0.93	31.24±0.70	55.90±0.65	36.99±1.16	28.24±0.85	43.93±3.68	13.57±0.67	40.22±0.85	22.52±0.42
Pretraining: DGI									
Full fine-tuning	88.03±1.32	38.51±0.88	51.10±1.60	31.97±4.97	34.92±1.27	49.58±9.00	33.40±13.94	51.24±0.49	14.07±6.25
All-in-One	88.25±0.30	40.19±0.40	49.61±3.29	32.53±0.75	37.65±0.71	49.54±13.69	90.50±0.26	54.00±0.57	10.99±0.16
EdgePrompt	29.11±17.34	8.20±5.94	52.61±3.24	26.66±5.82	29.15±4.41	54.15±27.81	107.48±0.04	51.81±0.58	10.72±0.74
EdgePrompt+	30.66±19.40	8.03±3.16	51.33±4.33	27.06±4.31	29.56±3.89	40.98±32.28	107.50±0.05	51.79±0.60	11.26±0.06
GPF	71.53±14.95	32.57±3.76	51.15±1.92	25.09±0.14	30.77±5.51	48.30±20.54	107.79±0.03	52.44±0.21	10.44±0.75
GPF+	57.07±25.72	29.31±1.69	51.78±1.82	27.29±4.91	34.33±3.29	44.83±28.65	107.79±0.03	51.87±1.23	9.98±0.87
GPPT	54.41±14.94	4.47±3.29	51.19±1.73	34.30±2.34	33.98±1.79	49.29±13.88	103.05±0.01	50.69±1.60	9.85±1.13
GraphPrompt	75.44±1.20	10.15±0.30	49.28±1.51	38.81±0.90	28.90±2.08	46.11±17.65	19.71±0.52	42.26±1.38	16.92±0.61
GraphPrompt+	80.85±3.62	20.56±1.46	50.40±2.45	36.86±1.59	31.47±3.49	55.36±8.78	31.04±1.18	43.02±1.45	16.76±1.01
ProNoG	62.37±2.86	18.41±0.77	55.13±1.29	27.87±3.02	30.68±1.17	54.36±12.76	14.30±1.16	40.63±1.08	16.47±0.44
Pretraining: EdgePred									
Full fine-tuning	85.45±0.43	38.07±1.61	55.65±2.55	25.52±1.98	32.70±3.66	49.51±16.90	36.65±20.82	52.35±1.07	10.22±0.93
All-in-One	31.96±3.65	23.40±2.71	50.91±4.74	36.58±2.83	35.49±1.22	47.87±18.53	101.71±0.87	56.28±0.38	25.67±0.32
EdgePrompt	9.44±0.69	1.33±1.80	51.93±0.61	31.39±4.71	25.63±5.09	52.44±10.36	108.43±0.04	52.94±0.34	9.88±0.03
EdgePrompt+	25.62±7.18	6.99±4.08	51.44±1.75	31.85±2.08	29.26±5.60	51.70±23.20	108.34±0.08	53.10±0.58	22.29±2.45
GPF	22.76±4.12	20.00±5.20	52.65±1.11	25.29±0.66	28.81±2.88	55.65±13.12	111.49±3.11	52.73±0.63	13.47±4.92
GPF+	24.24±6.29	22.97±5.65	50.99±1.60	27.32±1.84	29.79±1.08	51.77±12.77	110.37±0.04	52.80±0.61	12.02±1.15
GPPT	20.10±8.70	3.82±5.01	49.94±0.11	34.33±3.10	25.17±3.72	55.18±9.57	106.76±0.04	50.27±0.48	10.56±1.17
GraphPrompt	19.63±1.15	5.65±1.86	51.03±1.35	38.03±0.92	30.48±2.42	48.68±11.58	57.83±33.08	42.76±1.49	22.28±0.71
GraphPrompt+	33.18±3.52	13.12±1.35	52.91±2.25	34.78±3.25	29.06±4.15	50.13±8.83	36.92±0.63	43.69±1.70	19.27±0.98
ProNoG	39.30±0.54	7.42±4.25	56.13±0.72	29.57±2.99	29.71±2.90	61.66±12.23	13.56±0.74	41.30±0.97	22.87±0.53
Pretraining: GraphCL									
Full fine-tuning	83.00±3.61	21.15±2.22	55.44±6.74	25.72±1.38	35.26±1.28	38.43±18.70	62.89±24.16	52.42±1.17	18.70±7.91
All-in-One	77.80±2.32	30.50±1.93	53.05±1.34	25.16±2.32	37.86±1.00	47.56±18.66	107.23±0.05	53.59±1.20	9.72±0.37
EdgePrompt	57.40±4.60	17.41±0.83	53.35±2.92	24.94±0.00	35.99±1.58	52.44±6.82	113.76±3.05	51.43±0.80	10.09±0.66
EdgePrompt+	62.93±5.84	21.06±0.99	50.52±3.93	24.91±0.06	36.44±0.81	51.49±10.11	114.87±3.04	51.44±1.02	10.14±0.63
GPF	67.20±11.05	12.97±3.33	52.06±2.56	24.84±0.70	34.30±3.11	56.62±5.71	117.10±0.03	51.44±1.23	12.30±5.35
GPF+	70.50±4.83	11.54±2.19	49.14±3.21	25.90±1.48	35.03±2.65	52.70±15.87	112.17±0.07	52.12±0.65	10.33±1.02
GPPT	73.44±2.94	11.00±2.61	51.38±3.69	34.23±2.50	33.76±0.92	42.58±15.81	110.55±0.01	50.49±1.67	9.93±0.28
GraphPrompt	74.78±2.90	14.29±1.24	49.32±1.11	38.28±1.46	30.30±2.31	45.66±9.03	84.05±0.13	45.59±1.46	16.25±1.82
GraphPrompt+	80.75±1.92	15.11±0.95	51.67±2.65	38.33±1.19	31.40±4.99	52.68±13.15	33.00±2.84	45.36±1.32	17.86±0.88
ProNoG	80.55±1.23	30.51±0.36	53.92±0.61	36.56±2.49	31.81±0.66	46.53±12.63	13.55±0.37	45.69±1.34	16.18±0.19
Pretraining: InfoGraph									
Full fine-tuning	84.48±2.48	17.56±2.34	54.18±1.73	25.87±1.43	33.21±1.53	52.56±6.79	45.94±33.32	51.80±1.17	16.67±6.26
All-in-One	81.94±1.02	7.03±2.32	50.31±1.54	39.04±1.81	36.87±1.60	52.41±7.80	103.81±0.32	56.55±1.09	9.90±0.00
EdgePrompt	51.74±2.80	1.88±1.41	52.48±0.21	33.37±1.41	28.52±1.37	55.55±13.57	111.26±3.24	51.20±0.97	9.84±0.02
EdgePrompt+	59.12±2.03	4.55±2.11	52.54±0.25	31.04±1.94	27.37±0.48	53.01±12.80	109.86±0.06	51.28±0.99	12.61±4.02
GPF	75.99±0.82	7.51±4.53	52.97±1.06	25.54±0.39	30.40±3.87	47.85±12.44	117.07±0.02	51.80±1.19	11.87±3.79
GPF+	65.04±14.95	5.37±3.47	53.68±0.77	27.09±1.77	32.99±1.44	51.61±39.11	111.25±1.12	52.48±0.96	11.85±3.75
GPPT	68.86±5.47	5.55±6.02	49.86±0.10	35.04±6.00	25.66±5.35	57.40±19.11	109.31±0.03	50.42±0.79	9.99±0.39
GraphPrompt	73.49±1.59	4.72±1.91	51.34±1.53	37.06±1.48	31.01±4.88	52.46±10.50	68.60±0.10	44.03±1.43	18.62±1.32
GraphPrompt+	77.54±5.38	3.85±2.06	53.29±0.88	37.77±1.53	27.66±3.09	44.30±10.09	40.78±2.47	44.33±1.39	19.67±1.83
ProNoG	63.42±1.19	24.29±0.65	52.18±0.53	37.37±1.41	30.29±0.91	43.01±10.96	13.41±0.20	43.17±1.32	17.71±0.41